

Building A · FIRST FLOOR (Emergency light scene)

Room list



Building A · FIRST FLOOR (Emergency light scene)

Room list

201 STARIS

P_{total}
0.9 W**A_{Room}**
4.00 m²**Lighting power density**
0.22 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)	0.9 W	300 lm (100 %)

202 CORRIDOR

P_{total}
11.8 W**A_{Room}**
27.25 m²**Lighting power density**
0.43 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
2	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)	0.9 W	300 lm (100 %)
4	ETAP	-	K9R142/1-AX1 (_LTRevAB)	2.5 W	80 lm (100 %)

203 MEETING

P_{total}
0.9 W**A_{Room}**
6.59 m²**Lighting power density**
0.14 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Emergency light scene)

Room list

204 MANAGER

P_{total}
1.8 W

A_{Room}
24.86 m²

Lighting power density
0.07 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
2	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

206 TOILET

P_{total}
0.9 W

A_{Room}
5.03 m²

Lighting power density
0.18 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

207 OFFICE

P_{total}
0.9 W

A_{Room}
15.71 m²

Lighting power density
0.06 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Emergency light scene)

Room list

208 MEETING

P_{total}
0.9 W

A_{Room}
11.30 m²

Lighting power density
0.08 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

209 MEETING

P_{total}
0.9 W

A_{Room}
12.60 m²

Lighting power density
0.07 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

210 CAD

P_{total}
0.9 W

A_{Room}
34.98 m²

Lighting power density
0.03 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Emergency light scene)

Room list

211 CORRIDOR

P_{total}
18.8 W

A_{Room}
59.96 m²

Lighting power density
0.31 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
7	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)
5	ETAP	-	K9R142/1-AX1 (_LTRRevAB)	2.5 W	80 lm (100 %)

212 COMMON

P_{total}
0.9 W

A_{Room}
41.15 m²

Lighting power density
0.02 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

214 OFFICE

P_{total}
0.9 W

A_{Room}
13.60 m²

Lighting power density
0.07 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Emergency light scene)

Room list

215 MEETING

P_{total}
0.9 W

A_{Room}
12.81 m²

Lighting power density
0.07 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

216 MEETING

P_{total}
0.9 W

A_{Room}
12.65 m²

Lighting power density
0.07 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

219 MEETING

P_{total}
1.8 W

A_{Room}
8.23 m²

Lighting power density
0.22 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
2	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Emergency light scene)

Room list

220 OFFICE

 P_{total}
0.9 W A_{Room}
15.37 m²Lighting power density
0.06 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)	0.9 W	300 lm (100 %)

Building A · FIRST FLOOR (Light scene 1)

Room list



Building A · FIRST FLOOR (Light scene 1)

Room list

201 STARIS

P_{total}
33.0 W**A_{Room}**
4.00 m²**Lighting power density**
8.24 W/m² (Space)

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

202 CORRIDOR

P_{total}
165.0 W**A_{Room}**
27.25 m²**Lighting power density**
6.05 W/m² = 1.82 W/m²/100 lx (Space)**E_{perpendicular}** (Working plane)
332 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
5	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

203 MEETING

P_{total}
99.0 W**A_{Room}**
6.59 m²**Lighting power density**
15.03 W/m² = 1.96 W/m²/100 lx (Space)**E_{perpendicular}** (Working plane)
765 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
3	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

204 MANAGER

P_{total}
396.0 W**A_{Room}**
24.86 m²**Lighting power density**
15.93 W/m² = 1.57 W/m²/100 lx (Space)**E_{perpendicular} (Working plane)**
1013 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
12	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

205 BALCONY

P_{total}
70.0 W**A_{Room}**
7.38 m²**Lighting power density**
9.49 W/m² = 2.22 W/m²/100 lx (Space)**E_{perpendicular} (Working plane)**
427 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
2	ETAP	-	R3111/LEDW455DX1 (_LTRevAC)	35.0 W	5239 lm

206 TOILET

P_{total}
66.0 W**A_{Room}**
5.03 m²**Lighting power density**
13.12 W/m² = 2.43 W/m²/100 lx (Space)**E_{perpendicular} (Working plane)**
541 lx

pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
2	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

207 OFFICE

P_{total}
132.0 W

A_{Room}
15.71 m²

Lighting power density
8.40 W/m² = 1.33 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
632 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

208 MEETING

P_{total}
132.0 W

A_{Room}
11.30 m²

Lighting power density
11.68 W/m² = 1.50 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
780 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

209 MEETING

P_{total}
132.0 W

A_{Room}
12.60 m²

Lighting power density
10.48 W/m² = 1.45 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
724 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

210 CAD

P_{total} 297.0 W	A_{Room} 34.98 m ²	Lighting power density 8.49 W/m ² = 1.04 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 818 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
9	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

211 CORRIDOR

P_{total} 330.0 W	A_{Room} 59.96 m ²	Lighting power density 5.50 W/m ² = 1.79 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 307 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
10	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm

212 COMMON

P_{total} 210.0 W	A_{Room} 41.15 m ²	Lighting power density 5.10 W/m ² = 1.04 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 492 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
6	ETAP	-	R3111/LEDW455DX1 (_LTRevAC)	35.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

213 BALCONY

P_{total} 105.0 W	A_{Room} 10.84 m ²	Lighting power density 9.68 W/m ² = 2.76 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 350 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
3	ETAP	-	R3111/LEDW455DX1 (_LTRRevAC)	35.0 W	5239 lm

214 OFFICE

P_{total} 132.0 W	A_{Room} 13.60 m ²	Lighting power density 9.71 W/m ² = 1.31 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 743 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

215 MEETING

P_{total} 132.0 W	A_{Room} 12.81 m ²	Lighting power density 10.31 W/m ² = 1.37 W/m ² /100 lx (Space)	E_{perpendicular} (Working plane) 753 lx
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pcs.	Manufacturer	Article No.	Article name	P	Φ _{Luminaire}
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

216 MEETING

P_{total}
132.0 W

A_{Room}
12.65 m²

Lighting power density
10.43 W/m² = 1.43 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
732 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

217 TOILET

P_{total}
33.0 W

A_{Room}
5.03 m²

Lighting power density
6.56 W/m² = 2.32 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
283 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

218 TOILET

P_{total}
33.0 W

A_{Room}
4.60 m²

Lighting power density
7.18 W/m² = 2.40 W/m²/100 lx (Space)

$\bar{E}_{\text{perpendicular (Working plane)}}$
299 lx

pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR (Light scene 1)

Room list

219 MEETING

P_{total} 99.0 W	A_{Room} 8.23 m ²	Lighting power density 12.04 W/m ² = 1.61 W/m ² /100 lx (Space)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 747 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
3	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

220 OFFICE

P_{total} 132.0 W	A_{Room} 15.37 m ²	Lighting power density 8.59 W/m ² = 1.26 W/m ² /100 lx (Space)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 684 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm

Building A · FIRST FLOOR

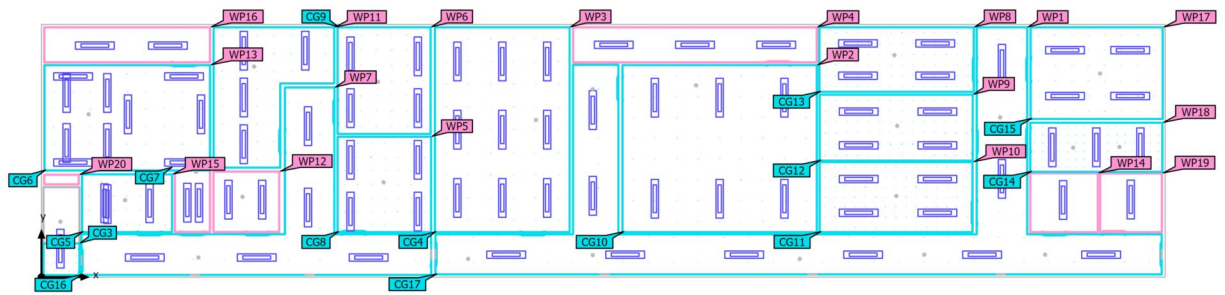
Luminaire list

Φ_{total} 459720 lm	P_{total} 2905.0 W	Luminous efficacy 158.3 lm/W	$\Phi_{\text{Emergency lighting}}$ 8220 lm	$P_{\text{Emergency lighting}}$ 45.0 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
25	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
9	ETAP	-	K9R142/1-AX1 (_LTRRevAB)	2.5 W	80 lm	32.4 lm/W
				 2.5 W	80 lm (100 %)	-
75	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W
11	ETAP	-	R3111/LEDW455DX1 (_LTRRevAC)	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR (Light scene 1)

Calculation objects



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (202 CORRIDOR) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	332 lx (≥ 500 lx) ✗	106 lx	476 lx	0.32 (≥ 0.60) ✗	0.22	WP7
Working plane (203 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	765 lx (≥ 500 lx) ✓	510 lx	1018 lx	0.67 (≥ 0.60) ✓	0.50	WP15
Working plane (204 MANAGER) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	1013 lx (≥ 500 lx) ✓	633 lx	1375 lx	0.62 (≥ 0.60) ✓	0.46	WP13
Working plane (204A DB) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	0.00 lx (≥ 500 lx) ✗	0.00 lx	0.00 lx	- (≥ 0.60)	-	WP20
Working plane (205 BALCONY) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	427 lx (≥ 500 lx) ✗	186 lx	766 lx	0.44 (≥ 0.60) ✗	0.24	WP16
Working plane (206 TOILET) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	541 lx (≥ 500 lx) ✓	447 lx	617 lx	0.83 (≥ 0.60) ✓	0.72	WP12
Working plane (207 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	632 lx (≥ 500 lx) ✓	331 lx	845 lx	0.52 (≥ 0.60) ✗	0.39	WP11
Working plane (208 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	780 lx (≥ 500 lx) ✓	572 lx	879 lx	0.73 (≥ 0.60) ✓	0.65	WP5
Working plane (209 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	724 lx (≥ 500 lx) ✓	508 lx	841 lx	0.70 (≥ 0.60) ✓	0.60	WP6
Working plane (210 CAD) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	818 lx (≥ 500 lx) ✓	432 lx	1043 lx	0.53 (≥ 0.60) ✗	0.41	WP3
Working plane (211 CORRIDOR) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	307 lx (≥ 500 lx) ✗	125 lx	448 lx	0.41 (≥ 0.60) ✗	0.28	WP1

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Working plane (212 COMMON) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	492 lx (≥ 500 lx) ✗	272 lx	702 lx	0.55 (≥ 0.60) ✗	0.39	WP2
Working plane (213 BALCONY) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	350 lx (≥ 500 lx) ✗	188 lx	588 lx	0.54 (≥ 0.60) ✗	0.32	WP4
Working plane (214 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	743 lx (≥ 500 lx) ✓	440 lx	965 lx	0.59 (≥ 0.60) ✗	0.46	WP10
Working plane (215 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	753 lx (≥ 500 lx) ✓	462 lx	940 lx	0.61 (≥ 0.60) ✓	0.49	WP9
Working plane (216 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	732 lx (≥ 500 lx) ✓	427 lx	928 lx	0.58 (≥ 0.60) ✗	0.46	WP8
Working plane (217 TOILET) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	283 lx (≥ 500 lx) ✗	227 lx	327 lx	0.80 (≥ 0.60) ✓	0.69	WP14
Working plane (218 TOILET) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	299 lx (≥ 500 lx) ✗	215 lx	340 lx	0.72 (≥ 0.60) ✓	0.63	WP19
Working plane (219 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	747 lx (≥ 500 lx) ✓	531 lx	908 lx	0.71 (≥ 0.60) ✓	0.58	WP18
Working plane (220 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	684 lx (≥ 500 lx) ✓	400 lx	861 lx	0.58 (≥ 0.60) ✗	0.46	WP17

Calculation surfaces

Properties	$\bar{E}$	E_{min}	E_{max}	$U_o (g_1)$	g_2	Index
201 Stair Upper Perpendicular illuminance Height: 0.000 m	331 lx	291 lx	356 lx	0.88	0.82	CG3

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

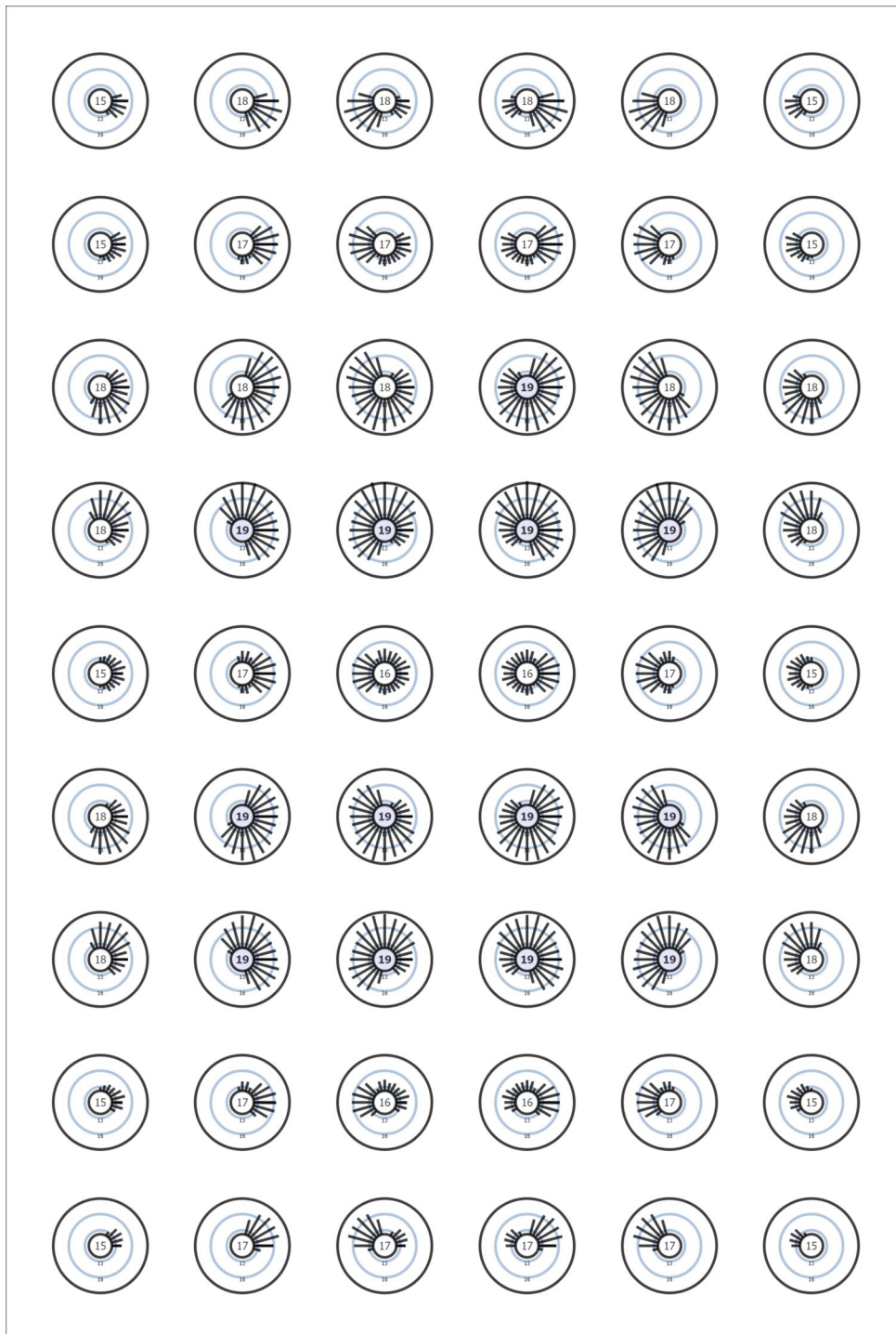
Glare plane 210 (RUG)

Strongest glare at	105°
max	19.3
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG4

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 210 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

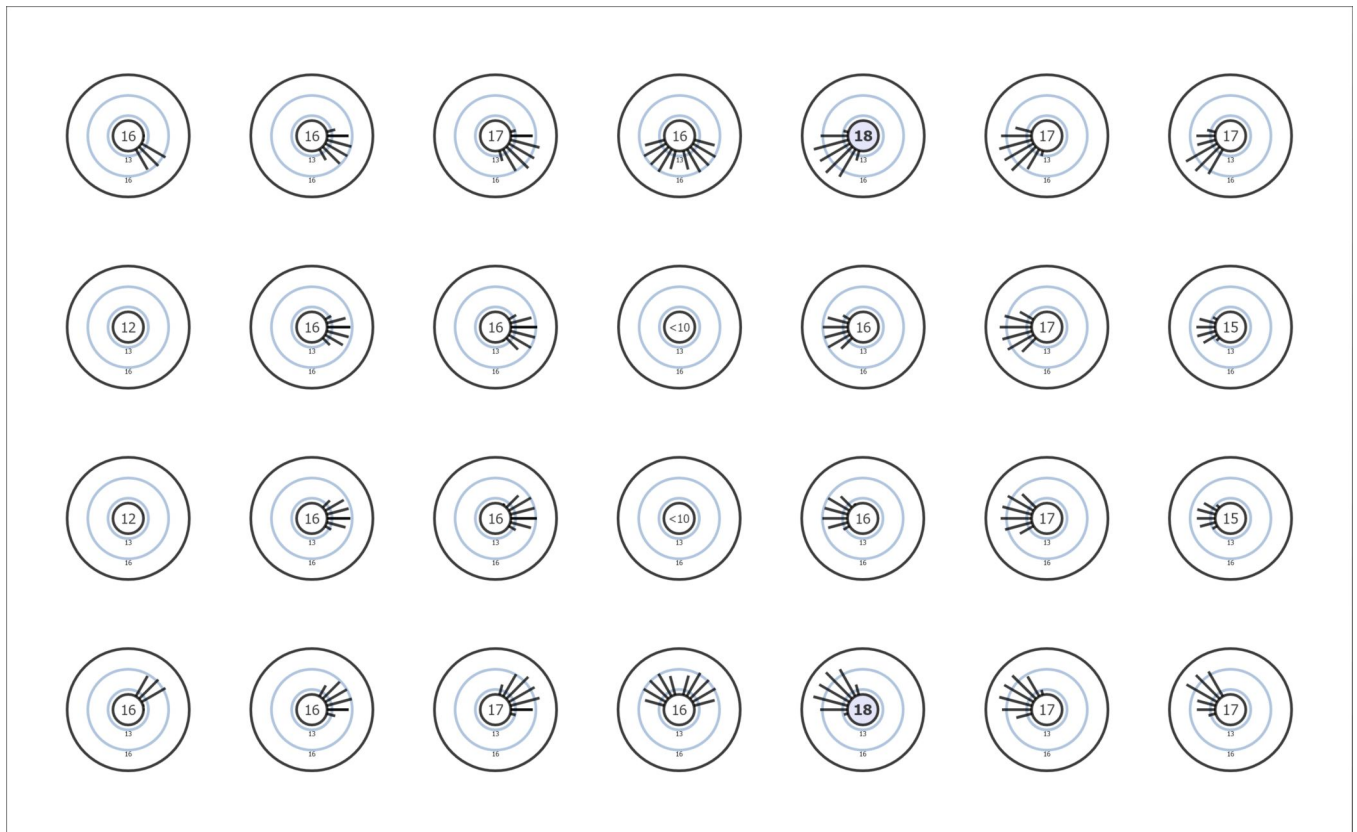
Glare plane 203 (RUG)

Strongest glare at	135°
max	17.7
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG5

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 203 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

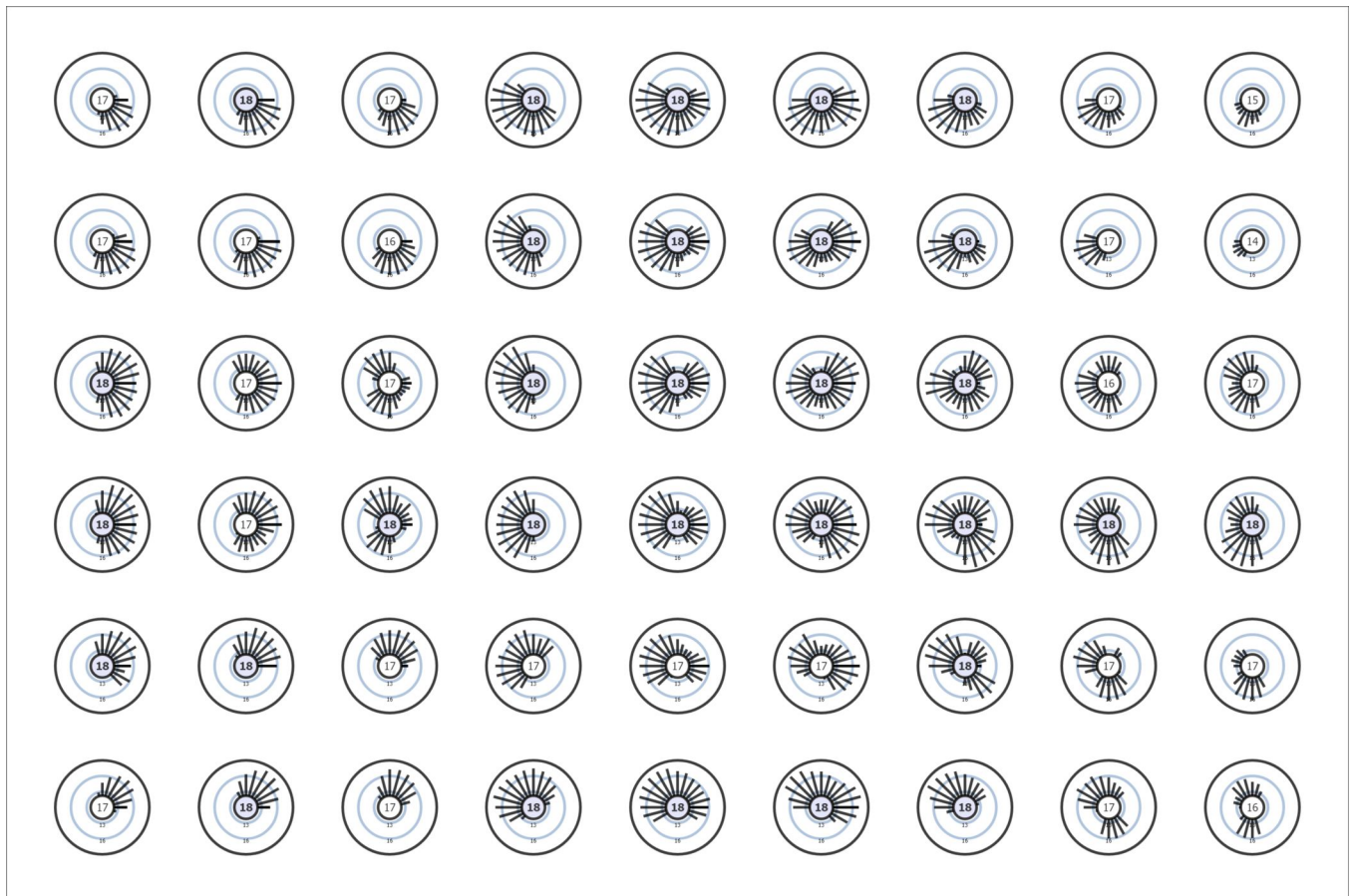
Glare Plane 204 (RUG)

Strongest glare at	285°
max	18.4
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG6

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare Plane 204 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

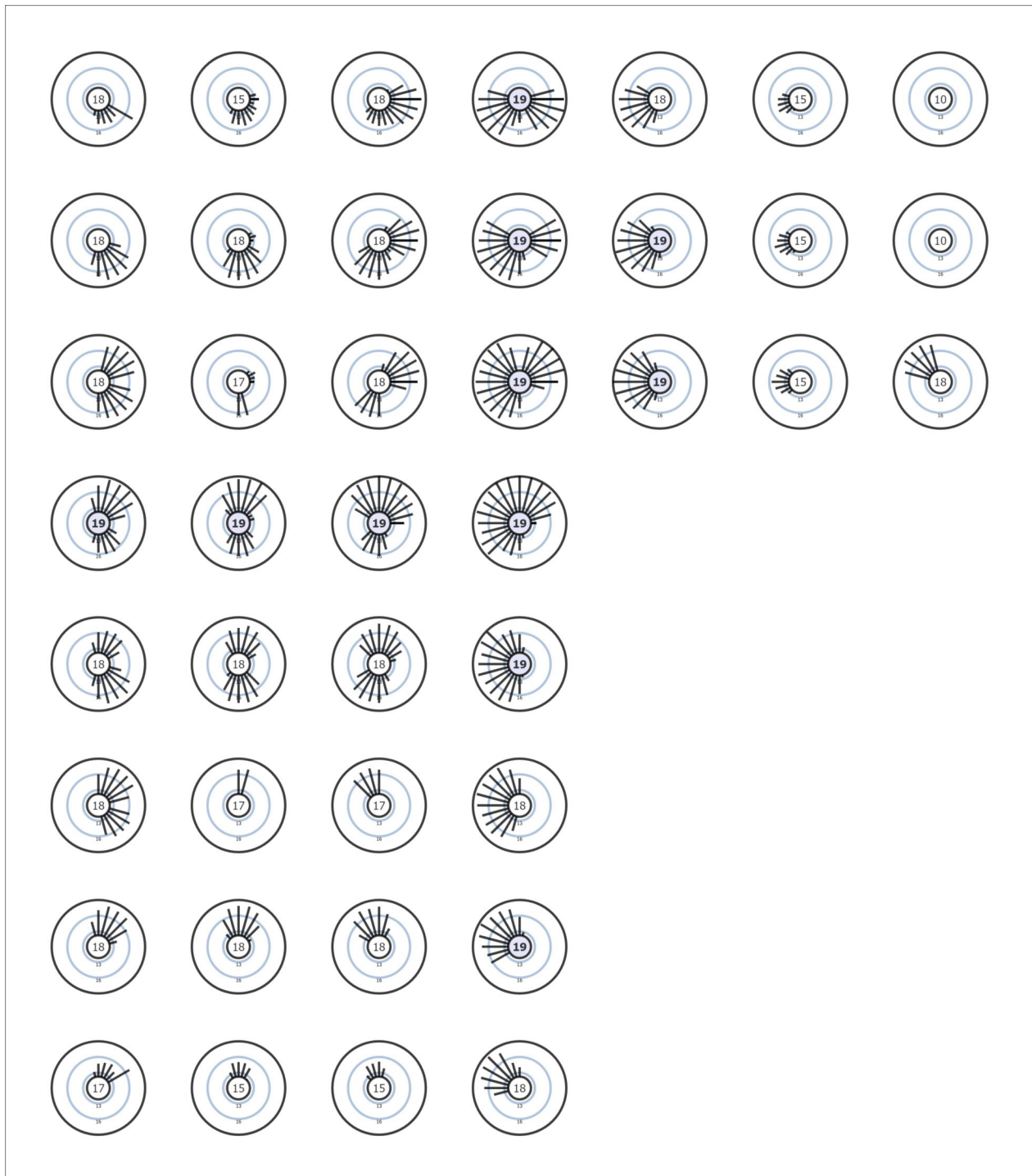
Glare Plane 207 (RUG)

Strongest glare at	45°
max	19.2
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG7

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare Plane 207 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

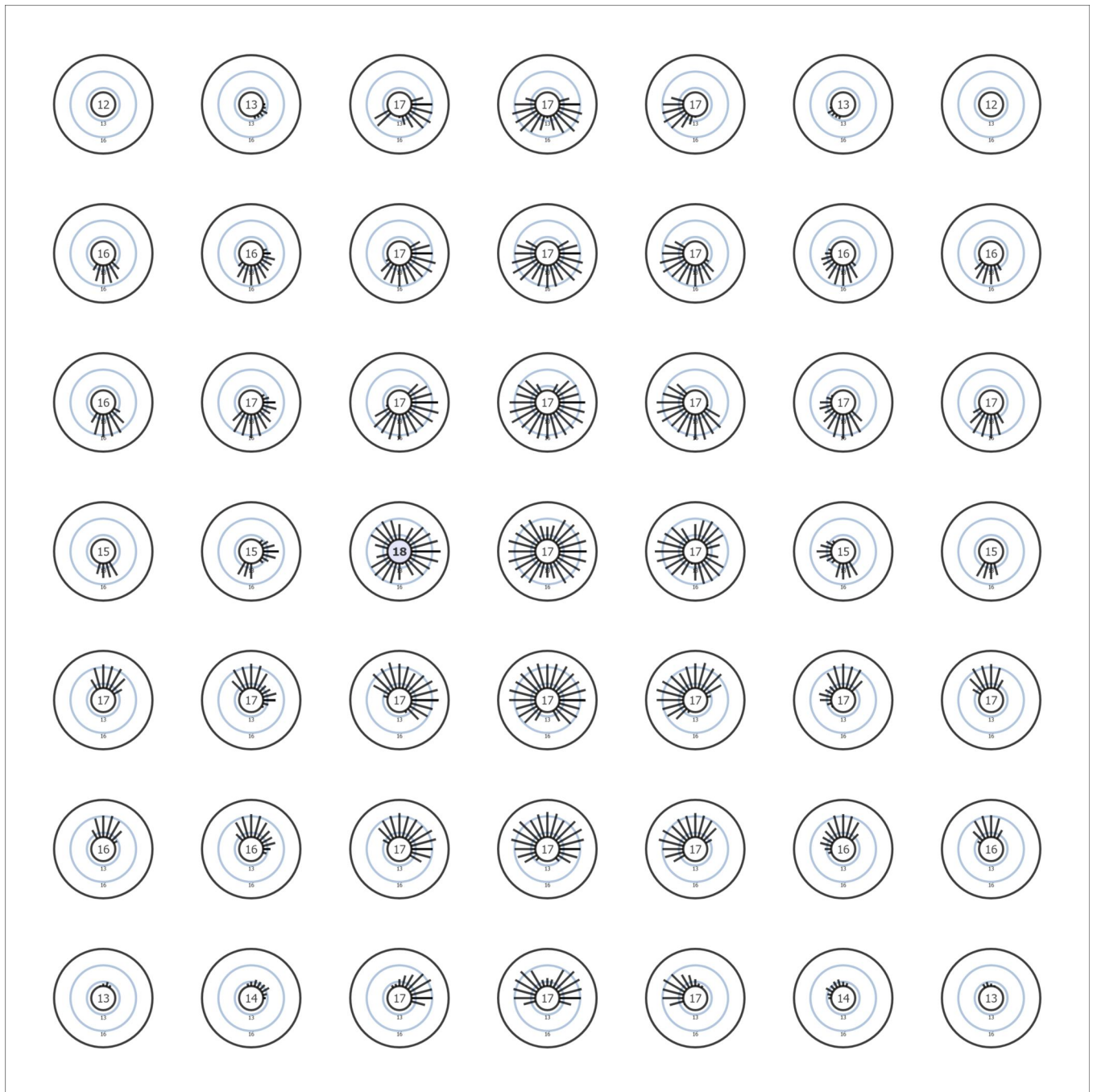
Glare Plane 208 (RUG)

Strongest glare at	0°
max	17.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG8

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare Plane 208 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

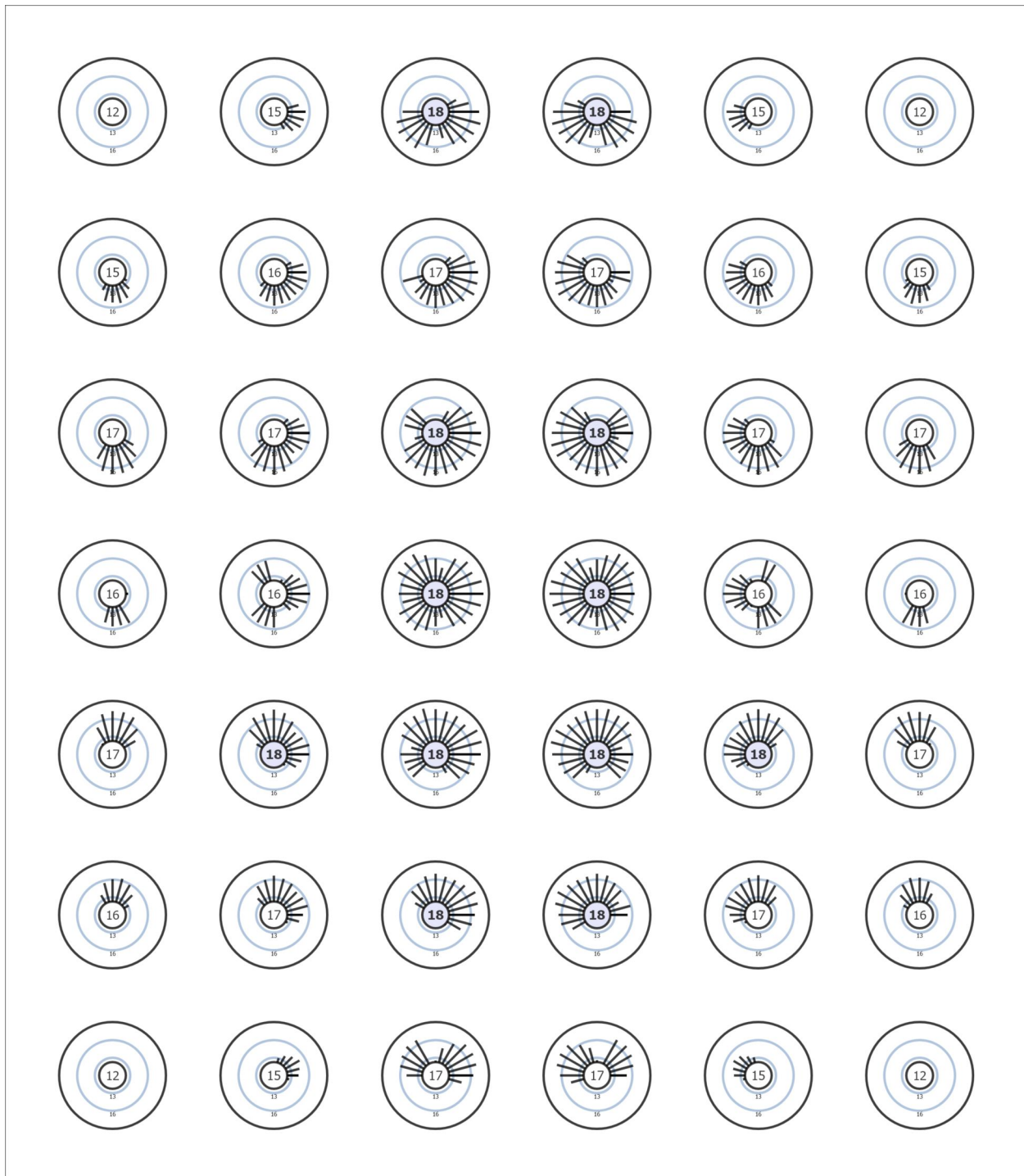
Glare Plane 209 (RUG)

Strongest glare at	0°
max	18.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG9

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare Plane 209 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

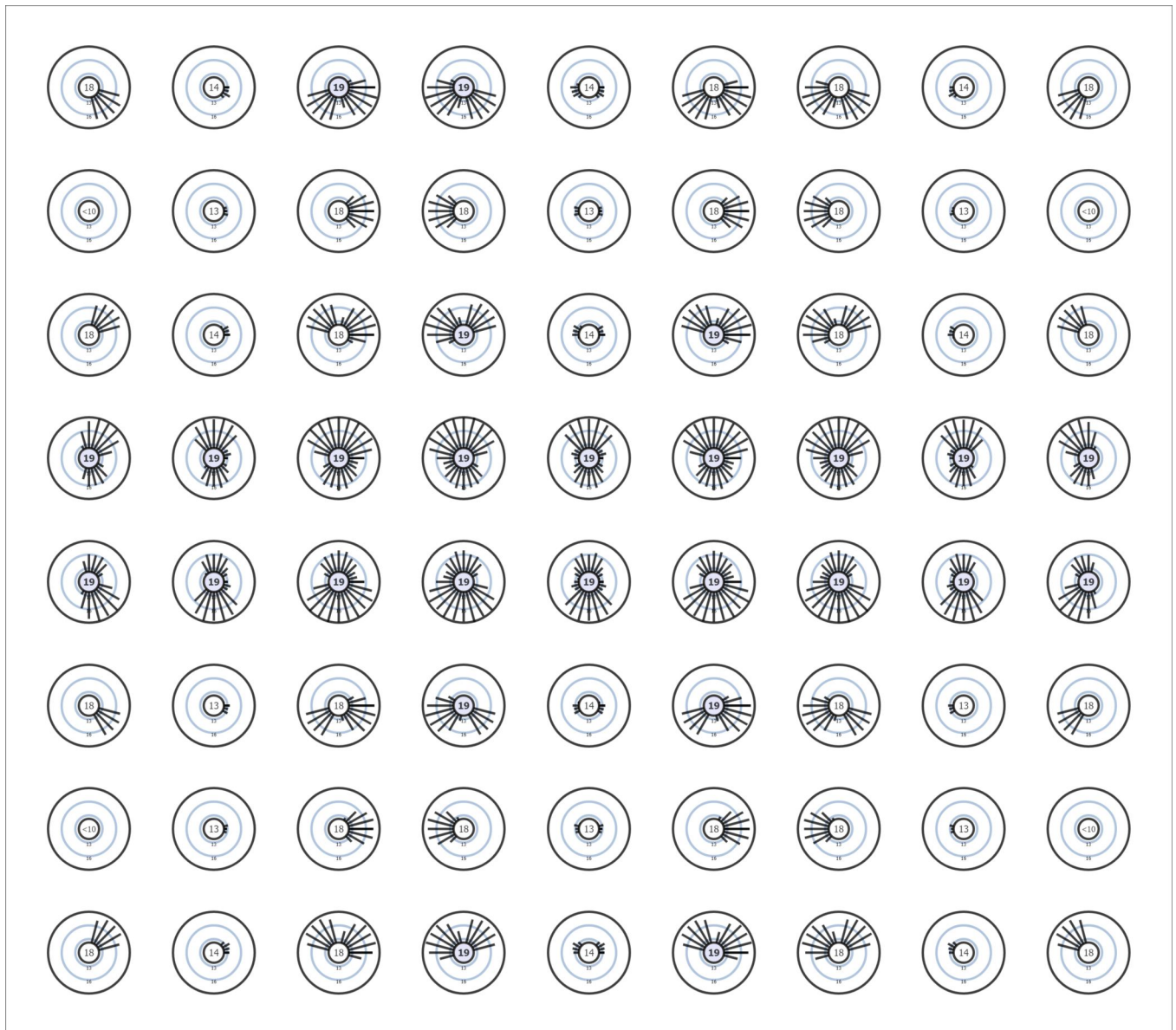
Glare plane 212 (RUG)

Strongest glare at	105°
max	19.2
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG10

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 212 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

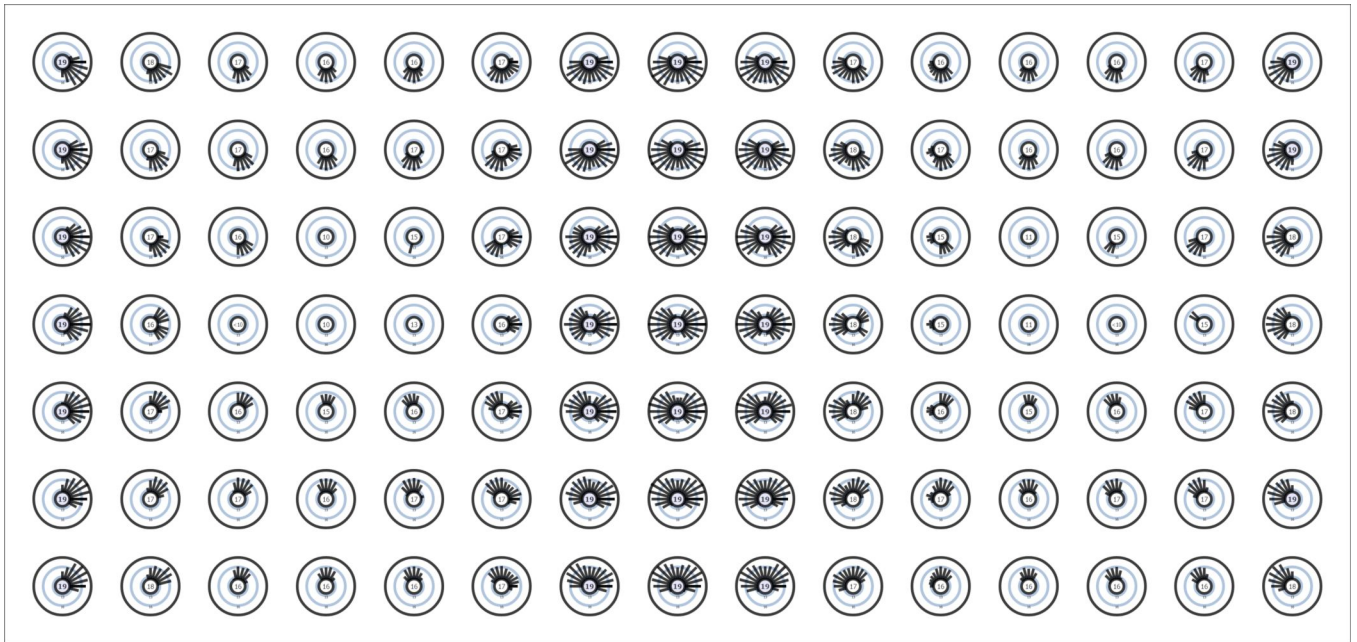
Glare plane 214 (RUG)

Strongest glare at	210°
max	19.4
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG11

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 214 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

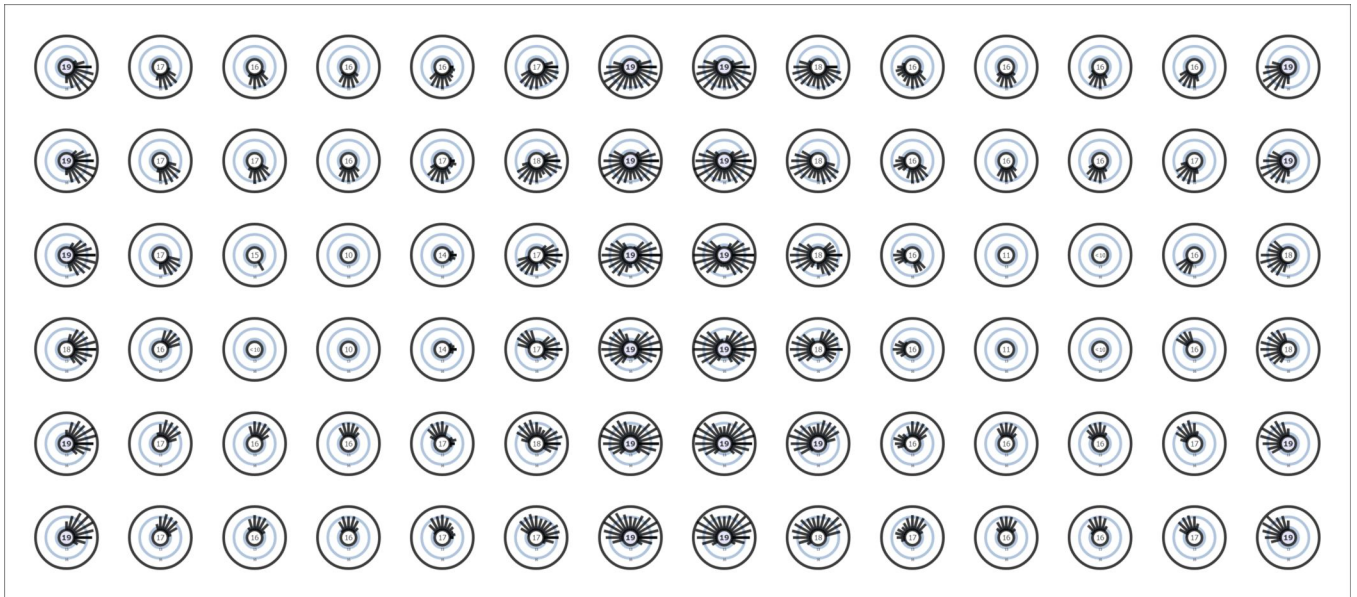
Glare plane 215 (RUG)

Strongest glare at	150°
max	19.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG12

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 215 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

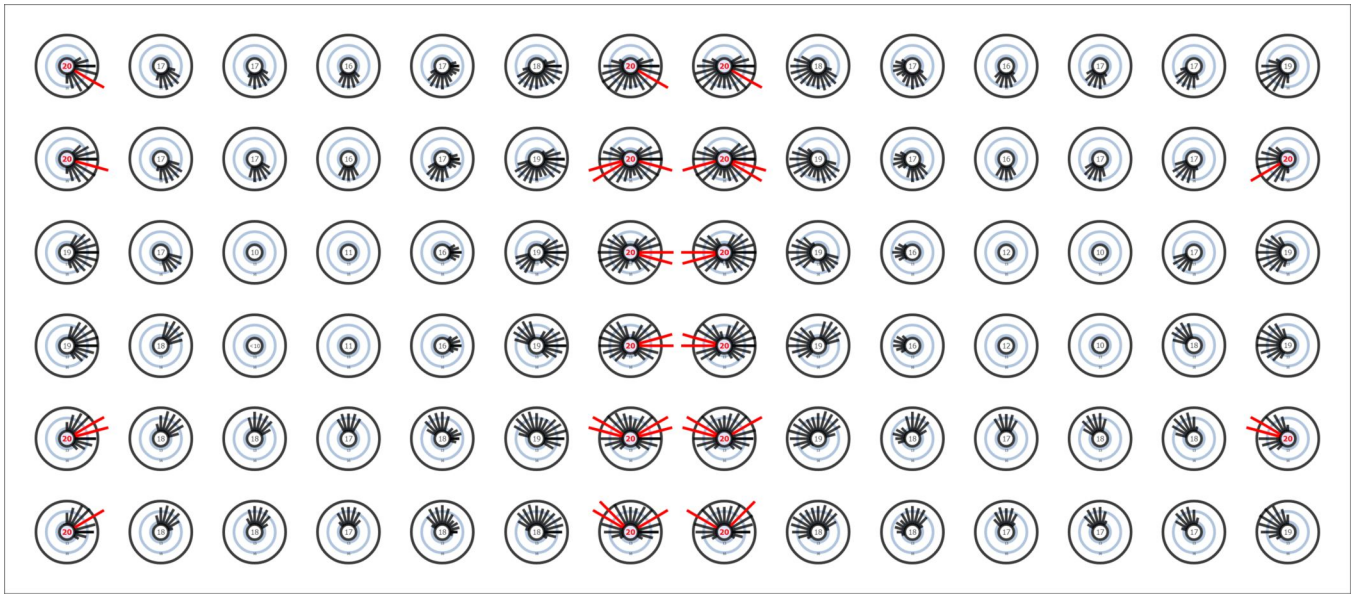
Glare plane 216 (RUG)

Strongest glare at	345°
max	20.0
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG13

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 216 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

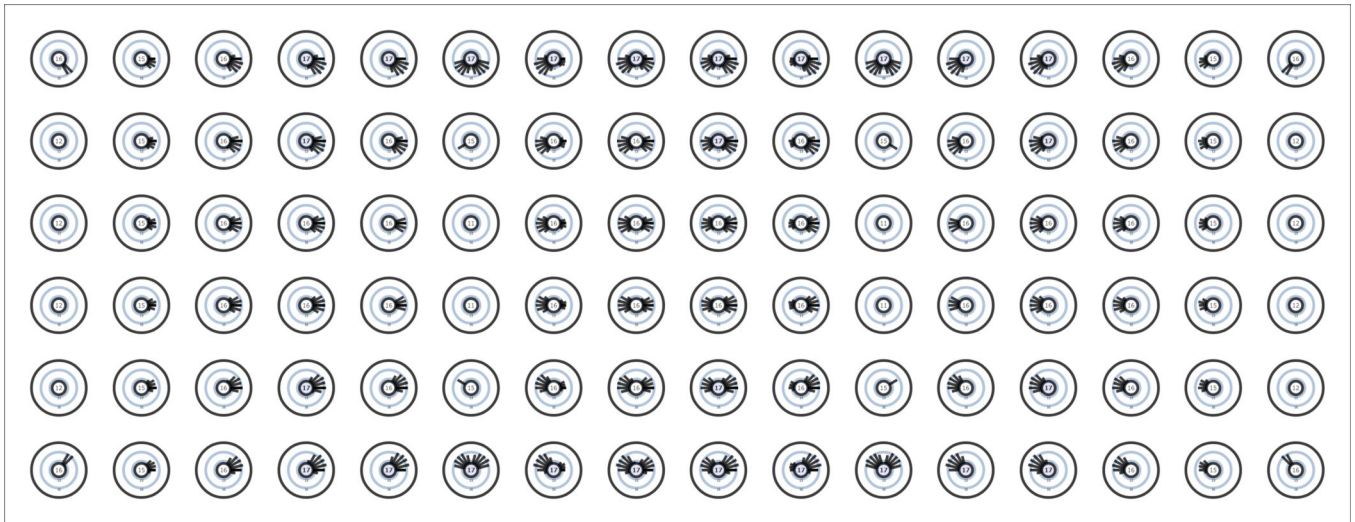
Glare plane 219 (RUG)

Strongest glare at	330°
max	17.0
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG14

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 219 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

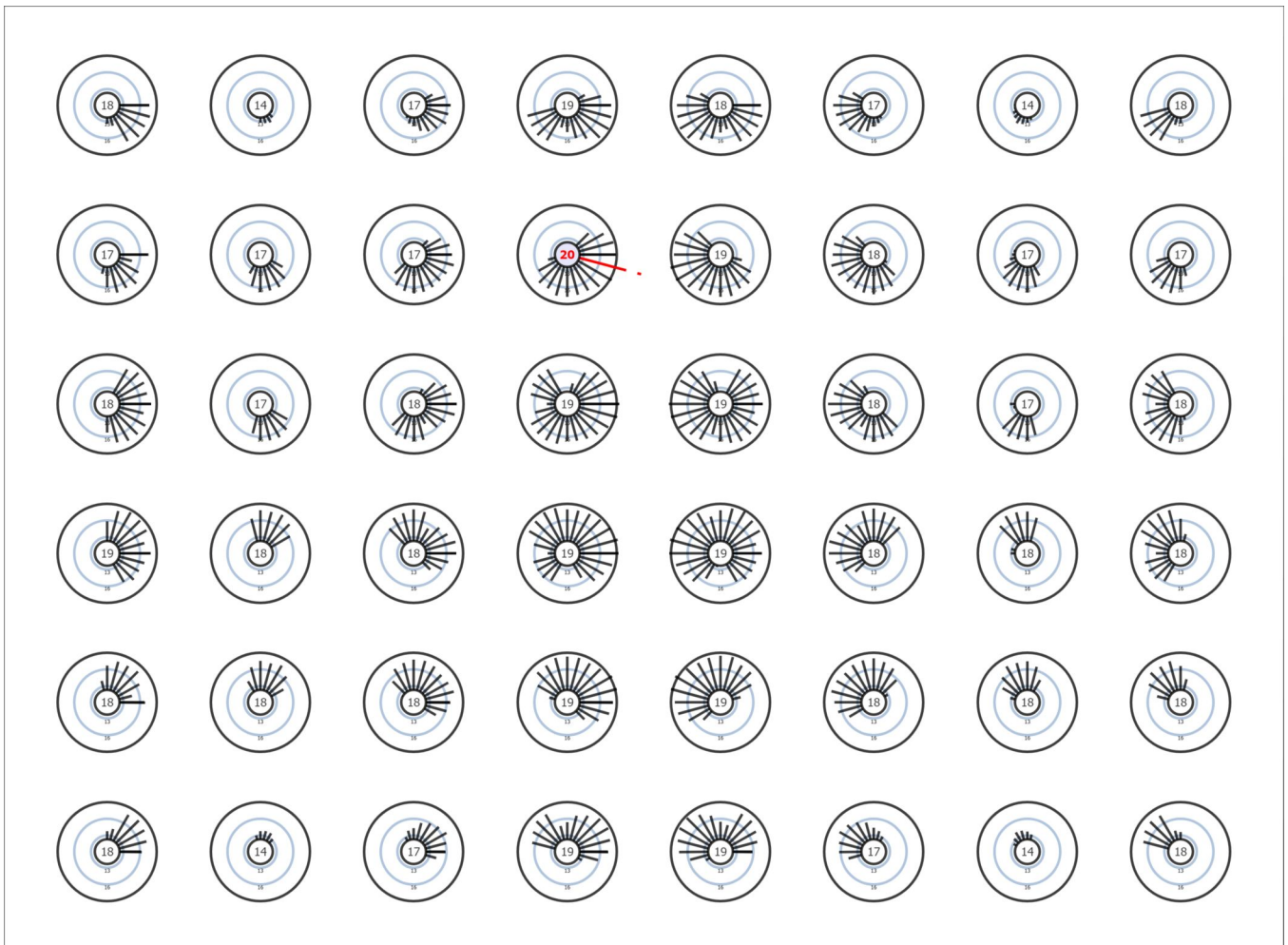
Glare plane 220 (RUG)

Strongest glare at	345°
max	19.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG15

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 220 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

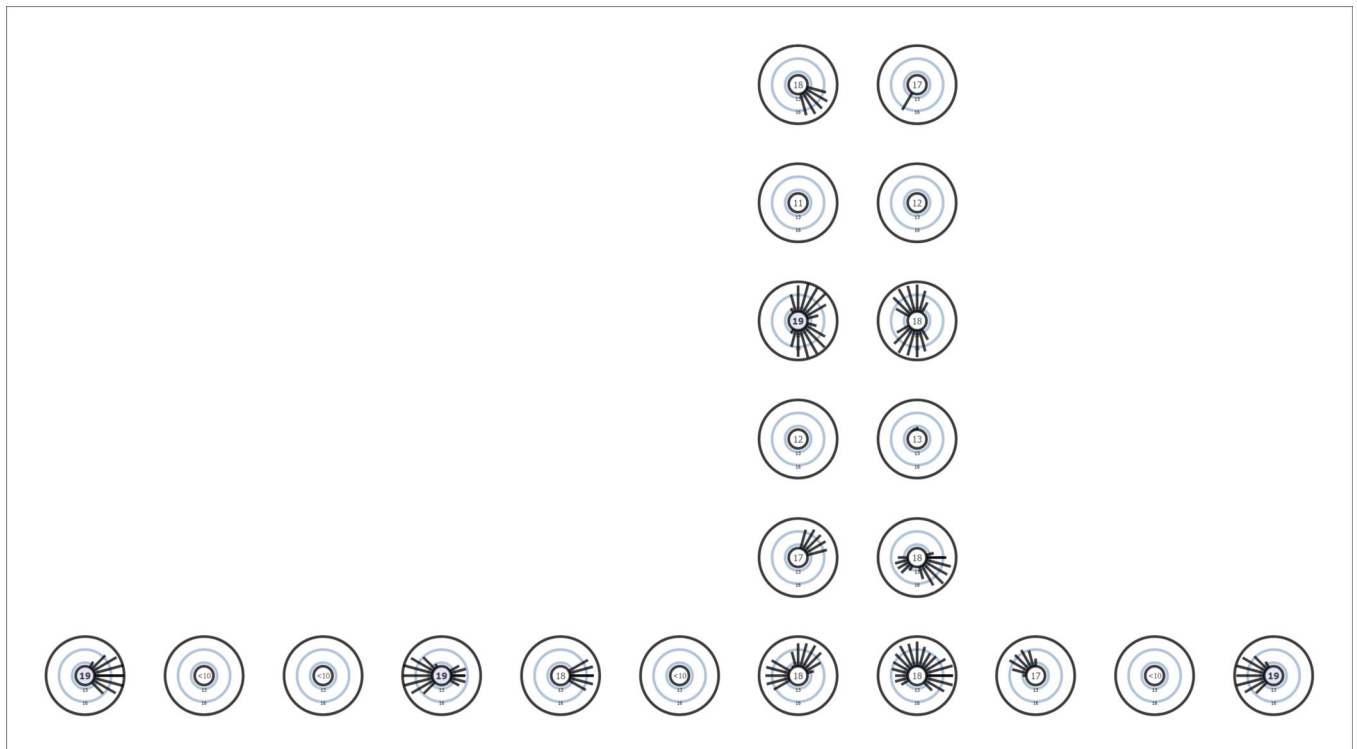
Glare plane 202 (RUG)

Strongest glare at	345°
max	19.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG16

Building A · FIRST FLOOR (Light scene 1)

Calculation objects

Glare plane 202 (RUG)



Building A · FIRST FLOOR (Light scene 1)

Calculation objects

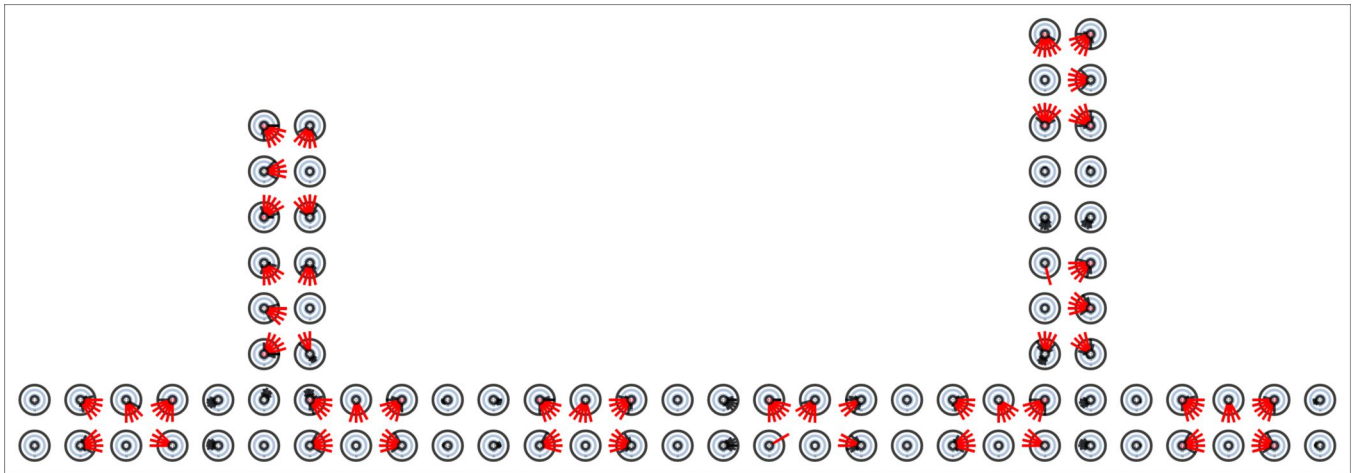
Glare plane 211 (RUG)

Strongest glare at	240°
max	22.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.800 m
Index	CG17

Building A · FIRST FLOOR (Light scene 1)

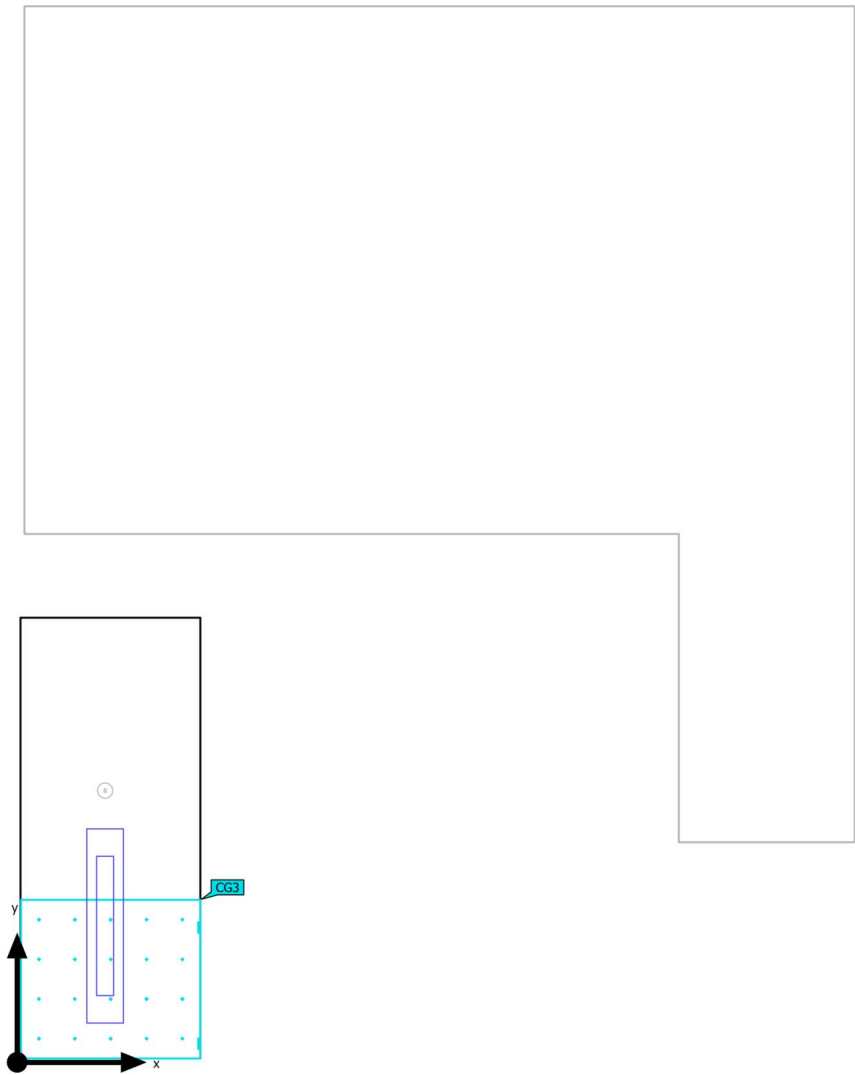
Calculation objects

Glare plane 211 (RUG)



Building A · FIRST FLOOR · 201 STARIS (Light scene 1)

Summary



Ground area	4.00 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m

Building A · FIRST FLOOR · 201 STARIS (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Glare valuation ⁽¹⁾	R _{UG, max}	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	81.7 kWh/a	max. 150 kWh/a	✓	
Space	Lighting power density	8.24 W/m ²	–		

(1) Based on a rectangular space of 1.278 m x 3.132 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

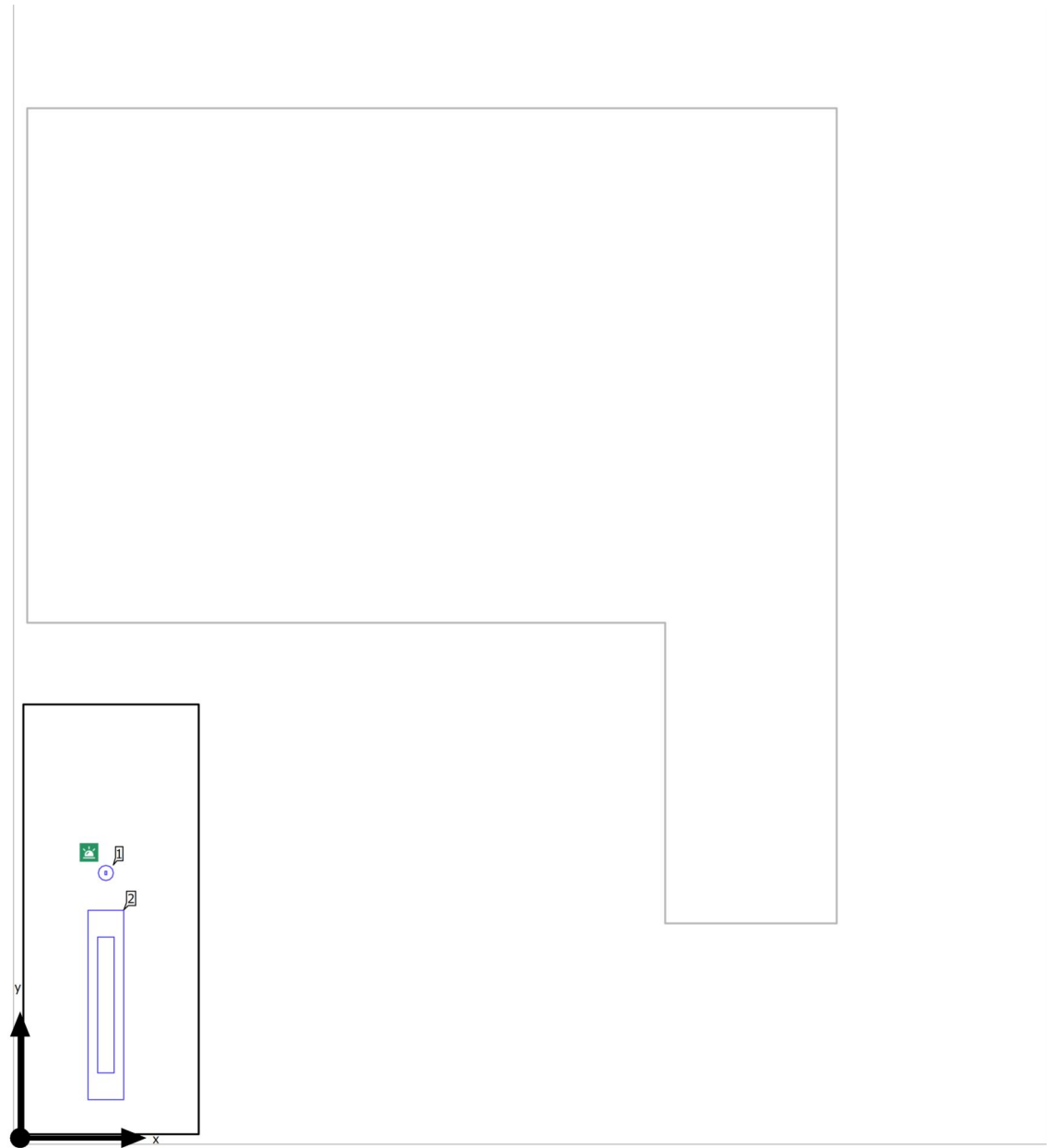
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

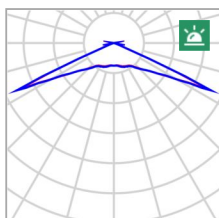
pcs.	Manufacturer	Article No.	Article name	R _{UG}	P	Φ	Luminous efficacy
1	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 201 STARIS

Luminaire layout plan



Building A · FIRST FLOOR · 201 STARIS

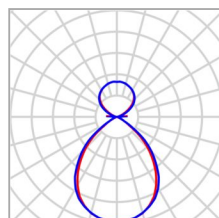
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
0.625 m	1.931 m	3.368 m	1

Building A · FIRST FLOOR · 201 STARIS

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

Individual luminaires

X	Y	Mounting height	Luminaire
0.625 m	0.969 m	2.800 m	2

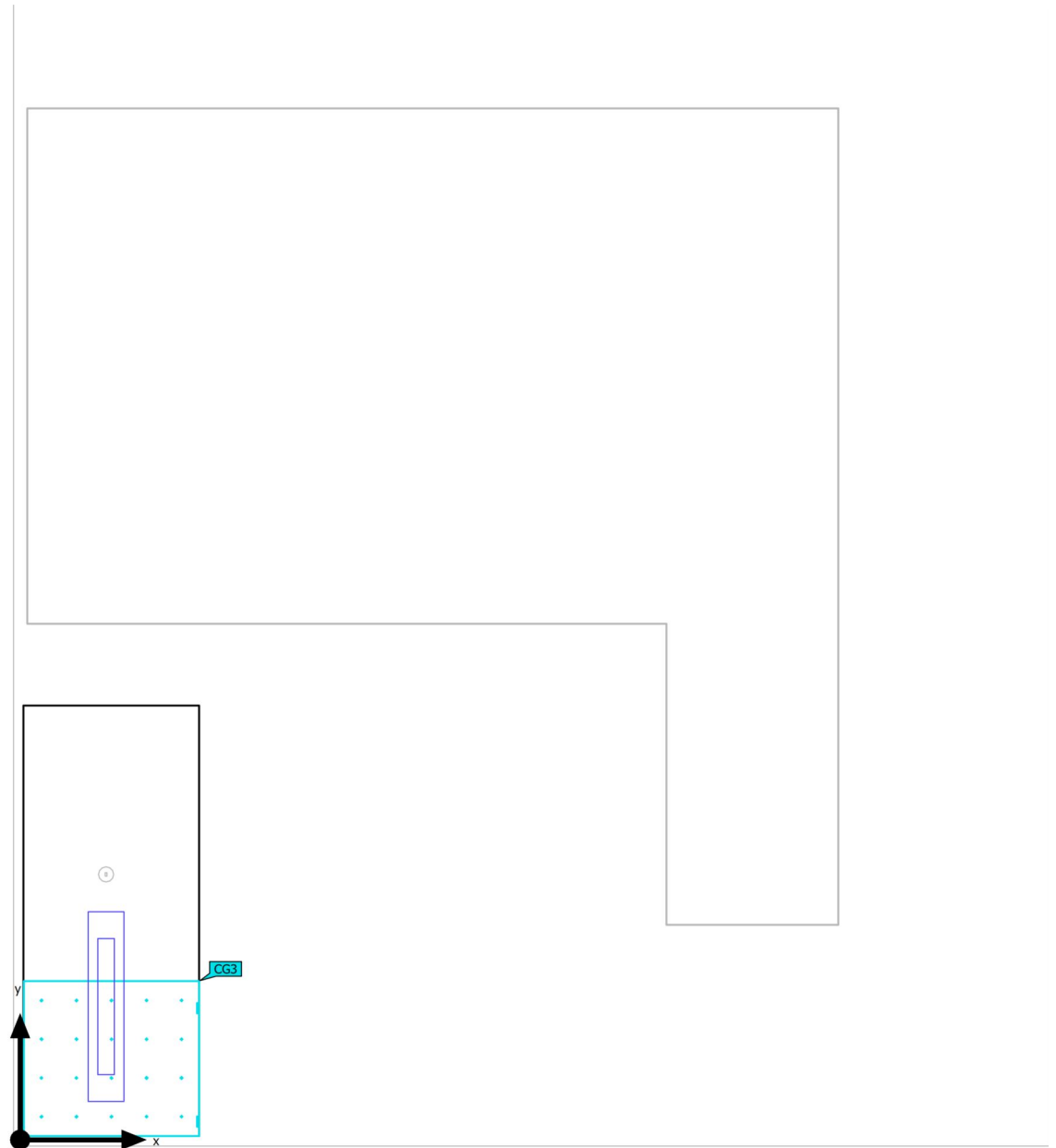
Building A · FIRST FLOOR · 201 STARIS

Luminaire list

Φ_{total} 5550 lm		P_{total} 33.9 W		Luminous efficacy 163.7 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)			0.9 W	300 lm	340.9 lm/W	
						0.9 W	300 lm (100 %)	-	
1	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 201 STARIS (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 201 STARIS (Light scene 1)

Calculation objects

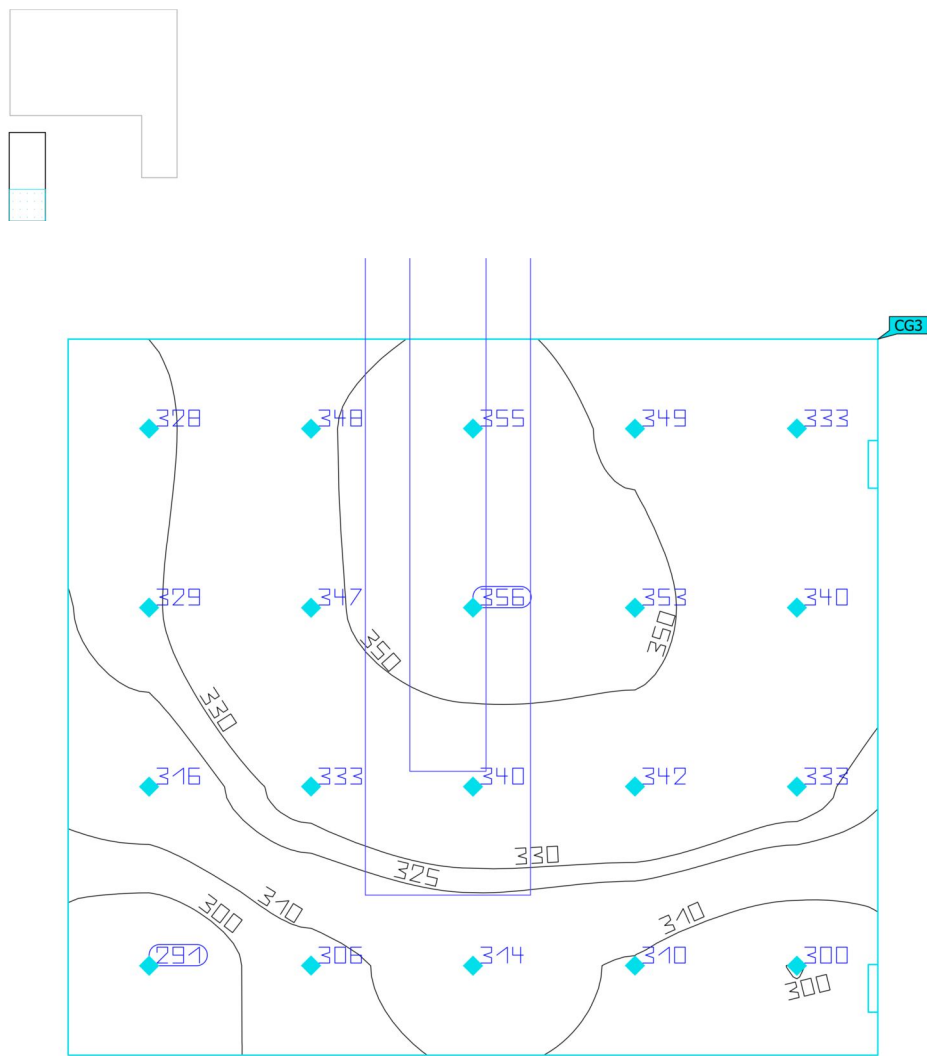
Calculation surfaces

Properties	$\bar{E}$	E_{min}	E_{max}	$U_o (g_1)$	g_2	Index
201 Stair Upper Perpendicular illuminance Height: 0.000 m	331 lx	291 lx	356 lx	0.88	0.82	CG3

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 201 STARIS (Light scene 1)

201 Stair Upper

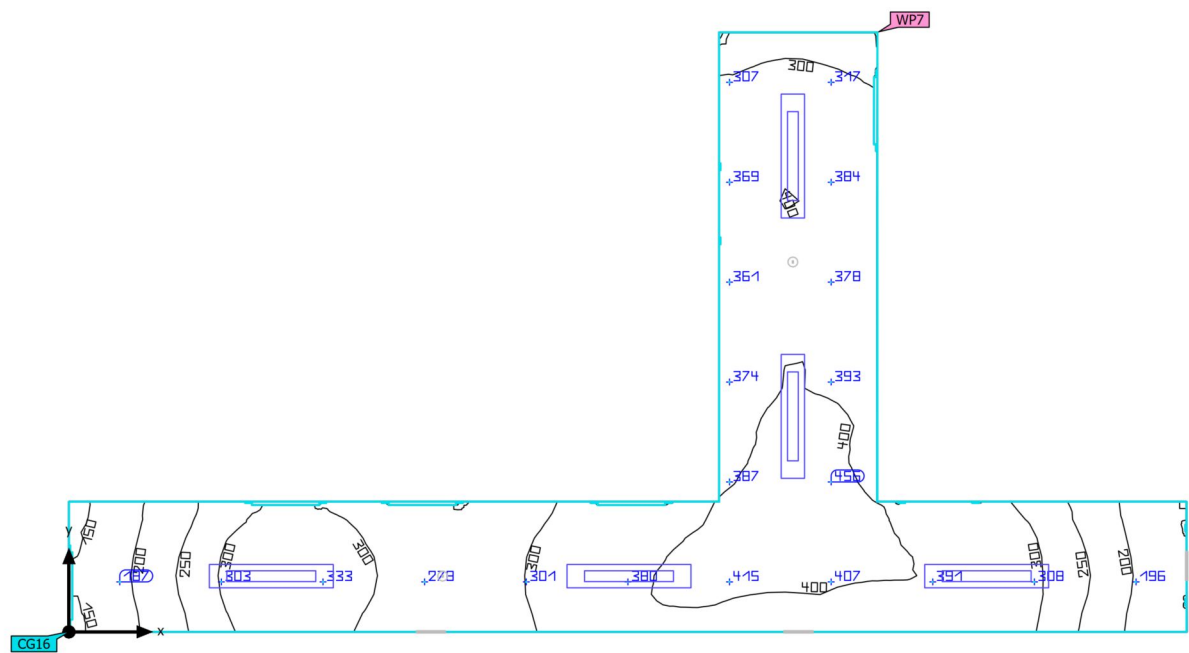


Properties	$\bar{E}$	E_{min}	E_{max}	$U_o (g_1)$	g_2	Index
201 Stair Upper Perpendicular illuminance Height: 0.000 m	331 lx	291 lx	356 lx	0.88	0.82	CG3

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Summary



Ground area	27.25 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.000 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	332 lx	$\geq 500 \text{ lx}$	✗	WP7
	$U_o (g_1)$	0.32	≥ 0.60	✗	WP7
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[257 - 408] kWh/a	max. 1000 kWh/a	✓	
Space	Lighting power density	6.05 W/m ²	–		
		1.82 W/m ² /100 lx	–		

(1) Based on a rectangular space of 12.450 m x 6.678 m and SHR of 0.25.

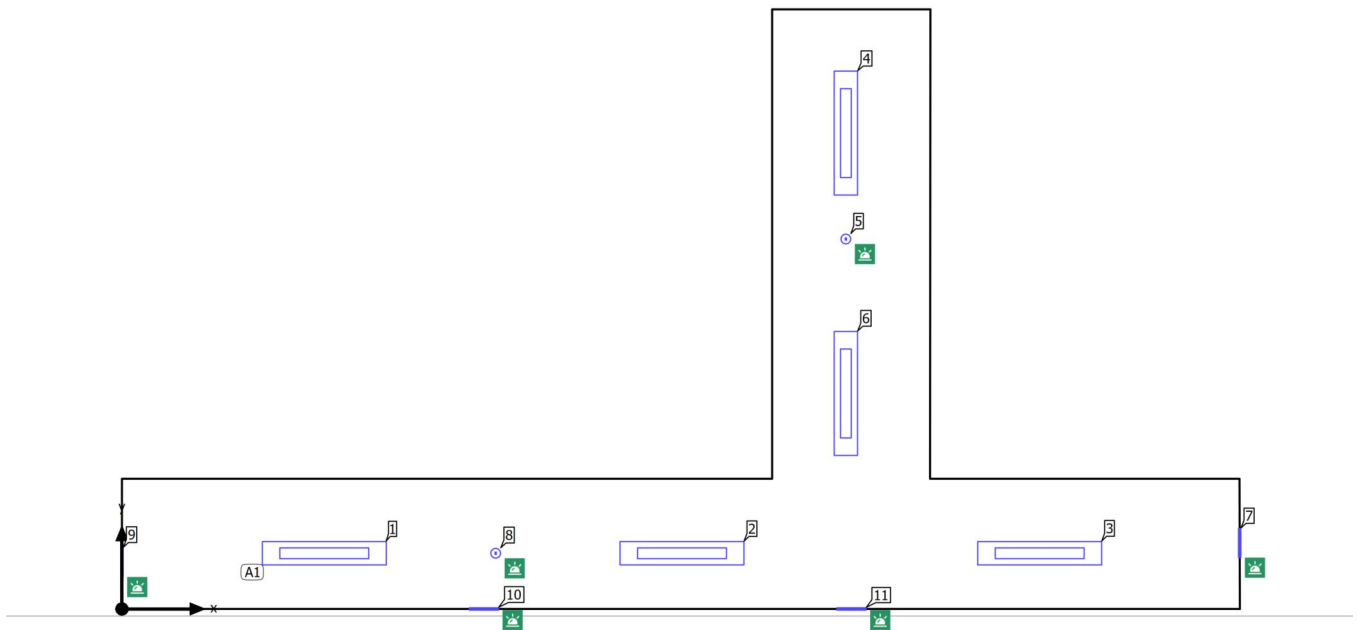
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

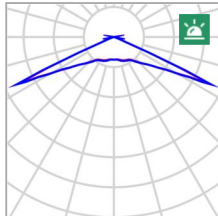
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
5	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 202 CORRIDOR

Luminaire layout plan

Building A · FIRST FLOOR · 202 CORRIDOR

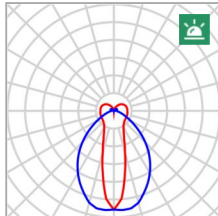
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
8.062 m	4.120 m	3.368 m	5
4.162 m	0.620 m	3.368 m	8

Building A · FIRST FLOOR · 202 CORRIDOR

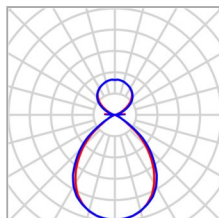
Luminaire layout plan

Manufacturer	ETAP	P	2.5 W
Article No.	-	P _{Emergency lighting}	2.5 W
Article name	K9R142/1-AX1 (_LTRevAB)	Φ _{Luminaire}	80 lm
Fitting	1x sign with picto	Φ _{Emergency lighting}	80 lm
		ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
12.448 m	0.735 m	3.300 m	7
0.000 m	0.518 m	3.300 m	9
4.031 m	0.000 m	3.300 m	10
8.125 m	0.000 m	3.300 m	11

Building A · FIRST FLOOR · 202 CORRIDOR

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

3 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	10.220 m / 0.620 m / 2.800 m	2.254 m	0.620 m	2.800 m	1
X-direction	3 pcs., Centre - centre, 3.983 m	6.237 m	0.620 m	2.800 m	2
Arrangement	A1	10.220 m	0.620 m	2.800 m	3

Individual luminaires

X	Y	Mounting height	Luminaire
8.062 m	5.300 m	2.800 m	4
8.062 m	2.400 m	2.800 m	6

Building A · FIRST FLOOR · 202 CORRIDOR

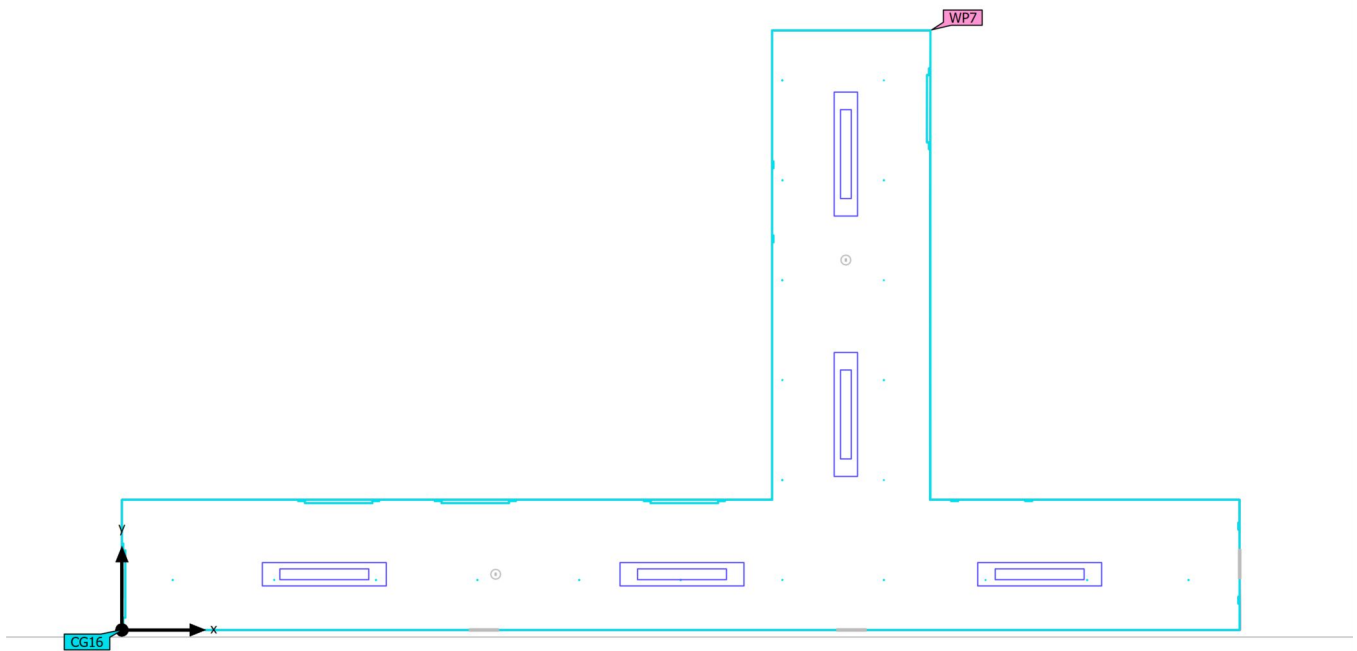
Luminaire list

Φ_{total} 27170 lm	P_{total} 176.8 W	Luminous efficacy 153.7 lm/W	$\Phi_{\text{Emergency lighting}}$ 920 lm	$P_{\text{Emergency lighting}}$ 11.8 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
4	ETAP	-	K9R142/1-AX1 (_LTRRevAB)	2.5 W	80 lm	32.4 lm/W
				 2.5 W	80 lm (100 %)	-
5	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (202 CORRIDOR) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	332 lx (≥ 500 lx) ✗	106 lx	476 lx	0.32 (≥ 0.60) ✗	0.22	WP7

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Calculation objects

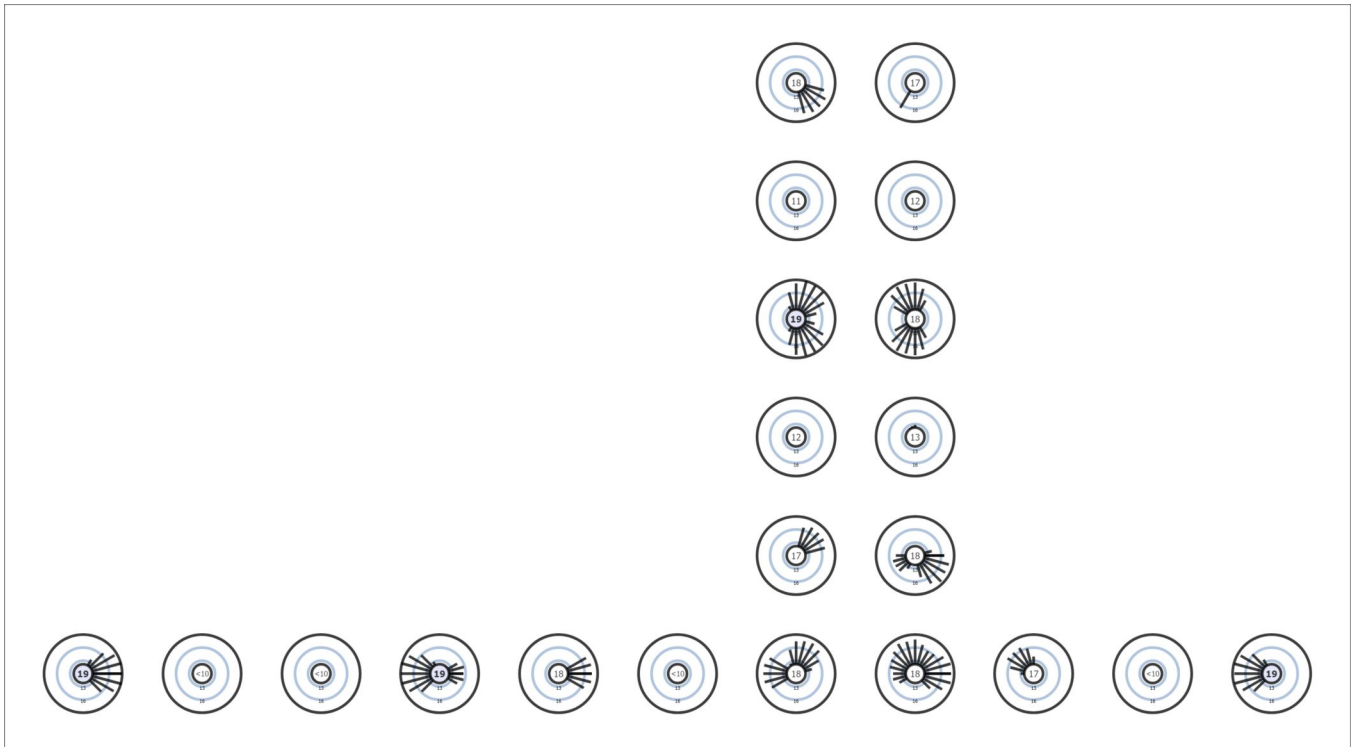
Glare plane 202 (RUG)

Strongest glare at	345°
max	19.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG16

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Calculation objects

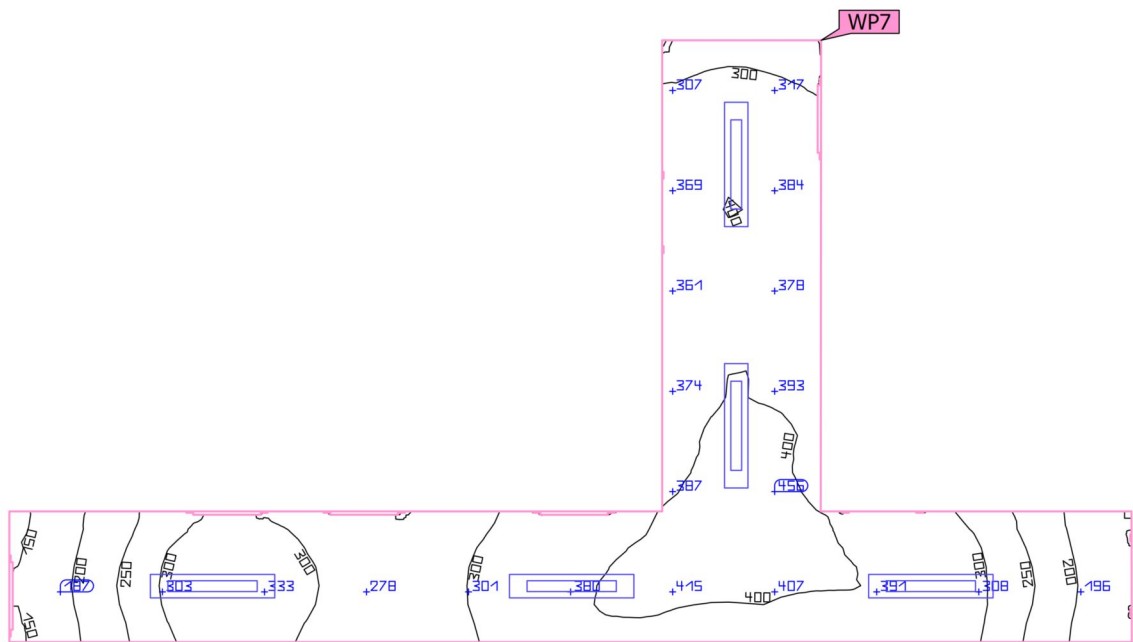
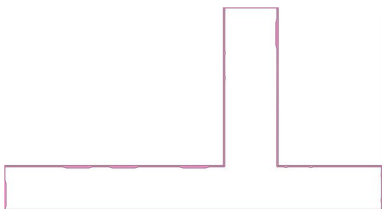
Glare plane 202 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 202 CORRIDOR (Light scene 1)

Working plane (202 CORRIDOR)

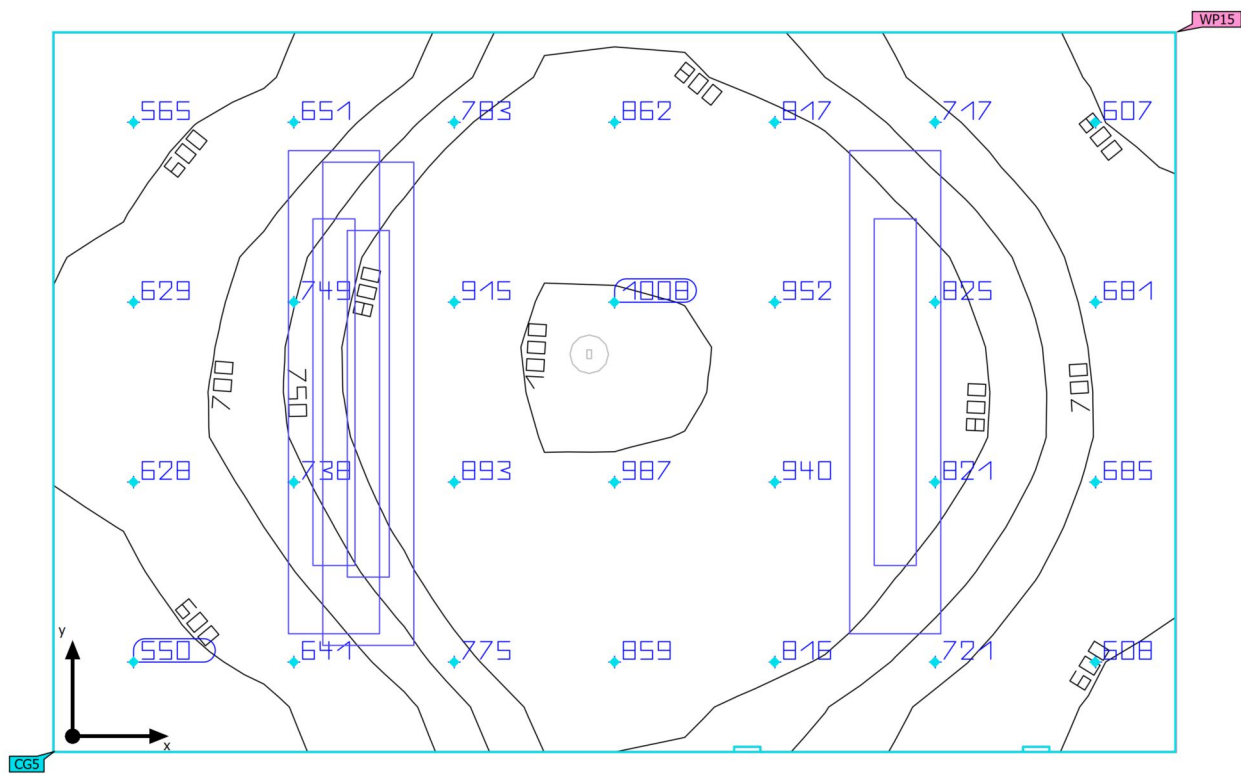


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (202 CORRIDOR)	332 lx	106 lx	476 lx	0.32	0.22	WP7
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.000 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

Summary



Ground area	6.59 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m – 3.300 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	765 lx	$\geq 500 \text{ lx}$	✓	WP15
	$U_o (g_1)$	0.67	≥ 0.60	✓	WP15
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	163 kWh/a	max. 250 kWh/a	✓	
Space	Lighting power density	15.03 W/m ²	–		
		1.96 W/m ² /100 lx	–		

(1) Based on a rectangular space of 2.056 m x 3.205 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

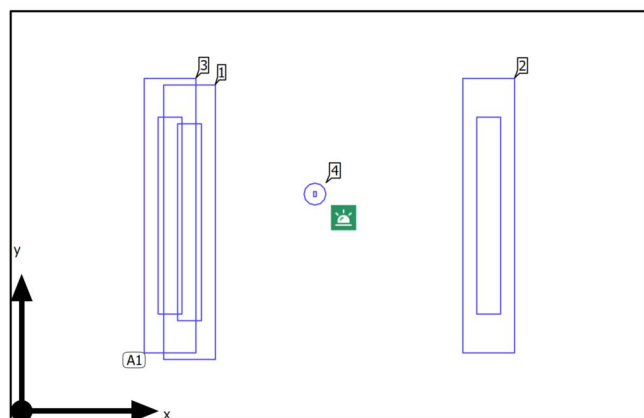
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
3	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

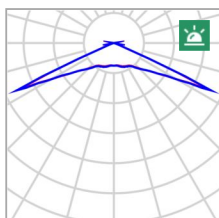
Building A · FIRST FLOOR · 203 MEETING

Luminaire layout plan

(A2)



Building A · FIRST FLOOR · 203 MEETING

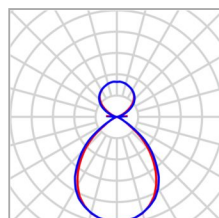
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.475 m	1.091 m	3.368 m	4

Building A · FIRST FLOOR · 203 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

2 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	2.349 m / 0.983 m / 2.800 m	2.349 m	0.983 m	2.800 m	2
X-direction	2 pcs., Centre - centre, 1.602 m	0.746 m	0.983 m	2.800 m	3
Y-direction	1 pcs., Centre - centre, 2.056 m				
Arrangement	A1				

7 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.844 m / 0.949 m / 3.300 m	0.844 m	0.949 m	3.300 m	1
X-direction	3 pcs., Centre - centre, Distances not equal				
Y-direction	3 pcs., Centre - centre, Distances not equal				

Building A · FIRST FLOOR · 203 MEETING

Luminaire layout plan

Arrangement A2

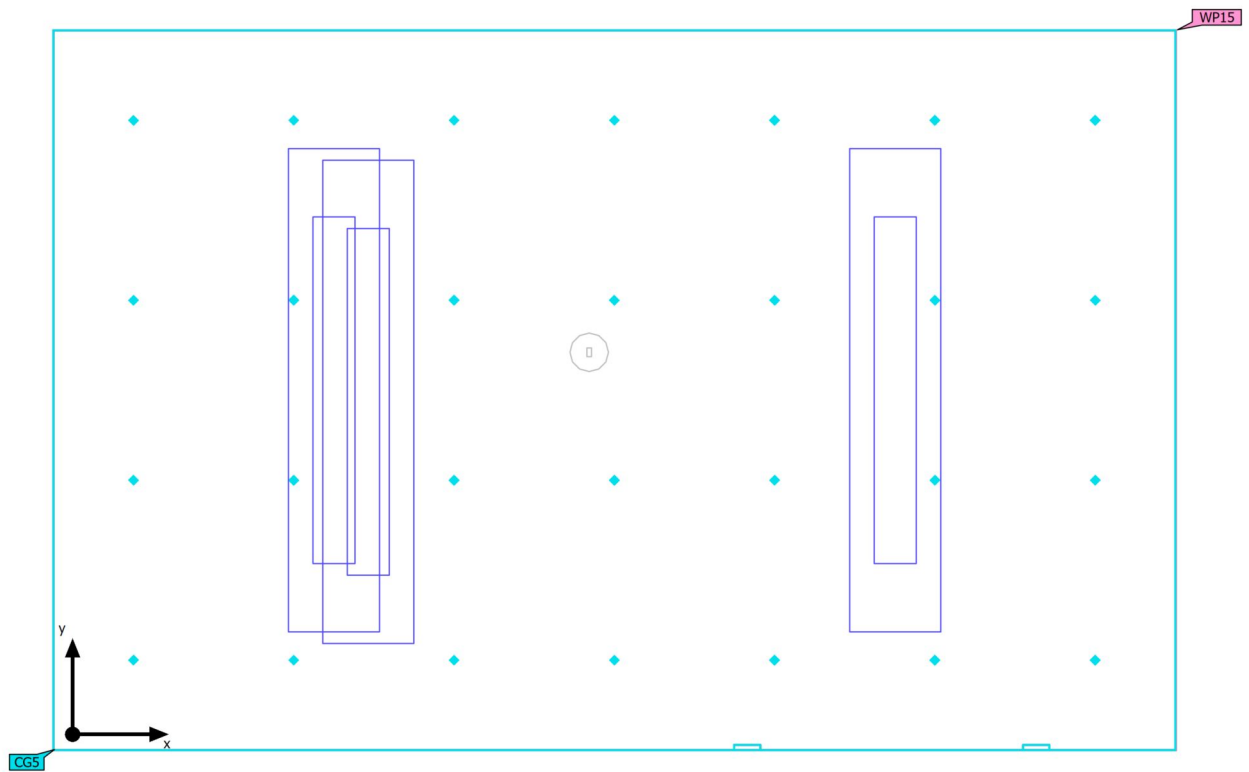
Building A · FIRST FLOOR · 203 MEETING

Luminaire list

Φ_{total} 16050 lm		P_{total} 99.9 W		Luminous efficacy 160.7 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)			0.9 W	300 lm	340.9 lm/W	
						0.9 W	300 lm (100 %)	-	
3	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (203 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	765 lx (≥ 500 lx) ✓	510 lx	1018 lx	0.67 (≥ 0.60) ✓	0.50	WP15

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

Calculation objects

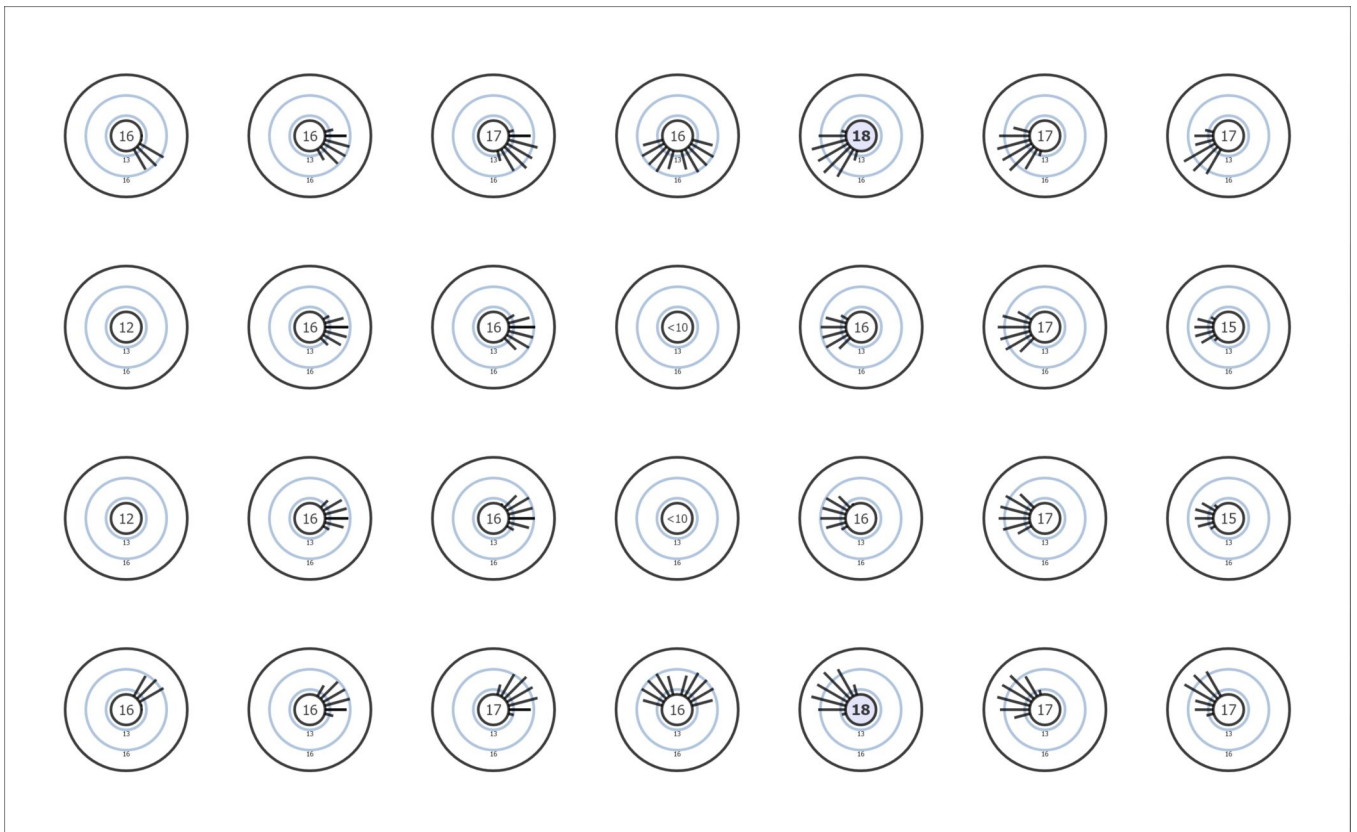
Glare plane 203 (RUG)

Strongest glare at	135°
max	17.7
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG5

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

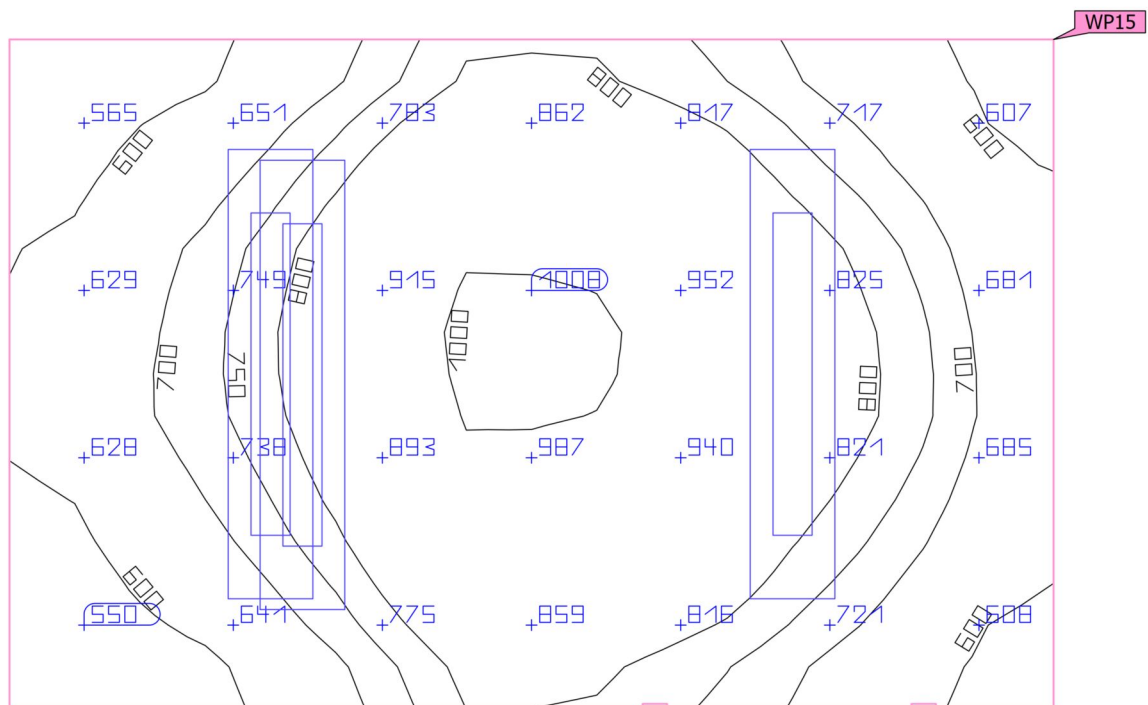
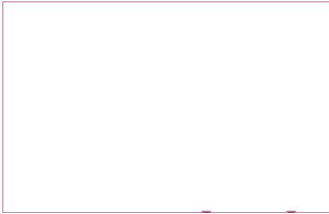
Calculation objects

Glare plane 203 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 203 MEETING (Light scene 1)

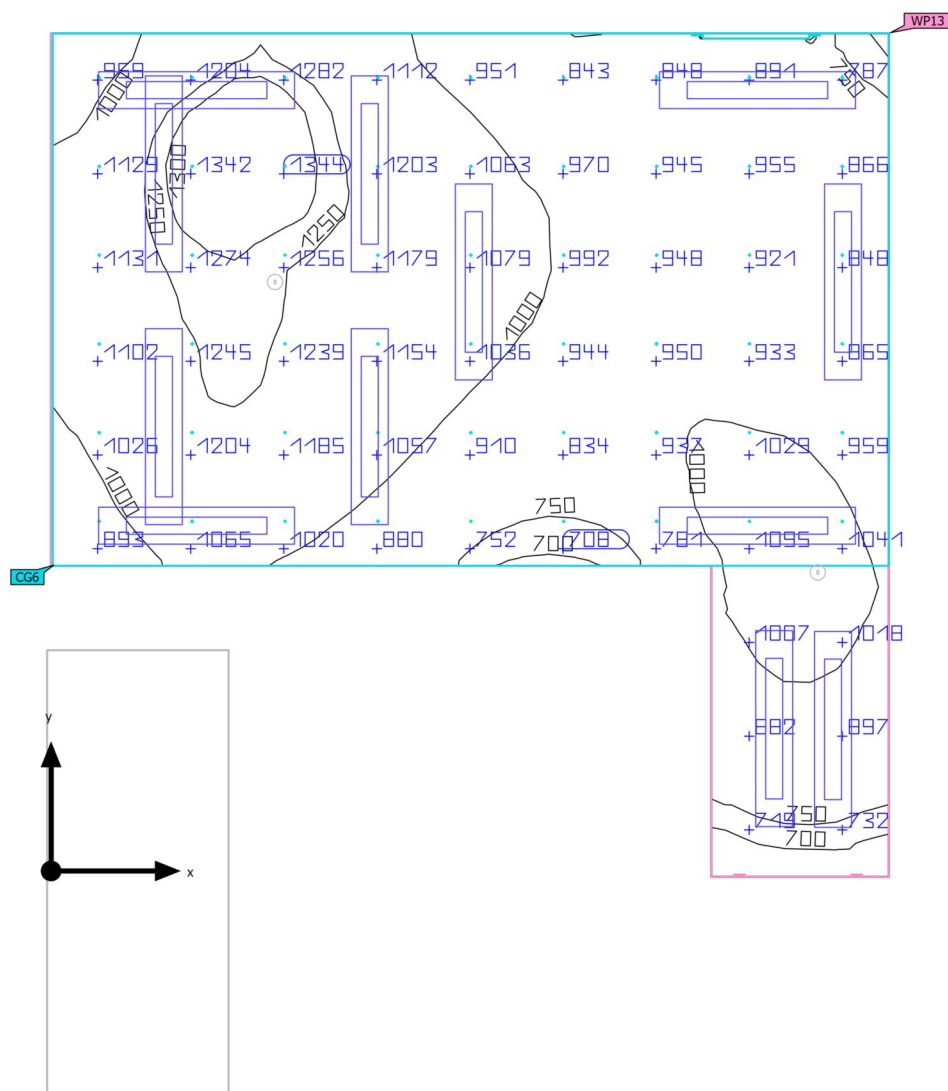
Working plane (203 MEETING)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (203 MEETING)	765 lx	510 lx	1018 lx	0.67	0.50	WP15
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.700 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

Summary



Ground area	24.86 m²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m – 3.300 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	1013 lx	$\geq 500 \text{ lx}$	✓	WP13
	$U_o (g_1)$	0.62	≥ 0.60	✓	WP13
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[472 - 653] kWh/a	max. 900 kWh/a	✓	
Space	Lighting power density	15.93 W/m ²	–		
		1.57 W/m ² /100 lx	–		

(1) Based on a rectangular space of 5.900 m x 5.941 m and SHR of 0.25.

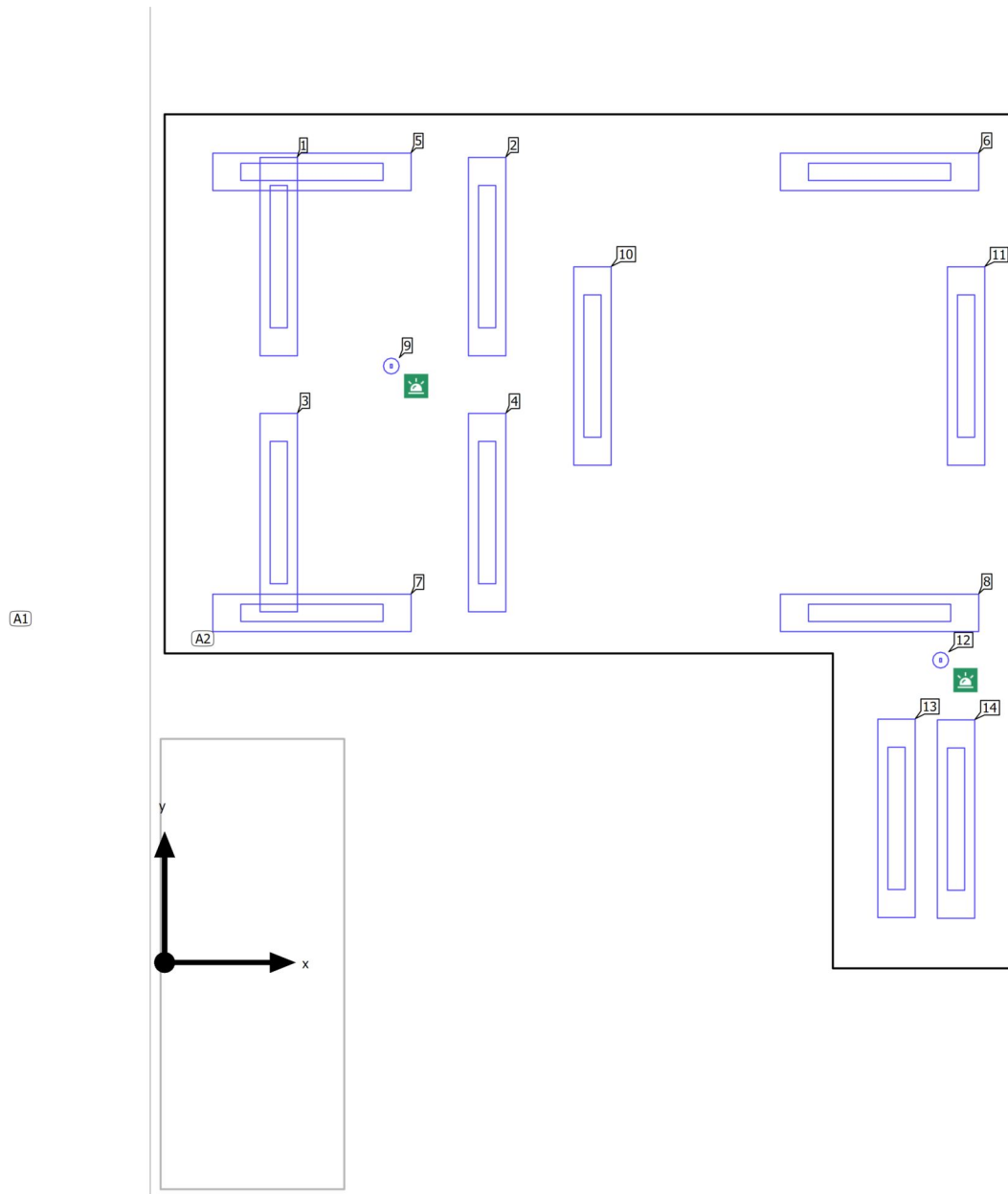
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

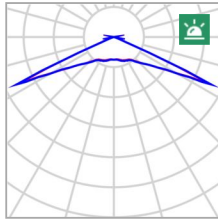
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
12	ETAP	-	R31111/LEDN455DX1 (_LTRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 204 MANAGER

Luminaire layout plan

Building A · FIRST FLOOR · 204 MANAGER

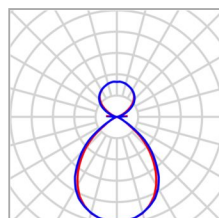
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.577 m	4.149 m	3.368 m	9
5.400 m	2.103 m	3.368 m	12

Building A · FIRST FLOOR · 204 MANAGER

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

7 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.794 m / 3.130 m / 3.300 m	0.794 m	4.910 m	3.300 m	1
X-direction	3 pcs., Centre - centre, Distances not equal	2.244 m	4.910 m	3.300 m	2
		0.794 m	3.130 m	3.300 m	3
Y-direction	3 pcs., Centre - centre, Distances not equal	2.244 m	3.130 m	3.300 m	4
Arrangement	A1				

4 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	4.975 m / 5.499 m / 2.800 m	1.025 m	5.499 m	2.800 m	5
X-direction	2 pcs., Centre - centre, Distances not equal	4.975 m	5.499 m	2.800 m	6
		1.025 m	2.432 m	2.800 m	7
		4.975 m	2.432 m	2.800 m	8

Building A · FIRST FLOOR · 204 MANAGER

Luminaire layout plan

Y-direction	2 pcs., Centre - centre, Distances not equal
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Arrangement	A2
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Individual luminaires

X	Y	Mounting height	Luminaire
2.977 m	4.149 m	3.300 m	10
5.577 m	4.149 m	3.300 m	11
5.093 m	1.003 m	3.300 m	13
5.507 m	0.997 m	3.300 m	14

Building A · FIRST FLOOR · 204 MANAGER

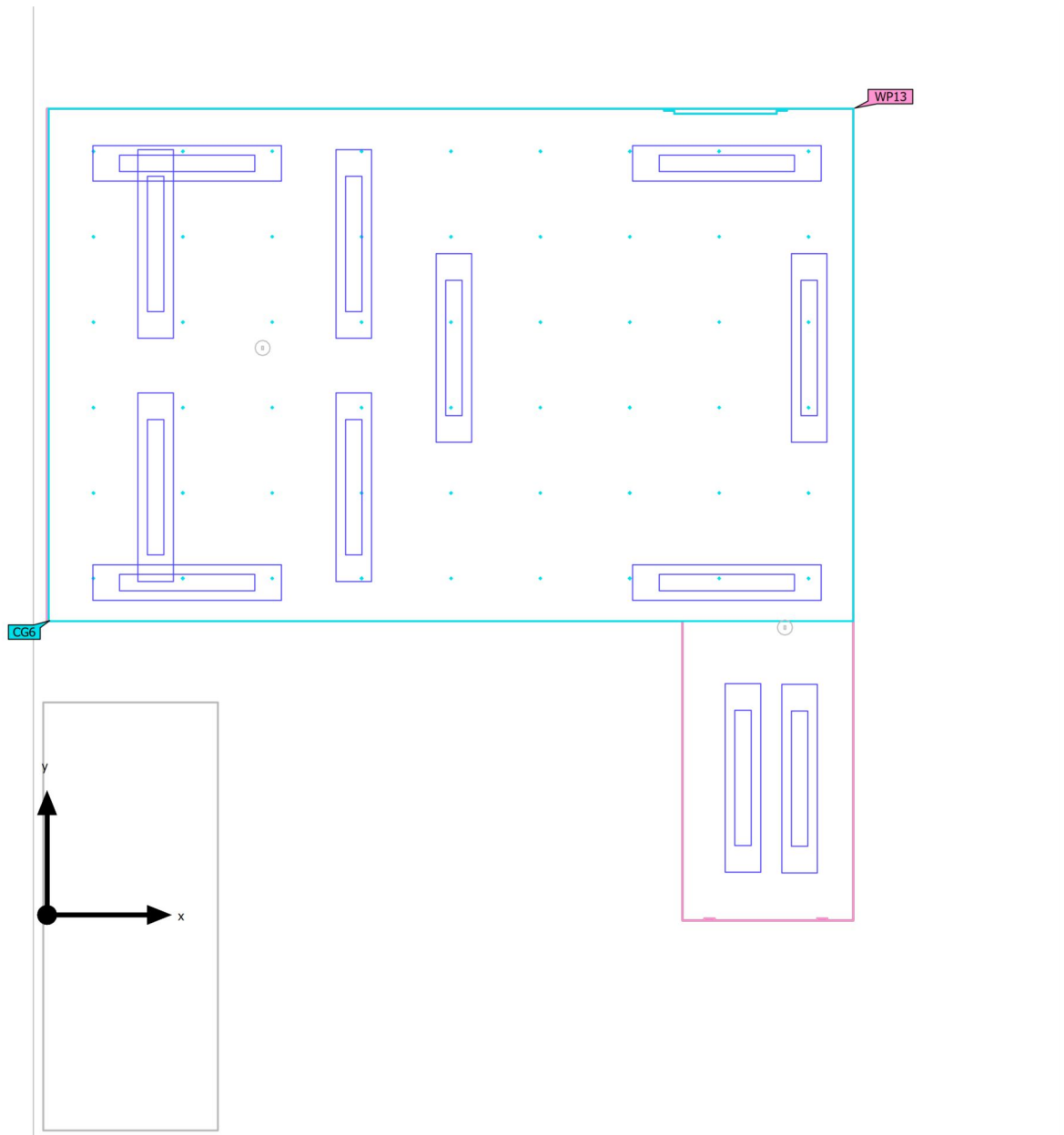
Luminaire list

Φ_{total} 63600 lm	P_{total} 397.8 W	Luminous efficacy 159.9 lm/W	$\Phi_{\text{Emergency lighting}}$ 600 lm	$P_{\text{Emergency lighting}}$ 1.8 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
12	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (204 MANAGER) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	1013 lx (≥ 500 lx) ✓	633 lx	1375 lx	0.62 (≥ 0.60) ✓	0.46	WP13

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

Calculation objects

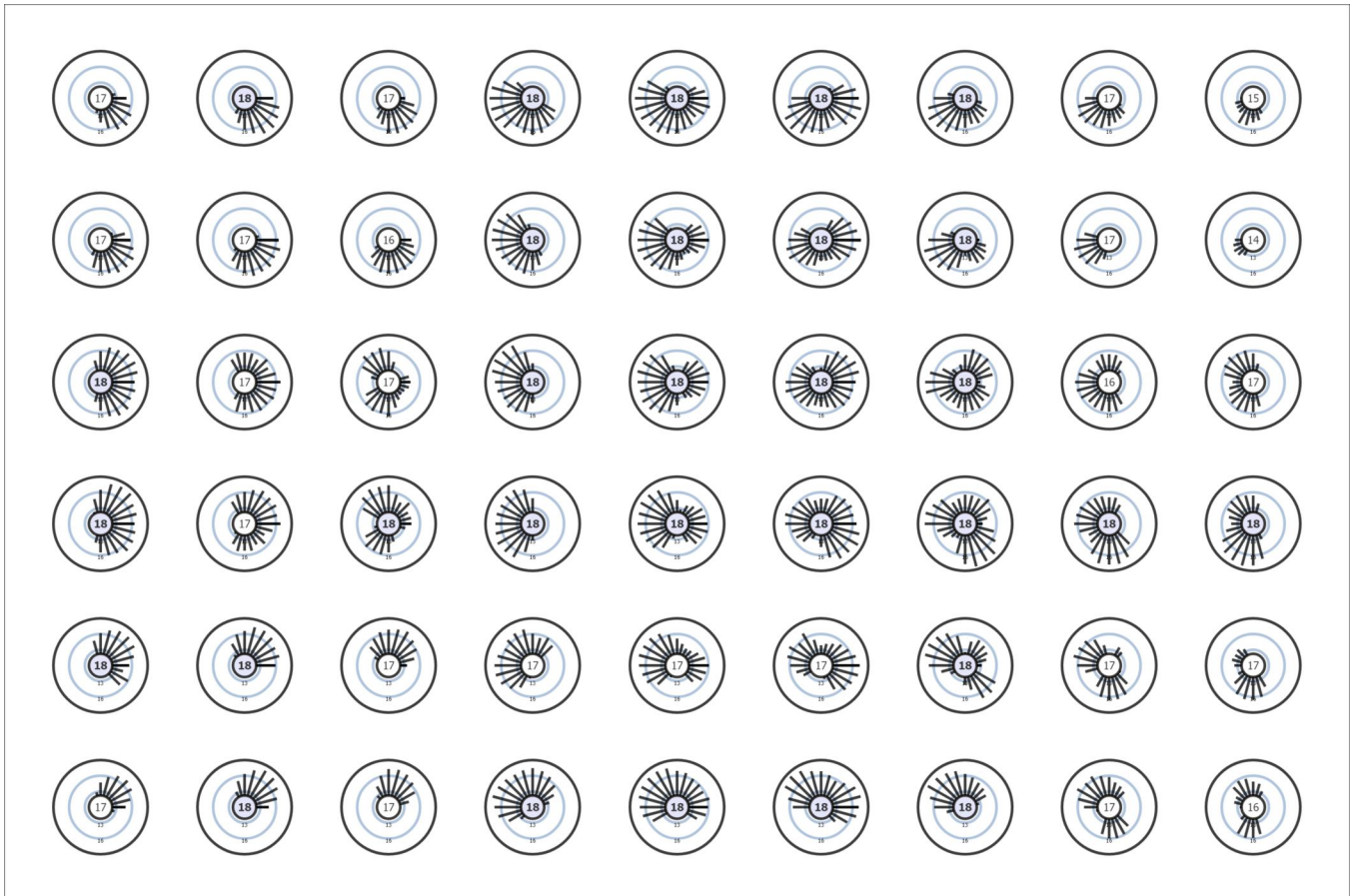
Glare Plane 204 (RUG)

Strongest glare at	285°
max	18.4
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG6

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

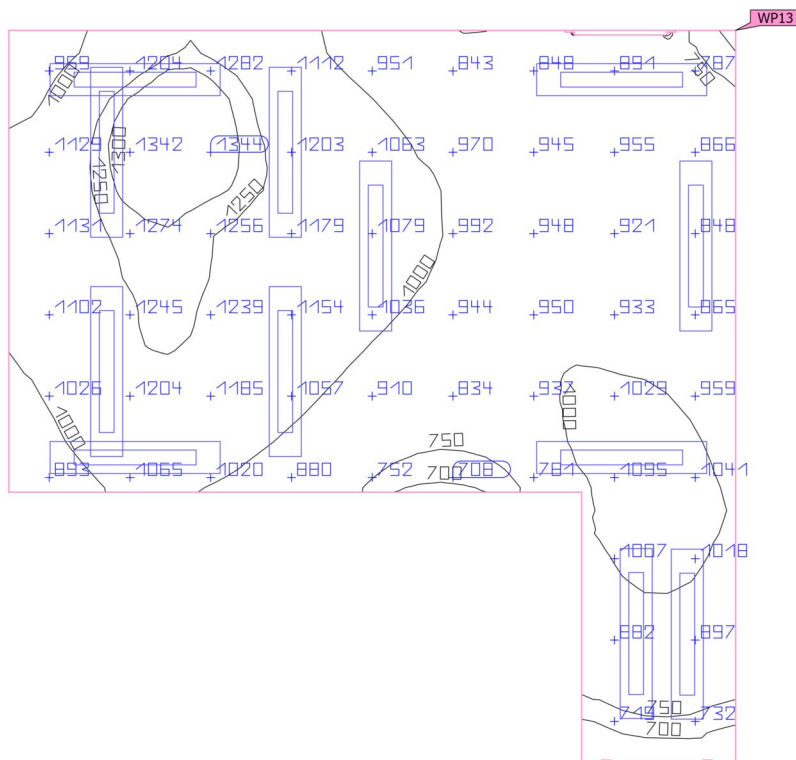
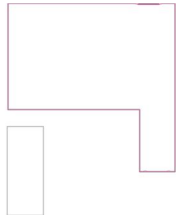
Calculation objects

Glare Plane 204 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 204 MANAGER (Light scene 1)

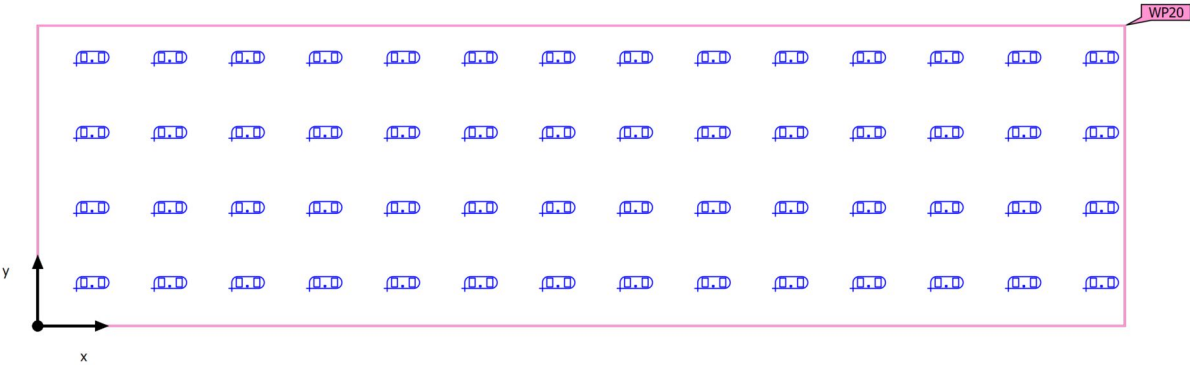
Working plane (204 MANAGER)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (204 MANAGER)	1013 lx	633 lx	1375 lx	0.62	0.46	WP13
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 204A DB (Light scene 1)

Summary



Ground area	0.42 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.80 (fixed)

Clearance height	0.000 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 204A DB (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	0.00 lx	$\geq 500 \text{ lx}$	✗	WP20
	$U_o (g_1)$	-	≥ 0.60		WP20
Energy estimation ⁽²⁾	Consumption	0.00 kWh/a	max. 50 kWh/a	✓	
Space	Lighting power density	0.00 W/m ²	-		

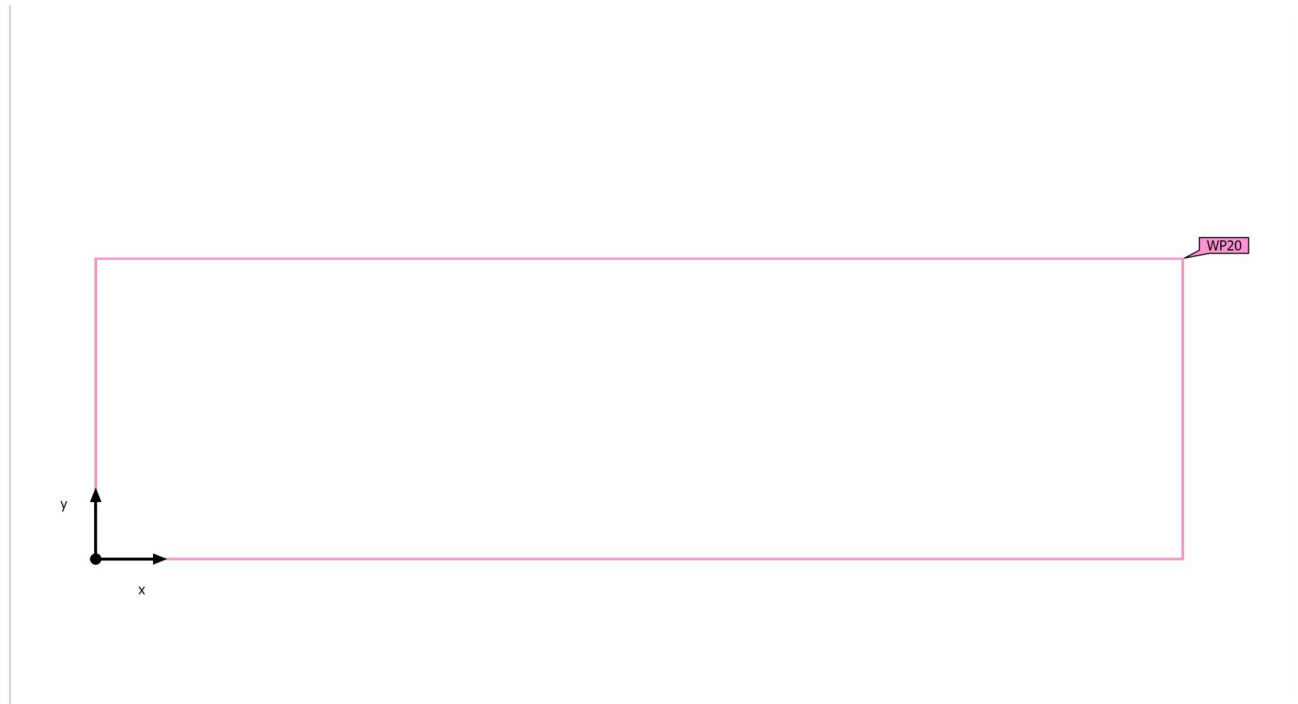
(1) Based on a rectangular space of 1.230 m x 0.340 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 204A DB (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 204A DB (Light scene 1)

Calculation objects

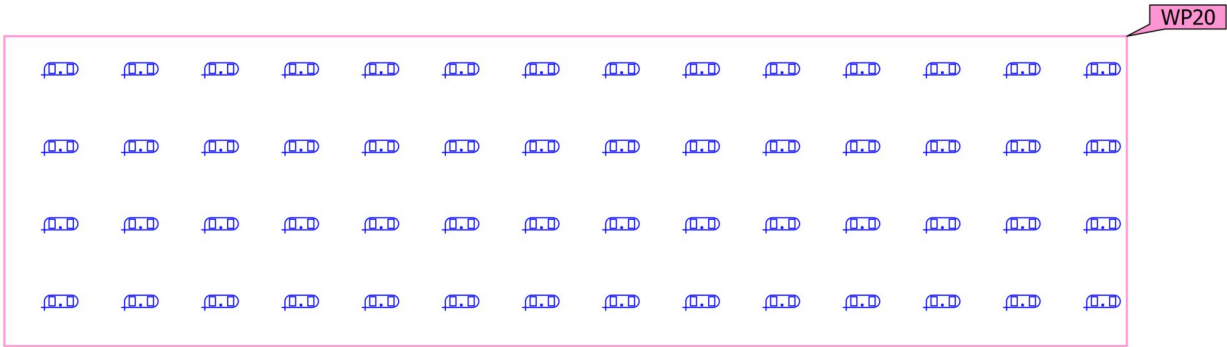
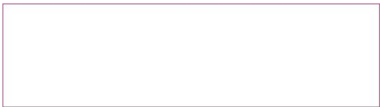
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (204A DB) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	0.00 lx (≥ 500 lx) ✗	0.00 lx	0.00 lx	- (≥ 0.60)	-	WP20

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 204A DB (Light scene 1)

Working plane (204A DB)

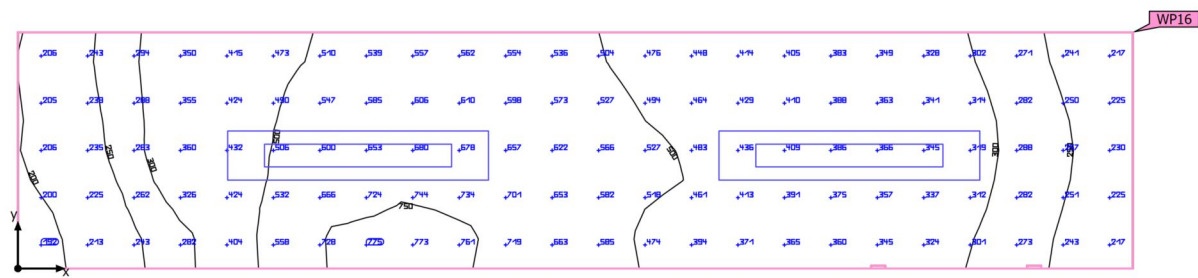


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (204A DB) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	0.00 lx (≥ 500 lx) ✗	0.00 lx	0.00 lx	- (≥ 0.60)	-	WP20

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 205 BALCONY (Light scene 1)

Summary



Ground area	7.38 m ²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	3.300 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 205 BALCONY (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	427 lx	$\geq 500 \text{ lx}$	✗	WP16
	$U_o (g_1)$	0.44	≥ 0.60	✗	WP16
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	173 kWh/a	max. 300 kWh/a	✓	
Space	Lighting power density	9.49 W/m ²	–		
		2.22 W/m ² /100 lx	–		

(1) Based on a rectangular space of 1.250 m x 5.900 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

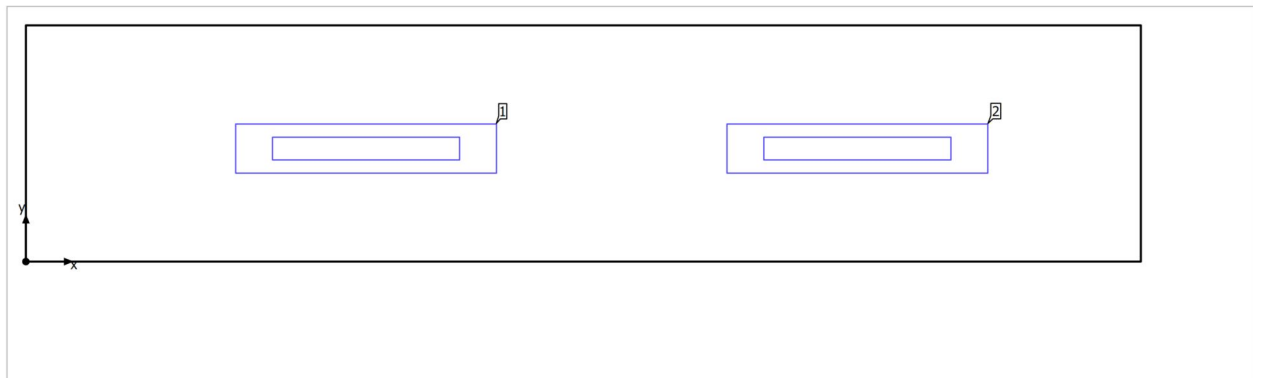
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

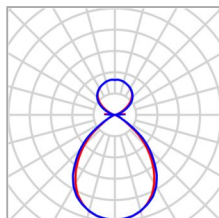
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
2	ETAP	-	R31111/LEDW455DX1 (_LTRRevAC)	16	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 205 BALCONY

Luminaire layout plan



Building A · FIRST FLOOR · 205 BALCONY

Luminaire layout plan

Manufacturer	ETAP	P	35.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDW455DX1 (_LTRevAC)		
Fitting	1x LED		

Individual luminaires

X	Y	Mounting height	Luminaire
1.800 m	0.598 m	3.300 m	1
4.400 m	0.598 m	3.300 m	2

Building A · FIRST FLOOR · 205 BALCONY

Luminaire list Φ_{total}

10500 lm

 P_{total}

70.0 W

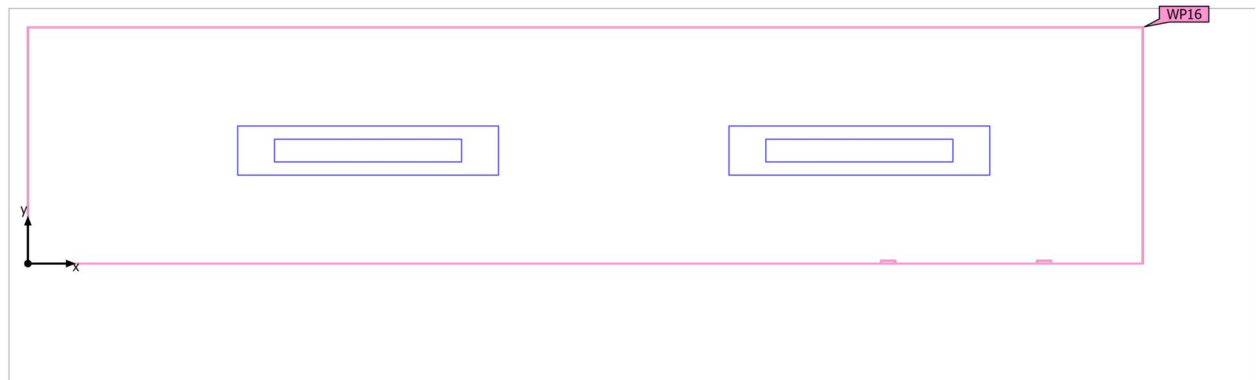
Luminous efficacy

150.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	ETAP	-	R3111/LEDW455DX1 (_LTRevAC)	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 205 BALCONY (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 205 BALCONY (Light scene 1)

Calculation objects

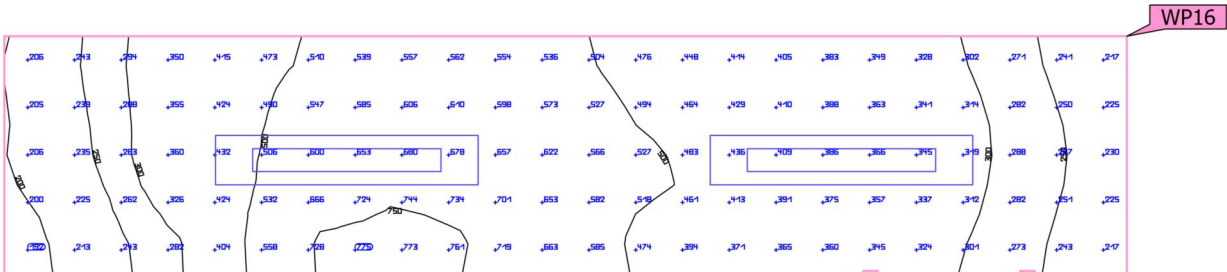
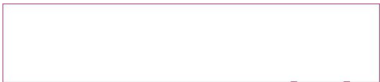
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (205 BALCONY)	427 lx	186 lx	766 lx	0.44	0.24	WP16
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 205 BALCONY (Light scene 1)

Working plane (205 BALCONY)

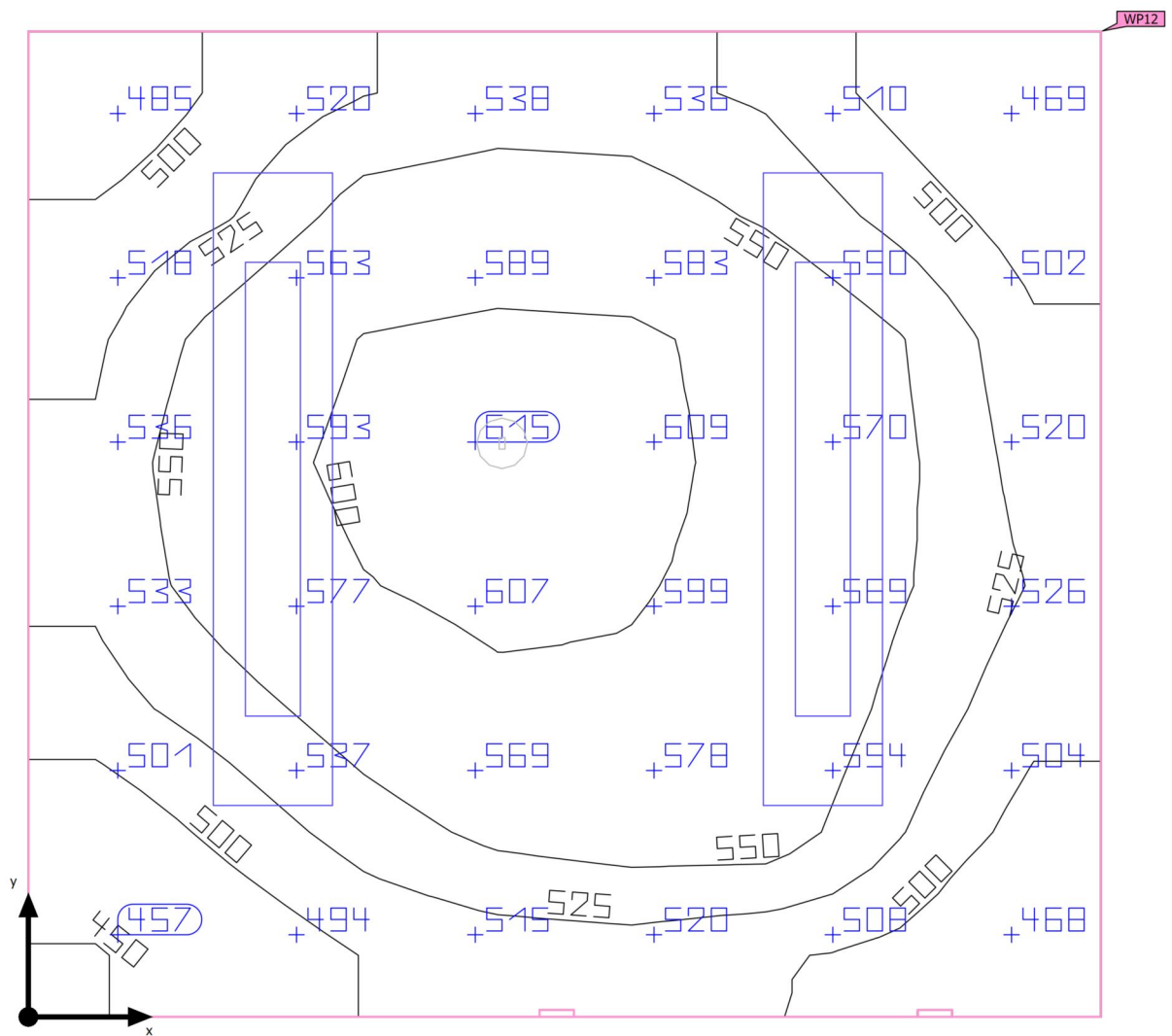


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (205 BALCONY) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	427 lx (≥ 500 lx) ✗	186 lx	766 lx	0.44 (≥ 0.60) ✗	0.24	WP16

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 206 TOILET (Light scene 1)

Summary



Ground area	5.03 m ²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.000 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 206 TOILET (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	541 lx	≥ 500 lx	✓	WP12
	$U_o (g_1)$	0.83	≥ 0.60	✓	WP12
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	163 kWh/a	max. 200 kWh/a	✓	
Space	Lighting power density	13.12 W/m ²	–		
		2.43 W/m ² /100 lx	–		

(1) Based on a rectangular space of 2.340 m x 2.150 m and SHR of 0.25.

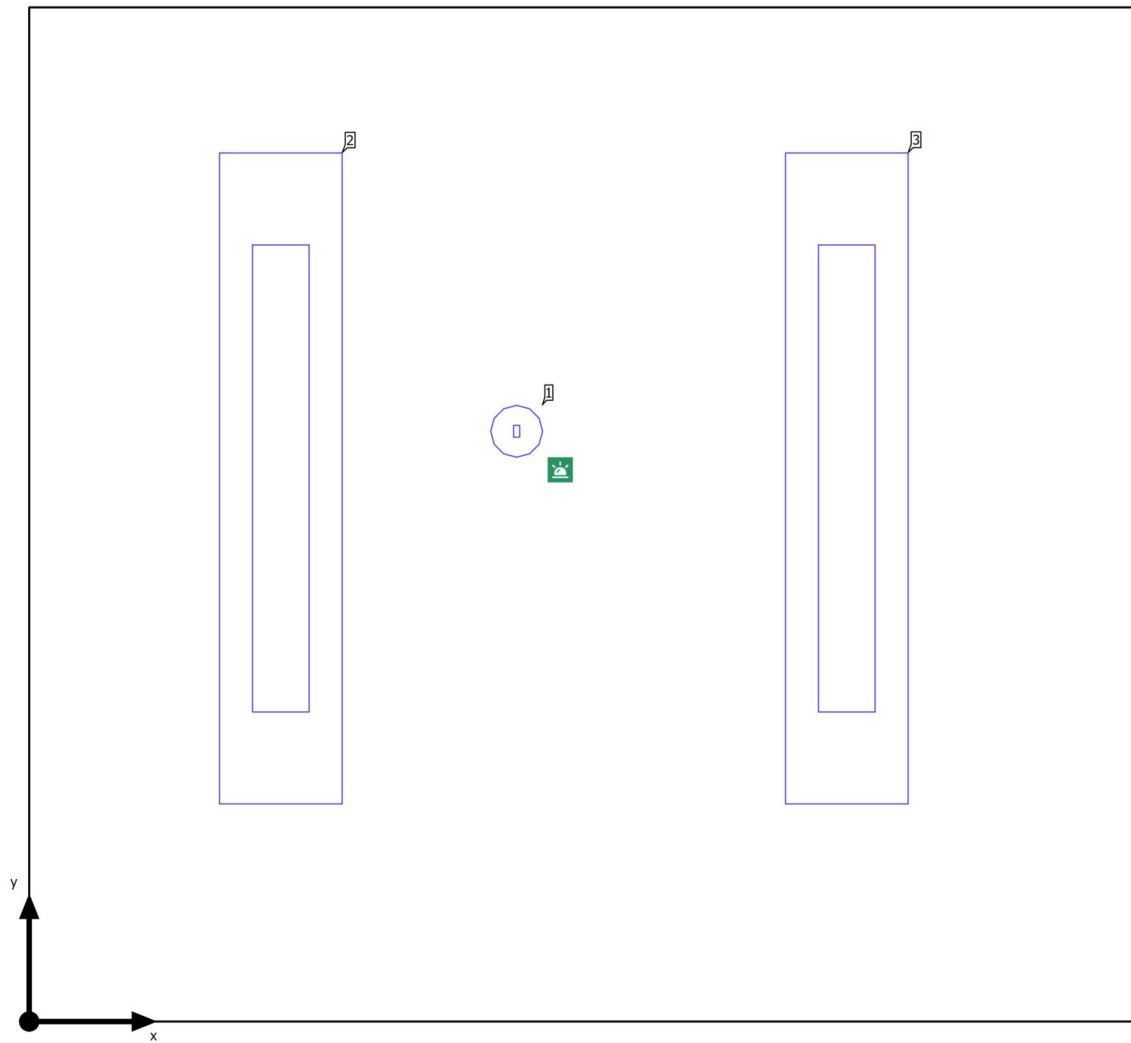
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

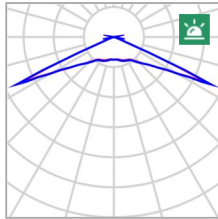
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
2	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 206 TOILET

Luminaire layout plan

Building A · FIRST FLOOR · 206 TOILET

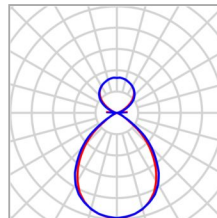
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.033 m	1.251 m	3.368 m	1

Building A · FIRST FLOOR · 206 TOILET

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

Individual luminaires

X	Y	Mounting height	Luminaire
0.533 m	1.151 m	2.800 m	2
1.733 m	1.151 m	2.800 m	3

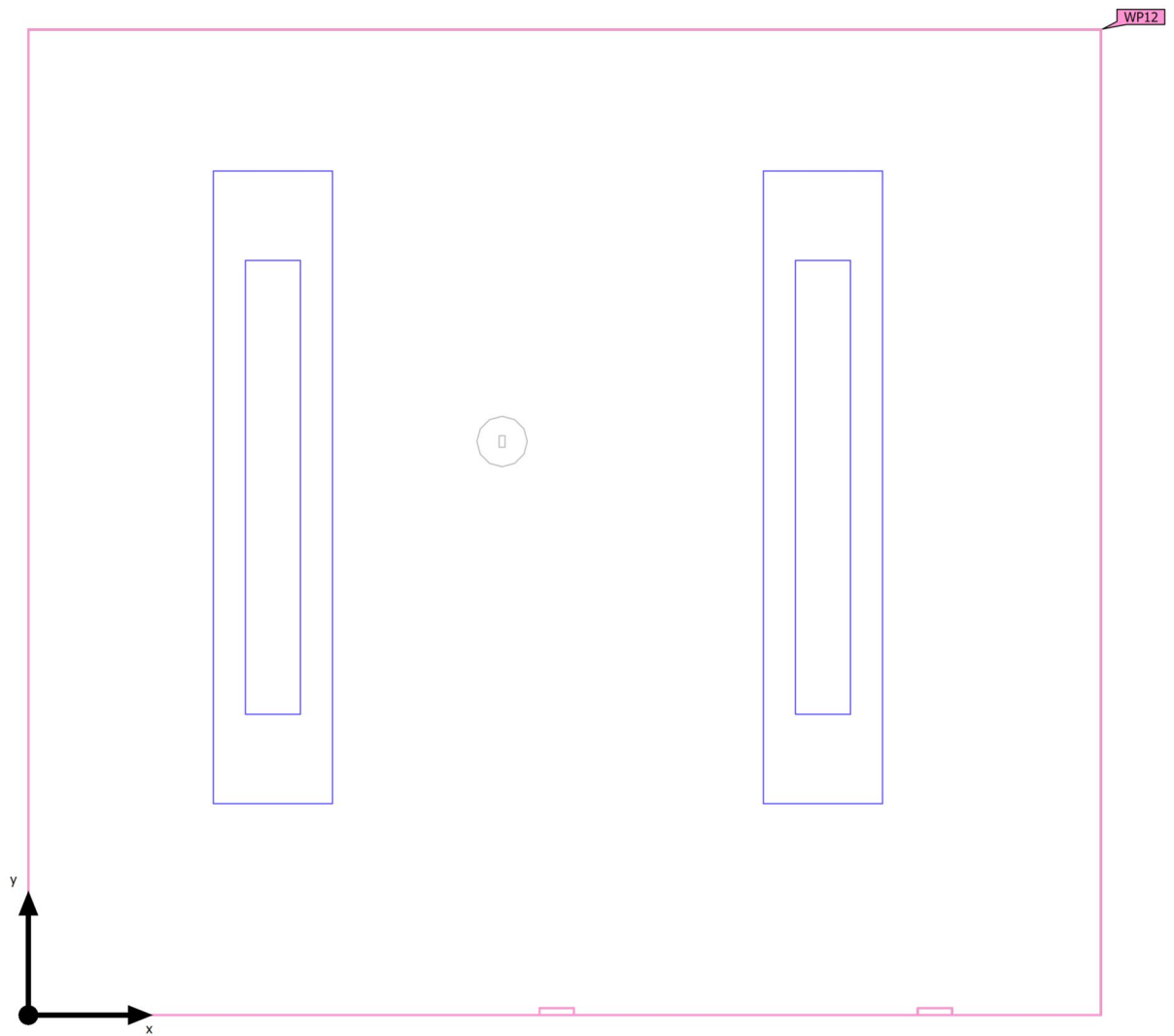
Building A · FIRST FLOOR · 206 TOILET

Luminaire list

Φ_{total} 10800 lm		P_{total} 66.9 W		Luminous efficacy 161.4 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)			0.9 W	300 lm	340.9 lm/W	
						 0.9 W	300 lm (100 %)	-	
2	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 206 TOILET (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 206 TOILET (Light scene 1)

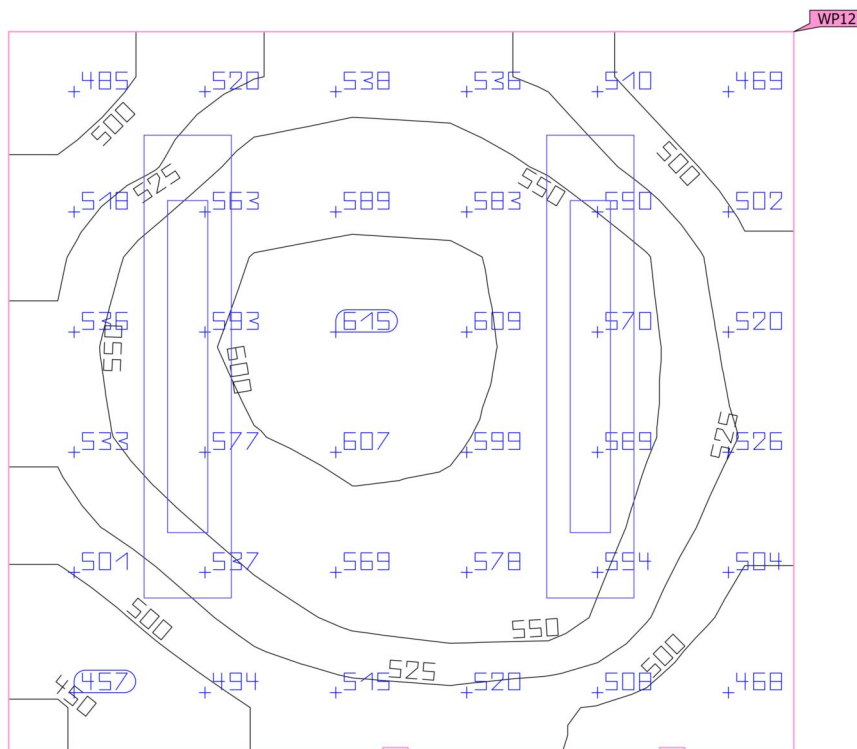
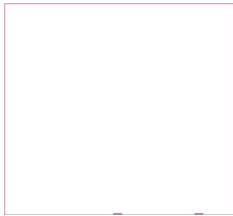
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (206 TOILET) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	541 lx (≥ 500 lx) ✓	447 lx	617 lx	0.83 (≥ 0.60) ✓	0.72	WP12

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 206 TOILET (Light scene 1)

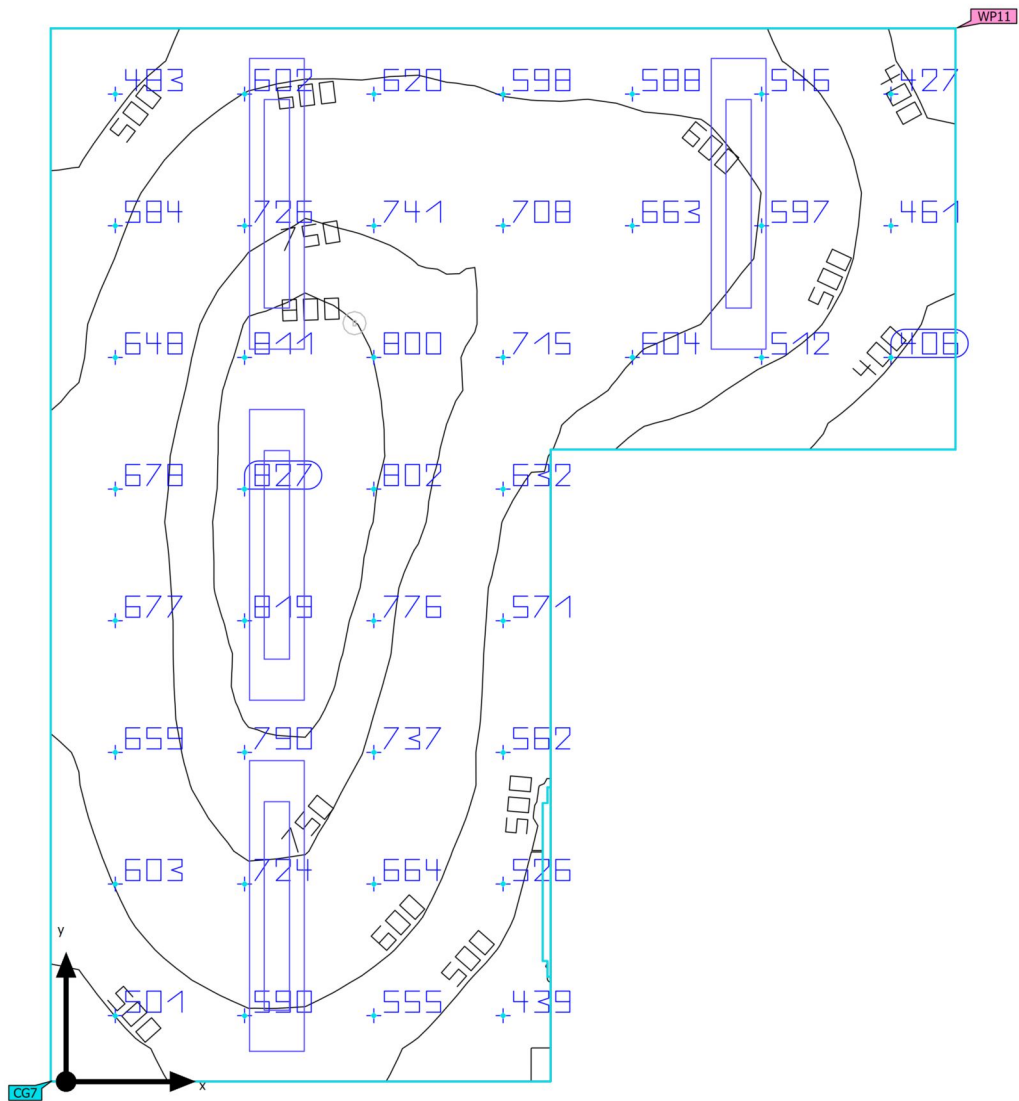
Working plane (206 TOILET)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (206 TOILET)	541 lx	447 lx	617 lx	0.83	0.72	WP12
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.000 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Summary



Ground area	15.71 m ²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	632 lx	$\geq 500 \text{ lx}$	✓	WP11
	$U_o (g_1)$	0.52	≥ 0.60	✗	WP11
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[206 - 327] kWh/a	max. 600 kWh/a	✓	
Space	Lighting power density	8.40 W/m ²	–		
		1.33 W/m ² /100 lx	–		

(1) Based on a rectangular space of 5.000 m x 4.296 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

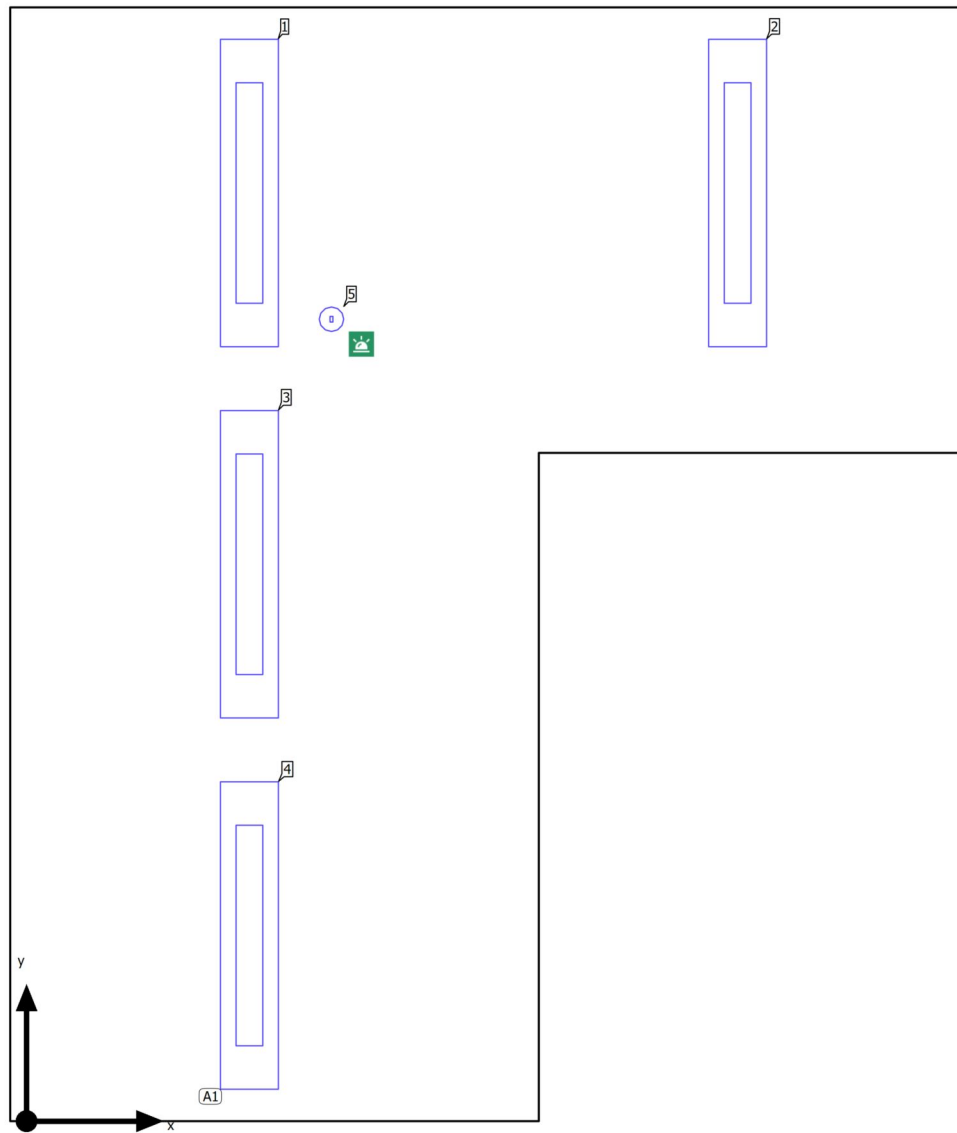
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

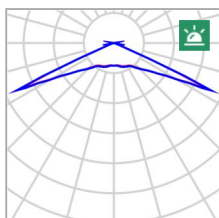
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 207 OFFICE

Luminaire layout plan



Building A · FIRST FLOOR · 207 OFFICE

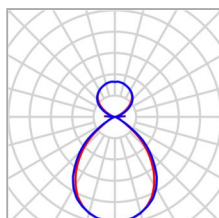
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.369 m	3.600 m	3.368 m	5

Building A · FIRST FLOOR · 207 OFFICE

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

4 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.001 m / 0.833 m / 2.800 m	1.001 m	4.167 m	2.800 m	1
X-direction	2 pcs., Centre - centre, Distances not equal	3.193 m	4.167 m	2.800 m	2
		1.001 m	2.500 m	2.800 m	3
Y-direction	3 pcs., Centre - centre, Distances not equal	1.001 m	0.833 m	2.800 m	4
Arrangement	A1				

Building A · FIRST FLOOR · 207 OFFICE

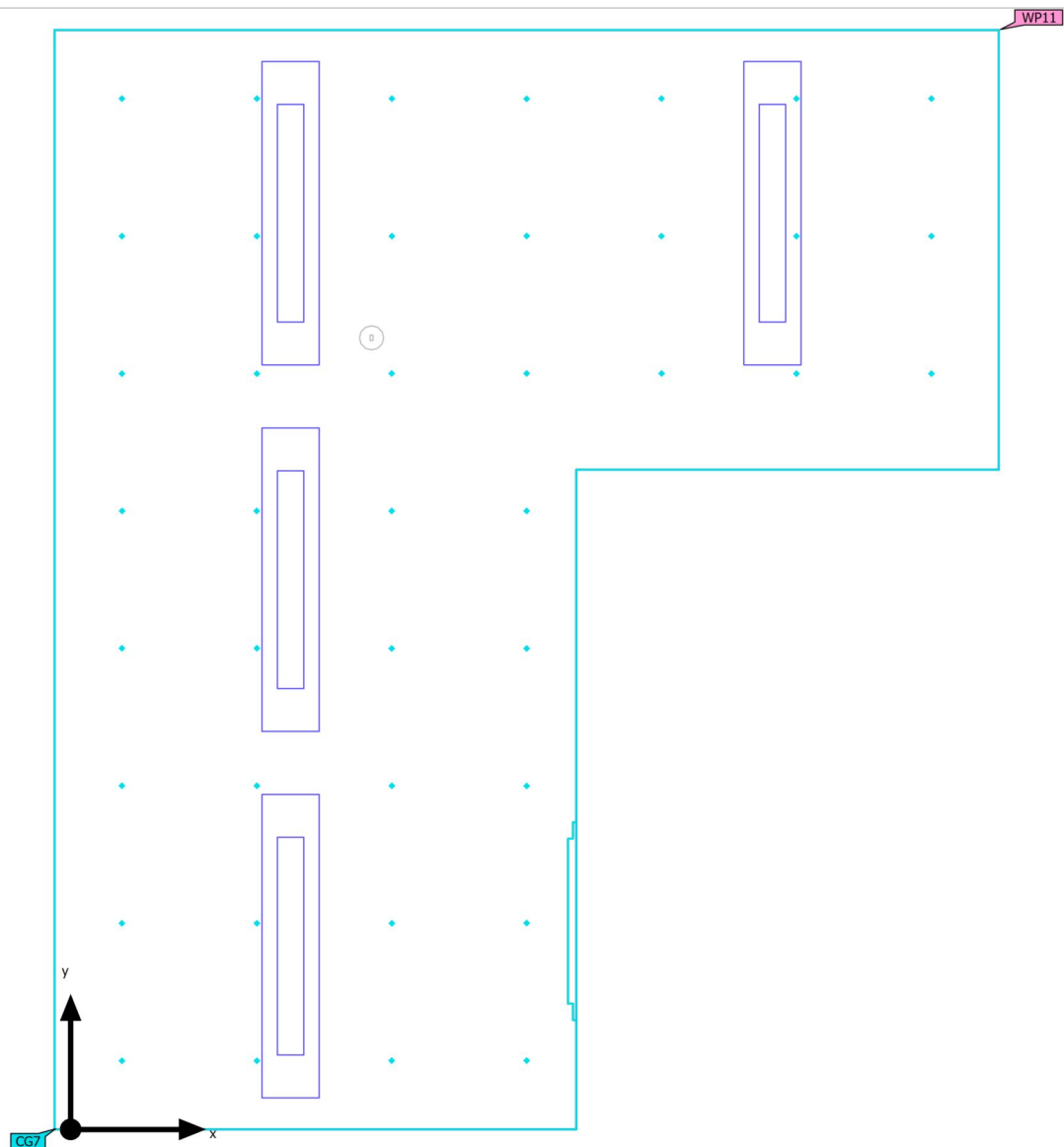
Luminaire list

Φ_{total} 21300 lm	P_{total} 132.9 W	Luminous efficacy 160.3 lm/W	$\Phi_{\text{Emergency lighting}}$ 300 lm	$P_{\text{Emergency lighting}}$ 0.9 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (207 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	632 lx (≥ 500 lx) ✓	331 lx	845 lx	0.52 (≥ 0.60) ✗	0.39	WP11

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Calculation objects

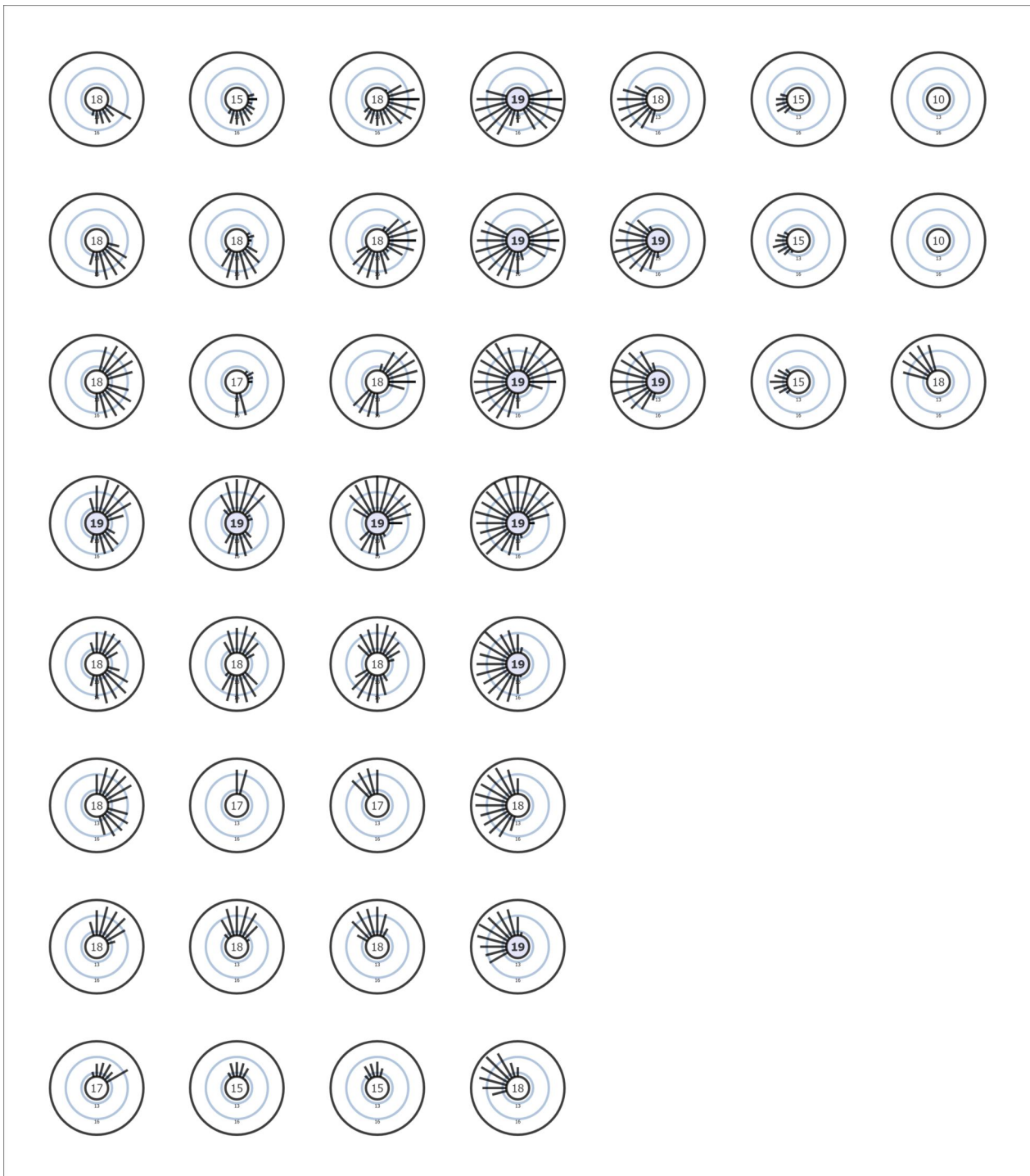
Glare Plane 207 (RUG)

Strongest glare at	45°
max	19.2
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG7

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Calculation objects

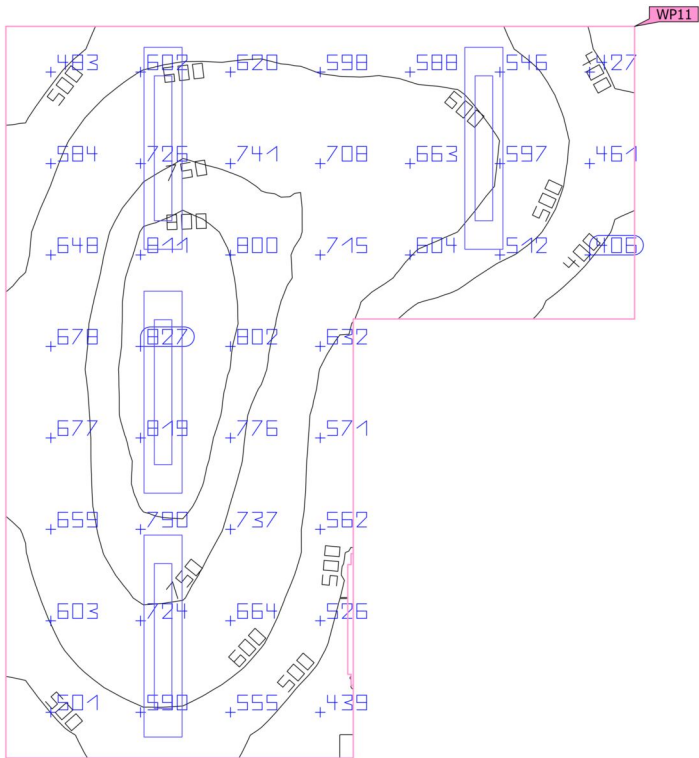
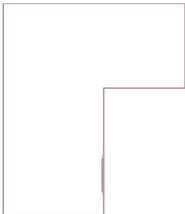
Glare Plane 207 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 207 OFFICE (Light scene 1)

Working plane (207 OFFICE)

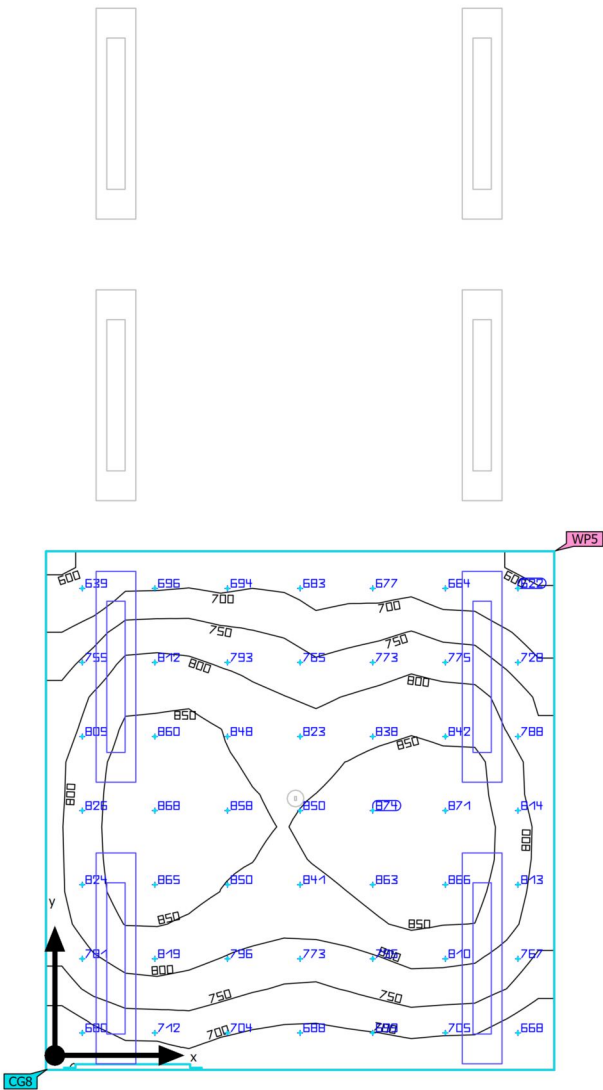


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (207 OFFICE)	632 lx	331 lx	845 lx	0.52	0.39	WP11
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

Summary



Ground area	11.30 m ²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	780 lx	$\geq 500 \text{ lx}$	✓	WP5
	$U_o (g_1)$	0.73	≥ 0.60	✓	WP5
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	327 kWh/a	max. 400 kWh/a	✓	
Space	Lighting power density	11.68 W/m ²	–		
		1.50 W/m ² /100 lx	–		

(1) Based on a rectangular space of 3.328 m x 3.395 m and SHR of 0.25.

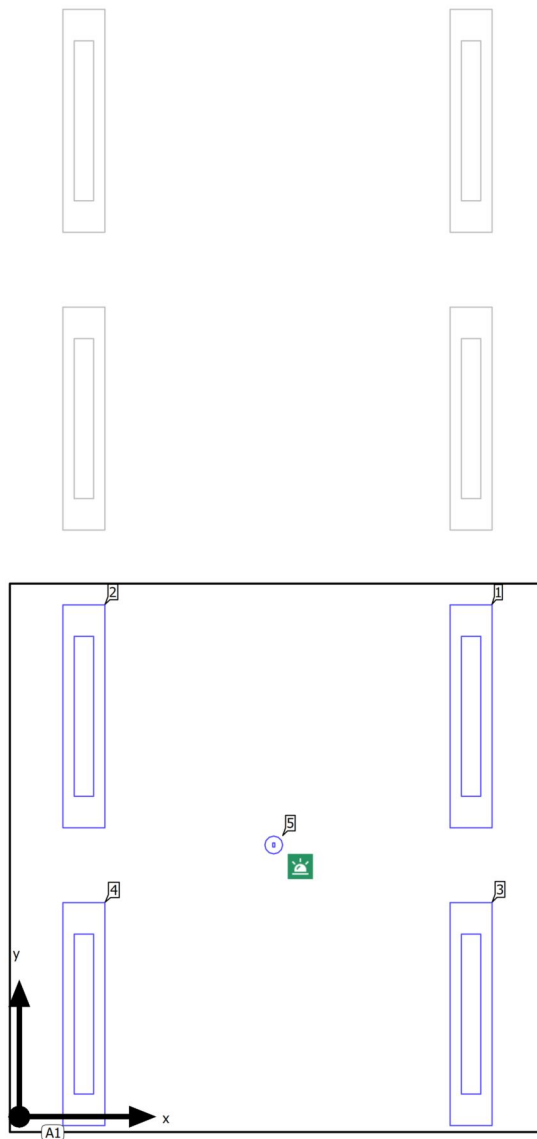
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

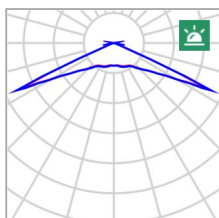
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R31111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 208 MEETING

Luminaire layout plan

Building A · FIRST FLOOR · 208 MEETING

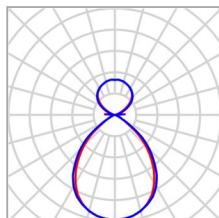
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.576 m	1.682 m	3.368 m	5

Building A · FIRST FLOOR · 208 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

8 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	2.797 m / 2.478 m / 2.800 m	2.797 m	2.478 m	2.800 m	1
X-direction	2 pcs., Centre - centre, 2.397 m	0.400 m	2.478 m	2.800 m	2
Y-direction	4 pcs., Centre - centre, 1.843 m	2.797 m	0.635 m	2.800 m	3
		0.400 m	0.635 m	2.800 m	4
Arrangement	A1				

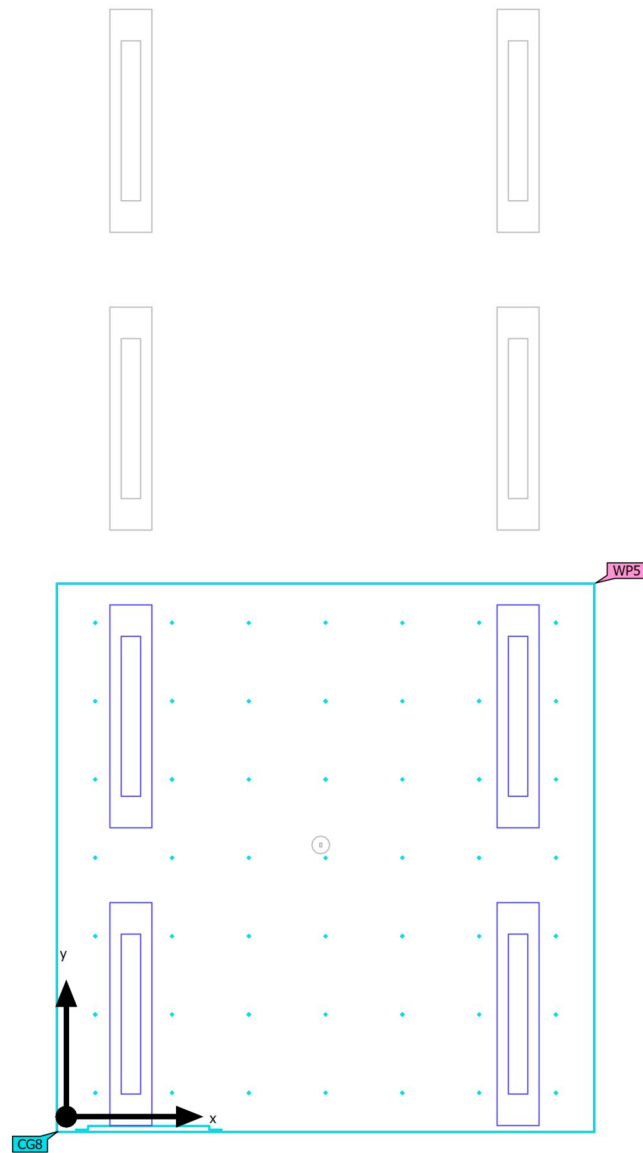
Building A · FIRST FLOOR · 208 MEETING

Luminaire list

Φ_{total} 21300 lm		P_{total} 132.9 W		Luminous efficacy 160.3 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)			0.9 W	300 lm	340.9 lm/W	
						 0.9 W	300 lm (100 %)	-	
4	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (208 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	780 lx (≥ 500 lx) ✓	572 lx	879 lx	0.73 (≥ 0.60) ✓	0.65	WP5

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

Calculation objects

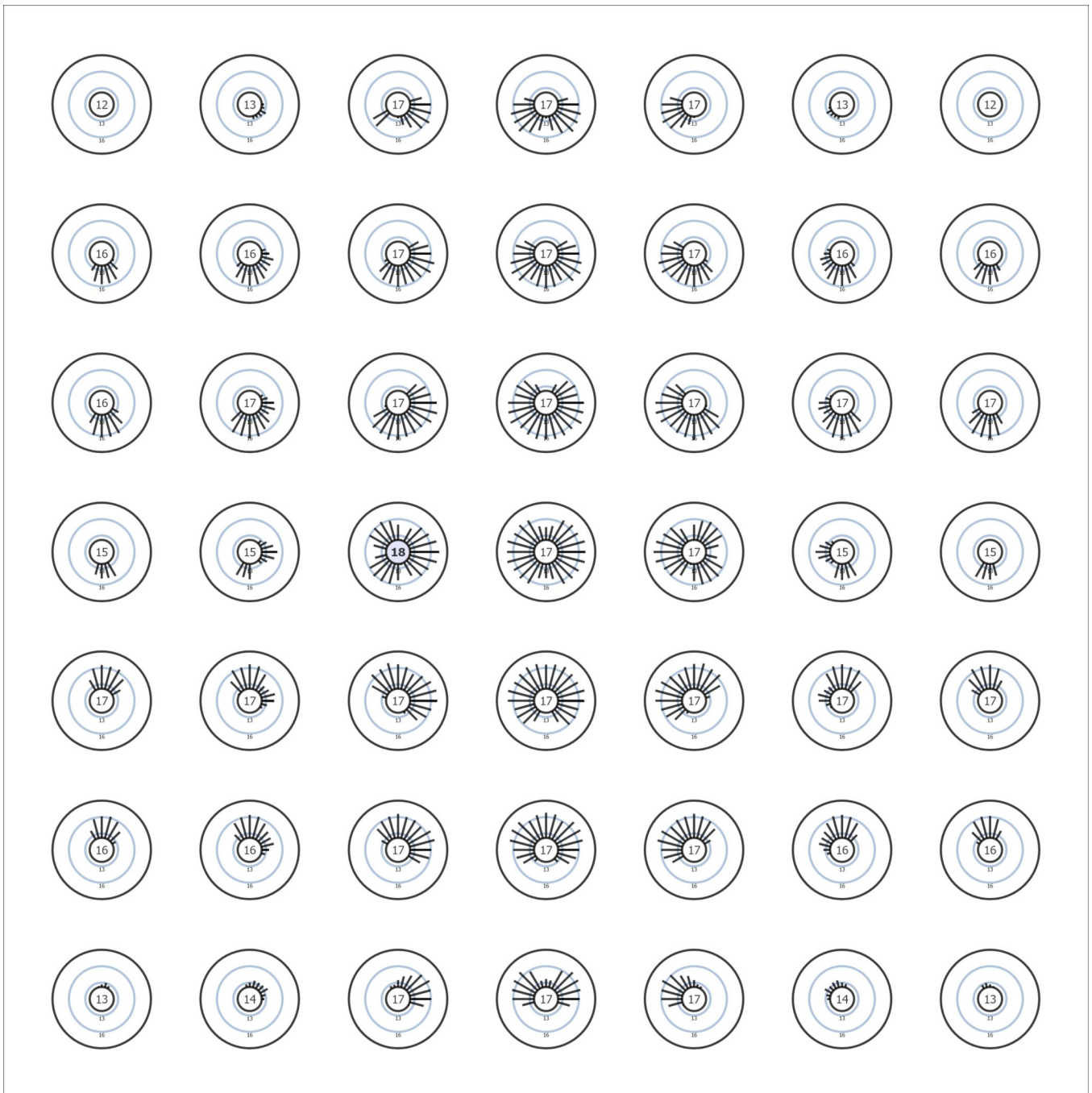
Glare Plane 208 (RUG)

Strongest glare at	0°
max	17.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG8

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

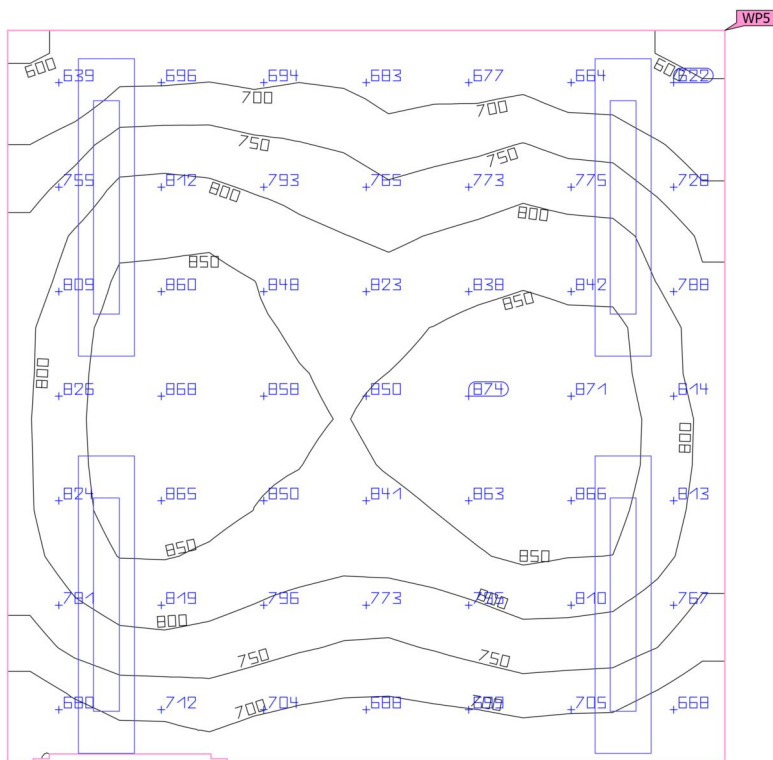
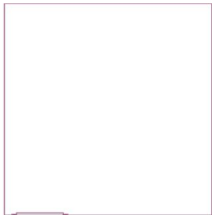
Calculation objects

Glare Plane 208 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 208 MEETING (Light scene 1)

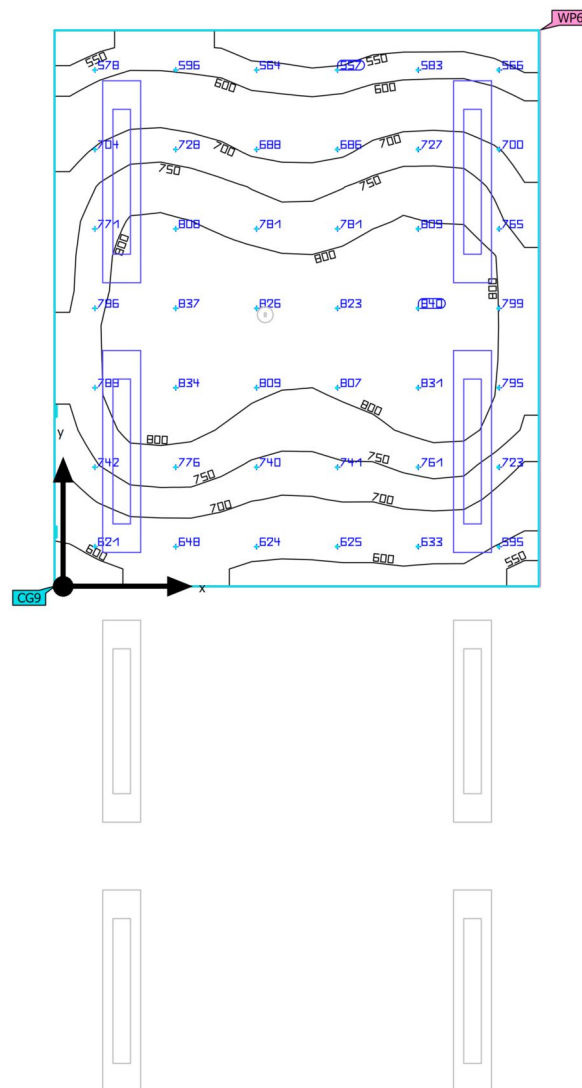
Working plane (208 MEETING)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (208 MEETING)	780 lx	572 lx	879 lx	0.73	0.65	WP5
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

Summary



Ground area	12.60 m ²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	724 lx	≥ 500 lx	✓	WP6
	$U_o (g_1)$	0.70	≥ 0.60	✓	WP6
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[206 - 327] kWh/a	max. 450 kWh/a	✓	
Space	Lighting power density	10.48 W/m ²	–		
		1.45 W/m ² /100 lx	–		

(1) Based on a rectangular space of 3.800 m x 3.316 m and SHR of 0.25.

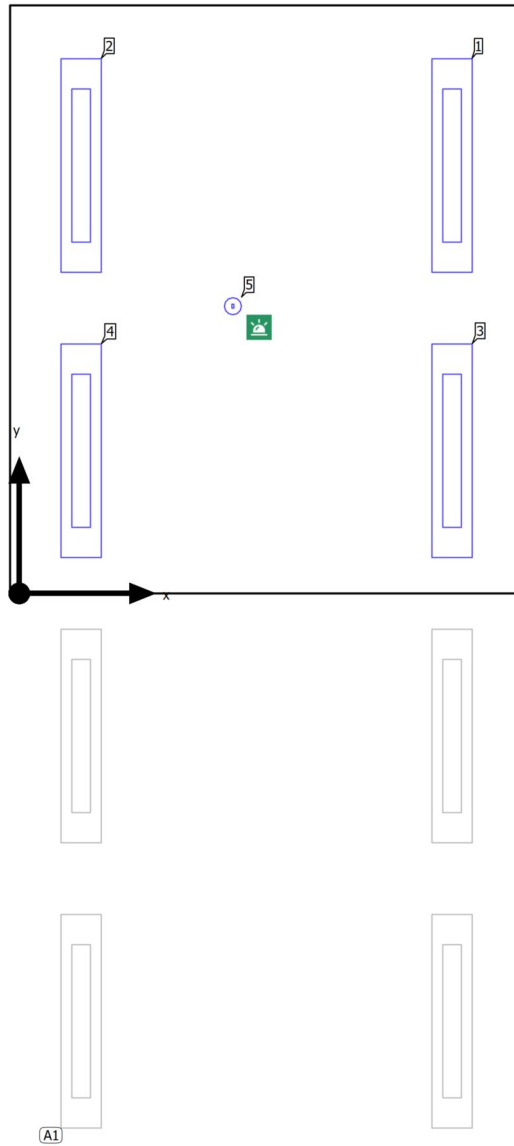
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

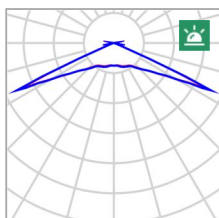
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 209 MEETING

Luminaire layout plan

Building A · FIRST FLOOR · 209 MEETING

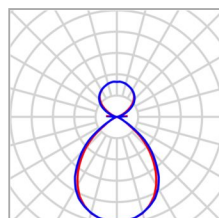
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
1.381 m	1.856 m	3.368 m	5

Building A · FIRST FLOOR · 209 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

8 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	2.797 m / 2.765 m / 2.800 m	2.797 m	2.765 m	2.800 m	1
X-direction	2 pcs., Centre - centre, 2.397 m	0.400 m	2.765 m	2.800 m	2
Y-direction	4 pcs., Centre - centre, 1.843 m	2.797 m	0.922 m	2.800 m	3
		0.400 m	0.922 m	2.800 m	4
Arrangement	A1				

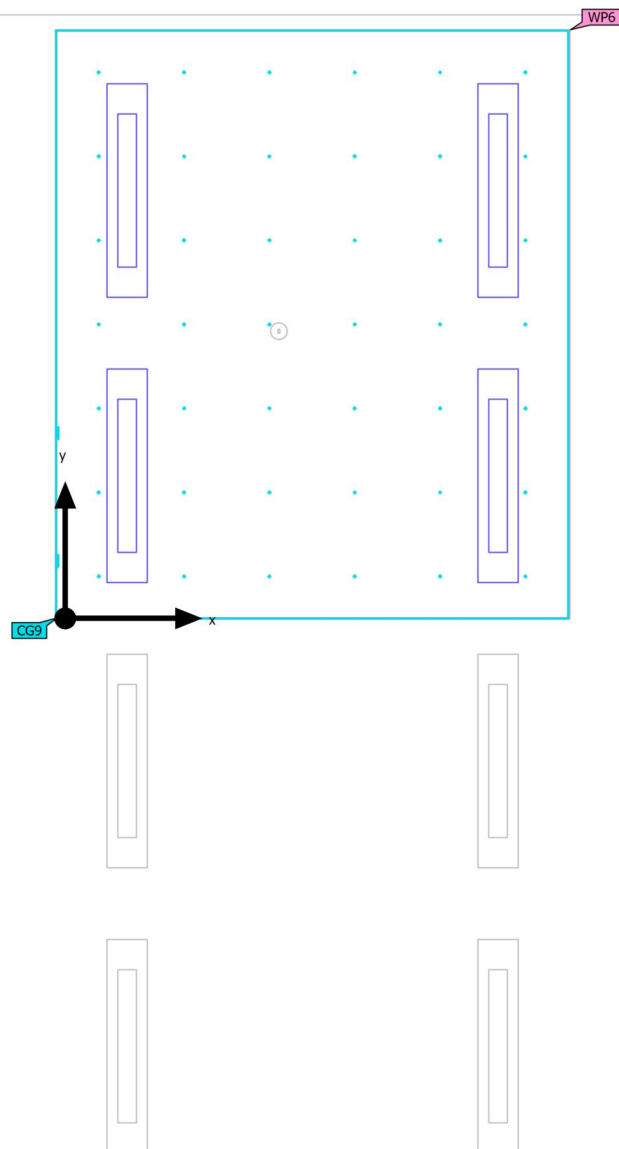
Building A · FIRST FLOOR · 209 MEETING

Luminaire list

Φ_{total} 21300 lm		P_{total} 132.9 W		Luminous efficacy 160.3 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)			0.9 W	300 lm	340.9 lm/W	
						 0.9 W	300 lm (100 %)	-	
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (209 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	724 lx (≥ 500 lx) ✓	508 lx	841 lx	0.70 (≥ 0.60) ✓	0.60	WP6

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

Calculation objects

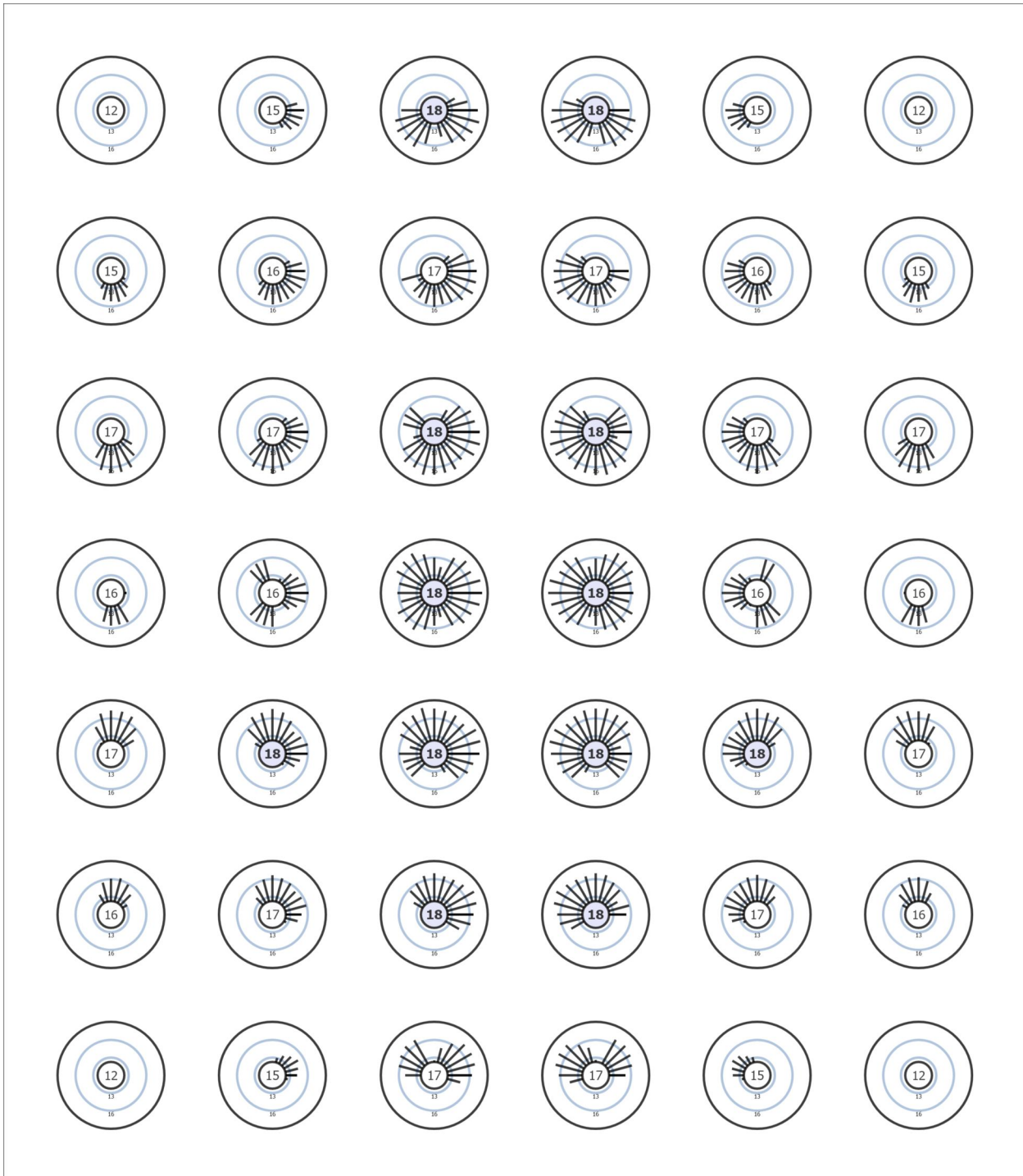
Glare Plane 209 (RUG)

Strongest glare at	0°
max	18.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG9

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

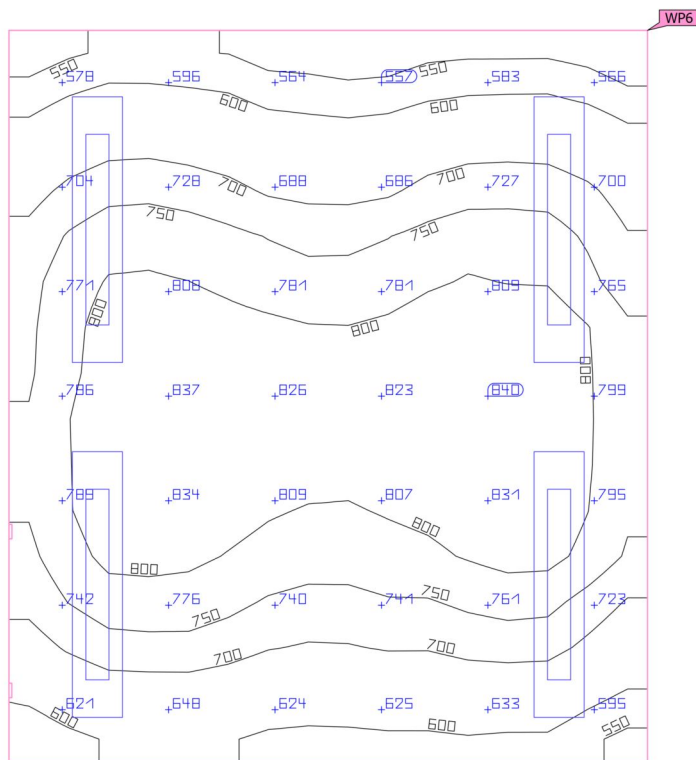
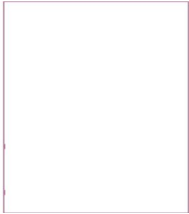
Calculation objects

Glare Plane 209 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 209 MEETING (Light scene 1)

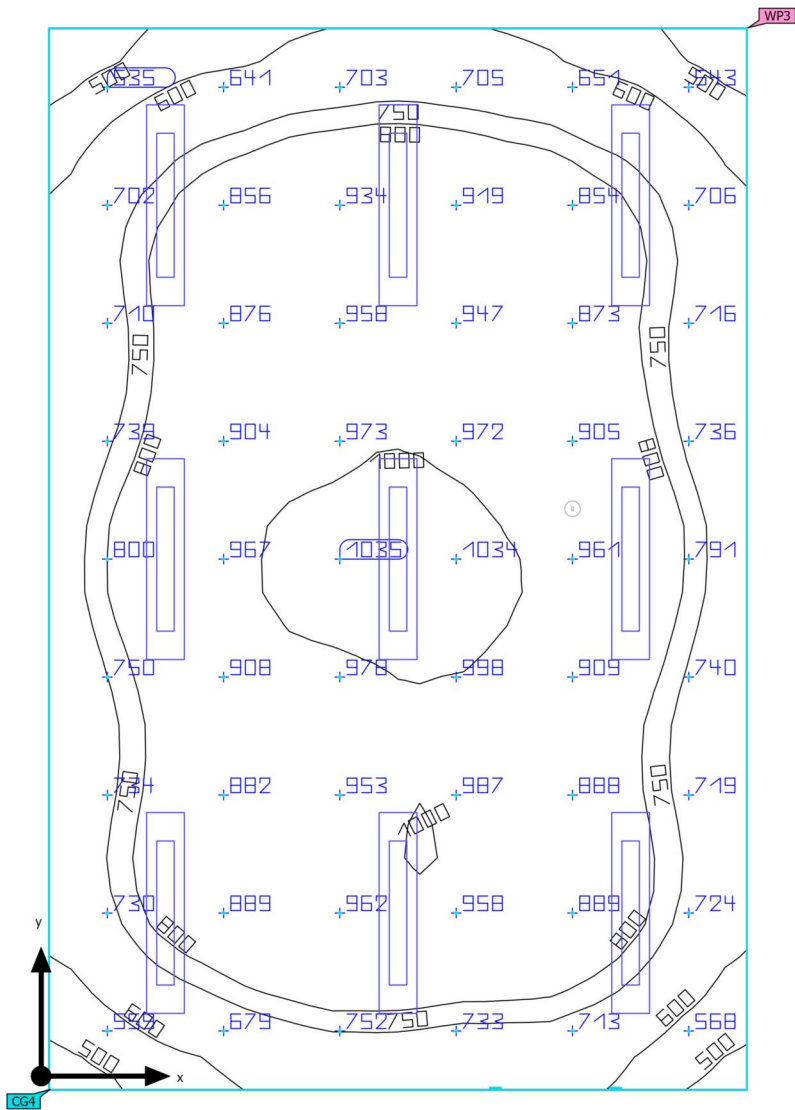
Working plane (209 MEETING)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (209 MEETING)	724 lx	508 lx	841 lx	0.70	0.60	WP6
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

Summary



Ground area	34.98 m ²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	818 lx	$\geq 500 \text{ lx}$	✓	WP3
	$U_o (g_1)$	0.53	≥ 0.60	✗	WP3
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[463 - 735] kWh/a	max. 1250 kWh/a	✓	
Space	Lighting power density	8.49 W/m ²	–		
		1.04 W/m ² /100 lx	–		

(1) Based on a rectangular space of 4.795 m x 7.295 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

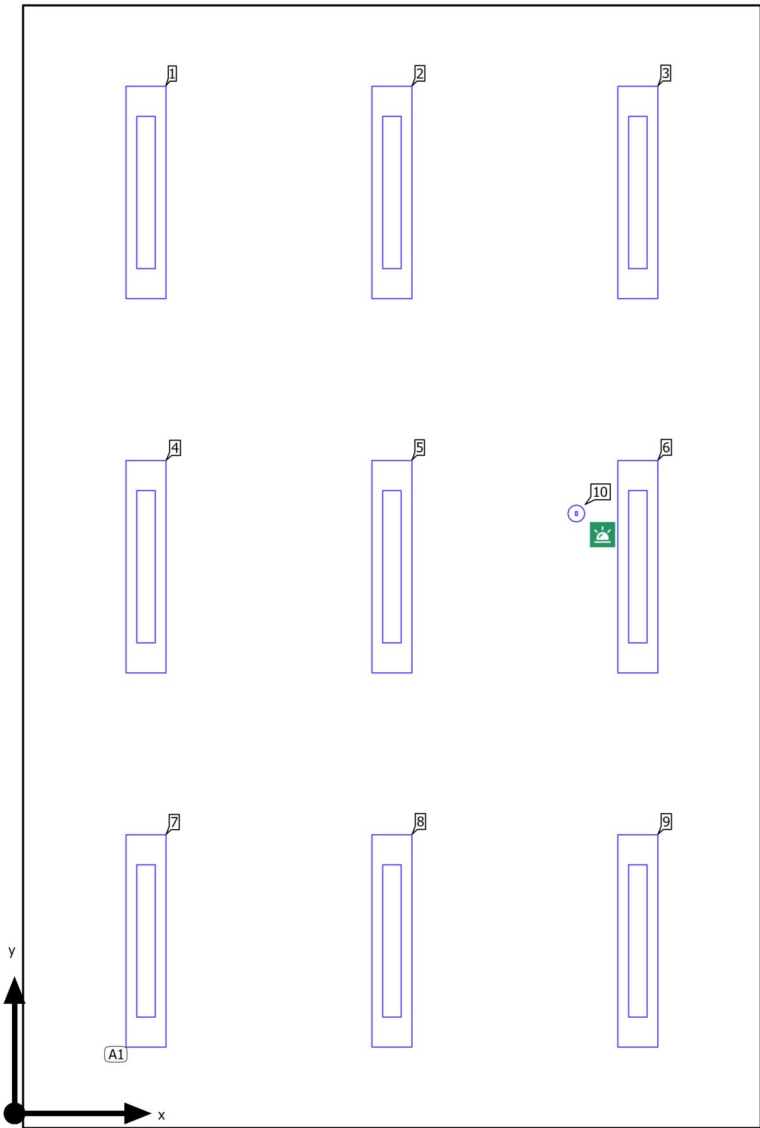
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

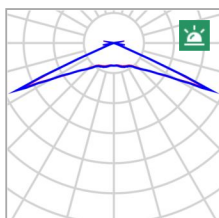
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
9	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 210 CAD

Luminaire layout plan



Building A · FIRST FLOOR · 210 CAD

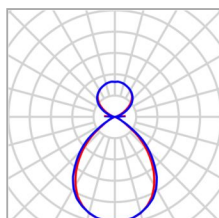
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
3.651 m	3.898 m	3.368 m	10

Building A · FIRST FLOOR · 210 CAD

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

9 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	0.854 m / 1.121 m / 2.800 m	0.854 m	5.984 m	2.800 m	1
X-direction	3 pcs., Centre - centre, 1.598 m	2.453 m	5.984 m	2.800 m	2
		4.051 m	5.984 m	2.800 m	3
Y-direction	3 pcs., Centre - centre, 2.432 m	0.854 m	3.552 m	2.800 m	4
		2.453 m	3.552 m	2.800 m	5
		4.051 m	3.552 m	2.800 m	6
Arrangement	A1	0.854 m	1.121 m	2.800 m	7
		2.453 m	1.121 m	2.800 m	8
		4.051 m	1.121 m	2.800 m	9

Building A · FIRST FLOOR · 210 CAD

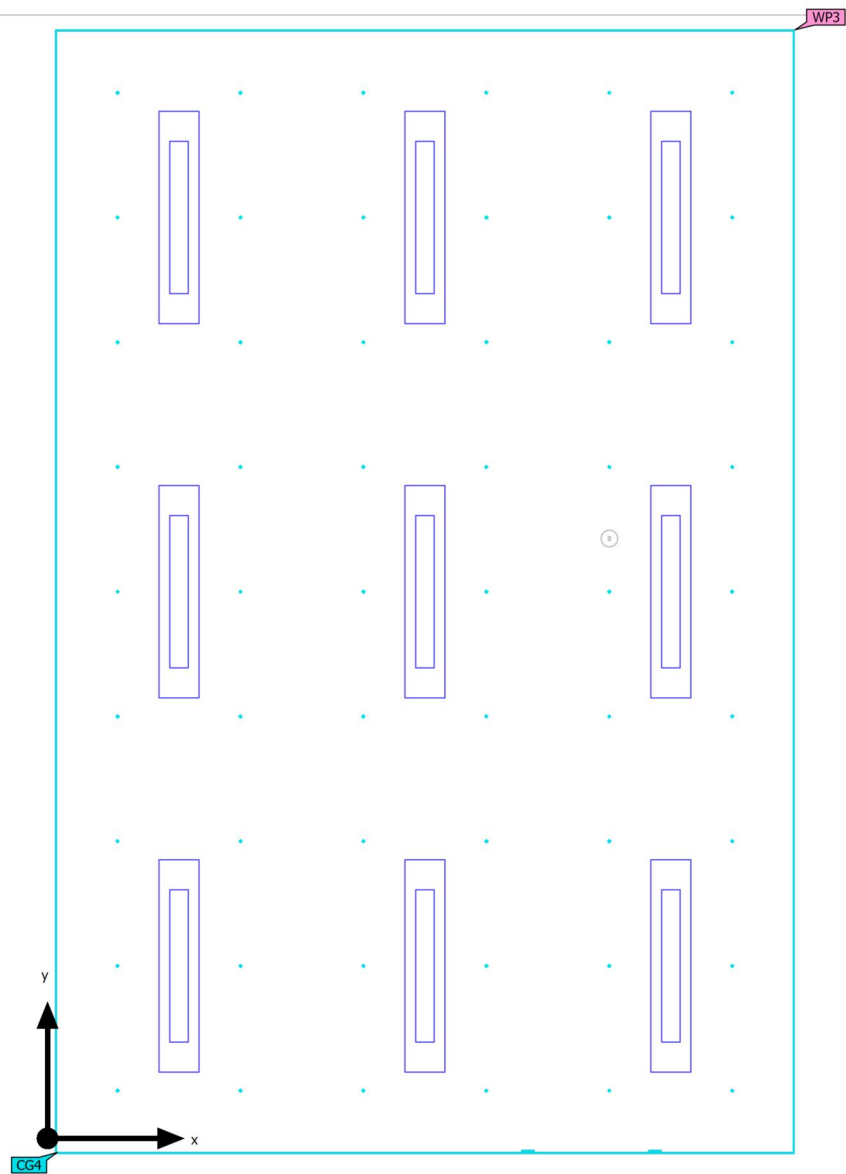
Luminaire list

Φ_{total} 47550 lm	P_{total} 297.9 W	Luminous efficacy 159.6 lm/W	$\Phi_{\text{Emergency lighting}}$ 300 lm	$P_{\text{Emergency lighting}}$ 0.9 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
9	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 210 CAD (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (210 CAD) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	818 lx (≥ 500 lx) ✓	432 lx	1043 lx	0.53 (≥ 0.60) ✗	0.41	WP3

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

Calculation objects

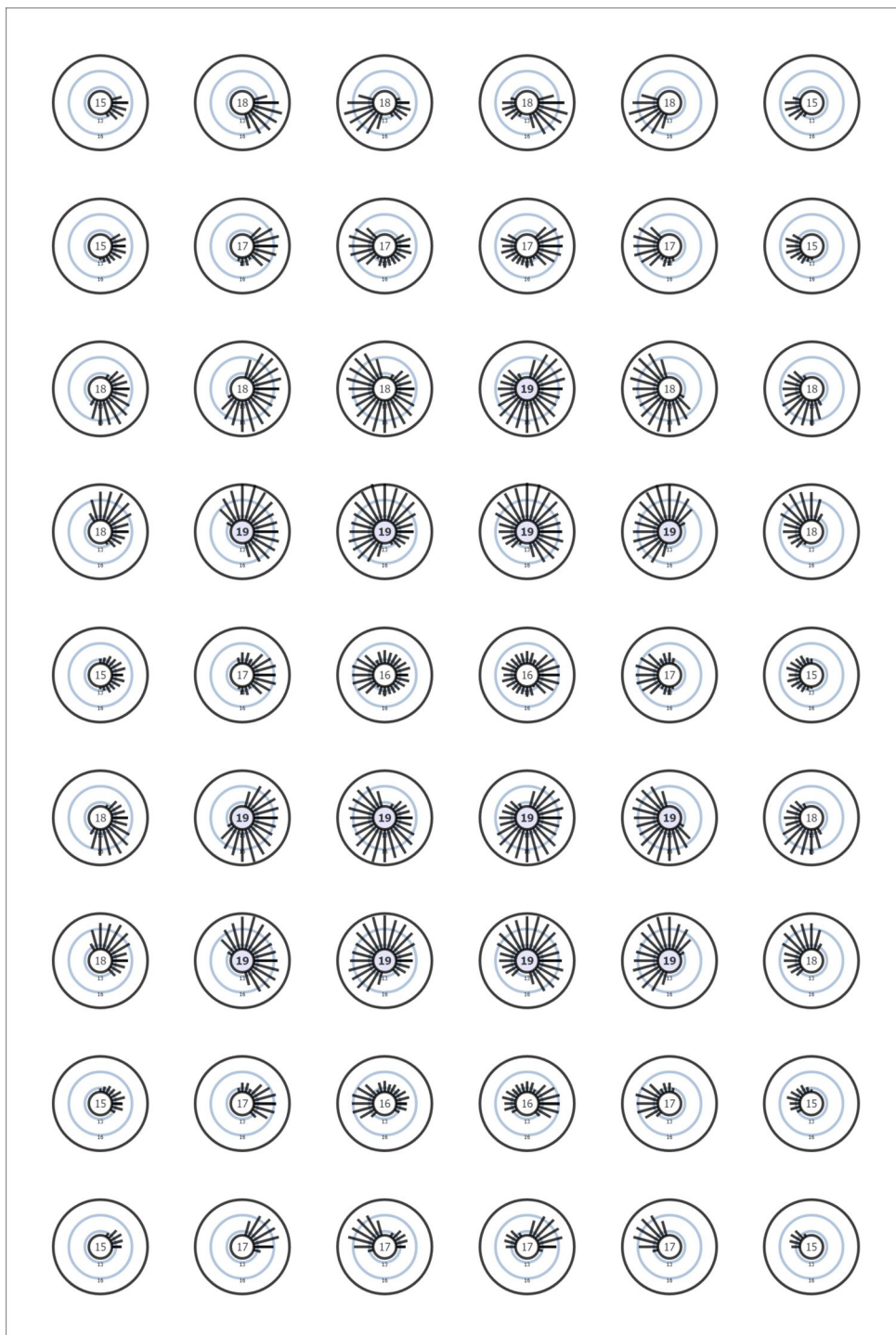
Glare plane 210 (RUG)

Strongest glare at	105°
max	19.3
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG4

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

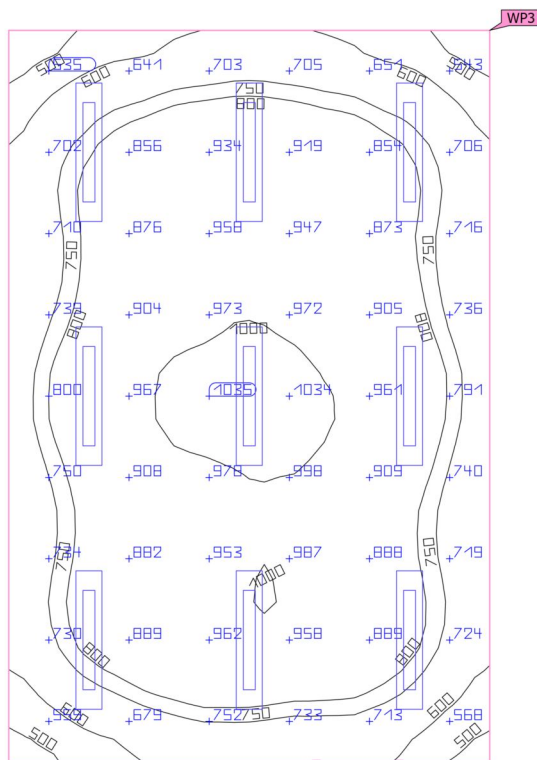
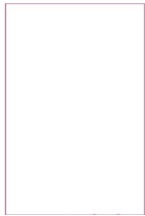
Calculation objects

Glare plane 210 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 210 CAD (Light scene 1)

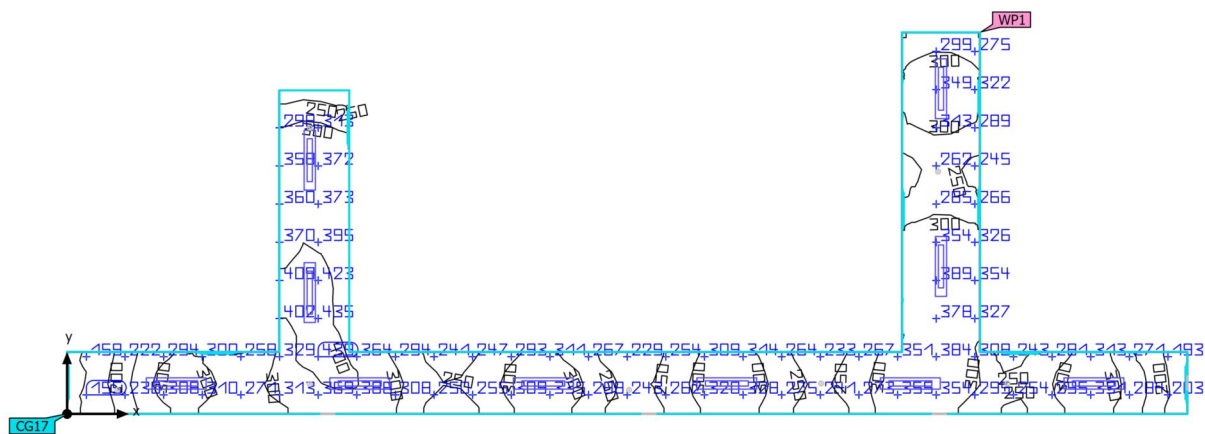
Working plane (210 CAD)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (210 CAD)	818 lx	432 lx	1043 lx	0.53	0.41	WP3
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Summary



Ground area	59.96 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.000 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	307 lx	≥ 500 lx	✗	WP1
	$U_o (g_1)$	0.41	≥ 0.60	✗	WP1
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[605 - 817] kWh/a	max. 2100 kWh/a	✓	
Space	Lighting power density	5.50 W/m ²	–		
		1.79 W/m ² /100 lx	–		

(1) Based on a rectangular space of 25.850 m x 8.803 m and SHR of 0.25.

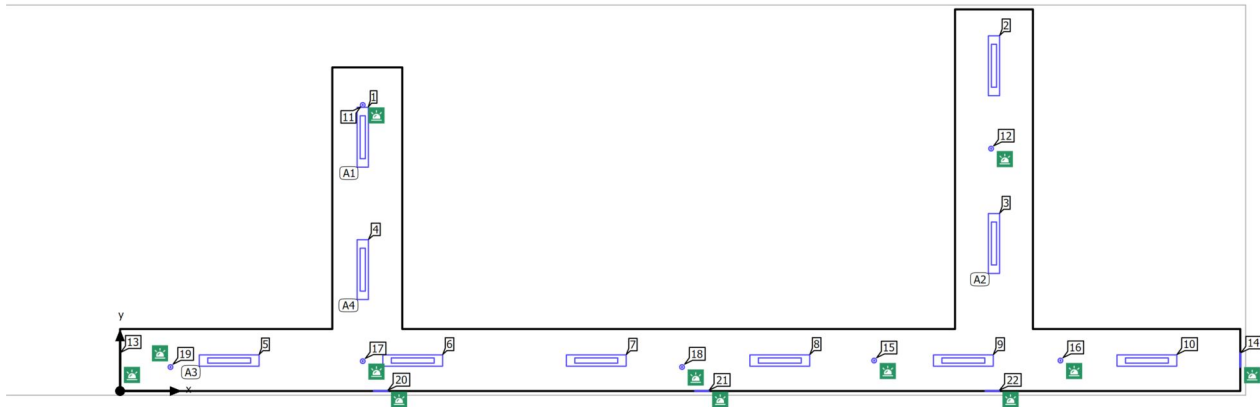
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

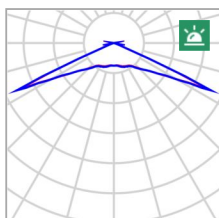
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
10	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 211 CORRIDOR

Luminaire layout plan

Building A · FIRST FLOOR · 211 CORRIDOR

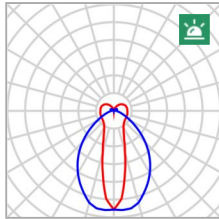
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
5.600 m	6.599 m	3.368 m	11
20.100 m	5.592 m	3.368 m	12
17.400 m	0.703 m	3.368 m	15
21.700 m	0.703 m	3.368 m	16
5.600 m	0.689 m	3.368 m	17
12.967 m	0.553 m	3.368 m	18
1.162 m	0.553 m	3.368 m	19

Building A · FIRST FLOOR · 211 CORRIDOR

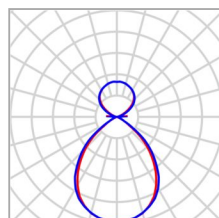
Luminaire layout plan

Manufacturer	ETAP	P	2.5 W
Article No.	-	P _{Emergency lighting}	2.5 W
Article name	K9R142/1-AX1 (_LTRevAB)	Φ _{Luminaire}	80 lm
Fitting	1x sign with picto	Φ _{Emergency lighting}	80 lm
		ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
0.000 m	0.720 m	3.300 m	13
25.850 m	0.715 m	3.300 m	14
6.014 m	0.000 m	3.300 m	20
13.425 m	0.000 m	3.300 m	21
20.125 m	0.000 m	3.300 m	22

Building A · FIRST FLOOR · 211 CORRIDOR

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRRevAC)		
Fitting	1x LED		

1 x ETAP R3111/LEDN455DX1 (_LTRRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	5.600 m / 5.854 m / 2.800 m	5.600 m	5.854 m	2.800 m	1
X-direction	1 pcs., Centre - centre, 5.850 m				
Arrangement	A1				

2 x ETAP R3111/LEDN455DX1 (_LTRRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	20.166 m / 3.403 m / 2.800 m	20.166 m	7.503 m	2.800 m	2
X-direction	2 pcs., Centre - centre, Distances not equal	20.166 m	3.403 m	2.800 m	3
Arrangement	A2				

6 x ETAP R3111/LEDN455DX1 (_LTRRevAC)

Building A · FIRST FLOOR · 211 CORRIDOR

Luminaire layout plan

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	19.459 m / 0.703 m / 2.800 m	2.517 m	0.703 m	2.800 m	5
X-direction	6 pcs., Centre - centre, 4.235 m	6.753 m	0.703 m	2.800 m	6
Arrangement	A3	10.988 m	0.703 m	2.800 m	7
		15.224 m	0.703 m	2.800 m	8
		19.459 m	0.703 m	2.800 m	9
		23.695 m	0.703 m	2.800 m	10

1 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Line arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	5.600 m / 2.800 m / 2.800 m	5.600 m	2.800 m	2.800 m	4
X-direction	1 pcs., Centre - centre, 5.850 m				
Arrangement	A4				

Building A · FIRST FLOOR · 211 CORRIDOR

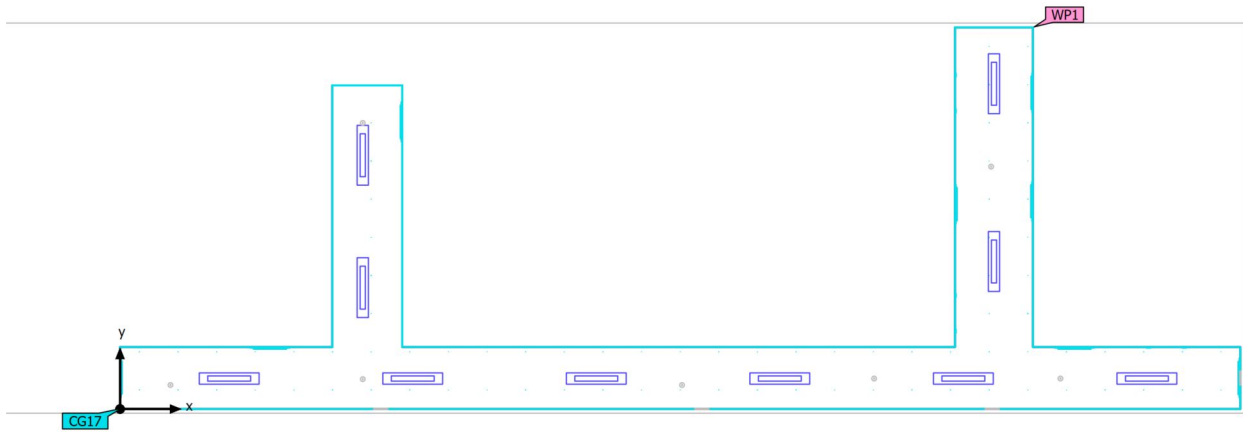
Luminaire list

Φ_{total} 55000 lm	P_{total} 348.8 W	Luminous efficacy 157.7 lm/W	$\Phi_{\text{Emergency lighting}}$ 2500 lm	$P_{\text{Emergency lighting}}$ 18.8 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
7	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
5	ETAP	-	K9R142/1-AX1 (_LTRRevAB)	2.5 W	80 lm	32.4 lm/W
				 2.5 W	80 lm (100 %)	-
10	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (211 CORRIDOR) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	307 lx (≥ 500 lx) ✗	125 lx	448 lx	0.41 (≥ 0.60) ✗	0.28	WP1

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Calculation objects

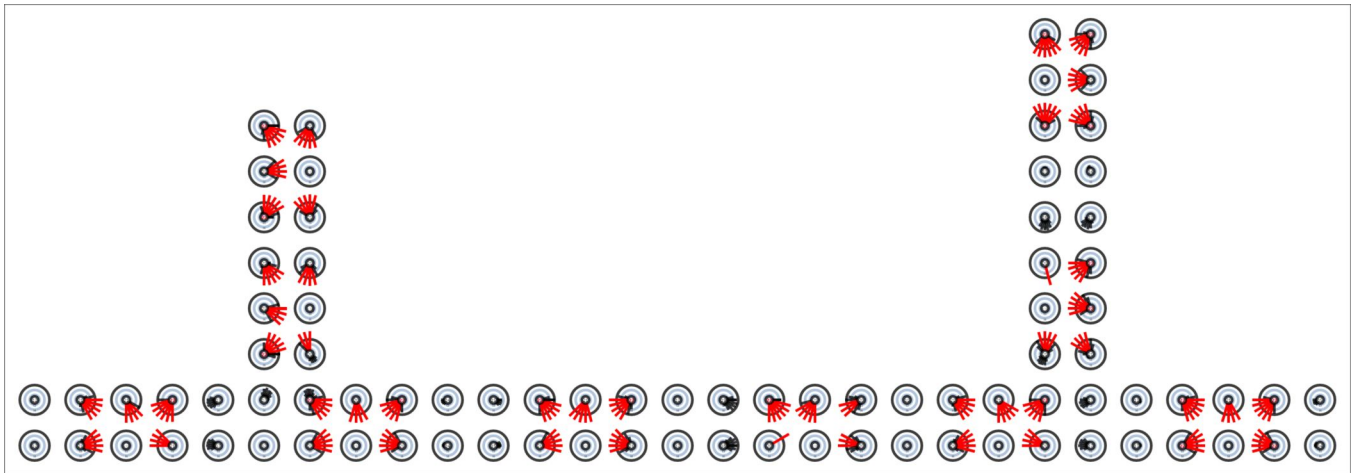
Glare plane 211 (RUG)

Strongest glare at	240°
max	22.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.800 m
Index	CG17

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Calculation objects

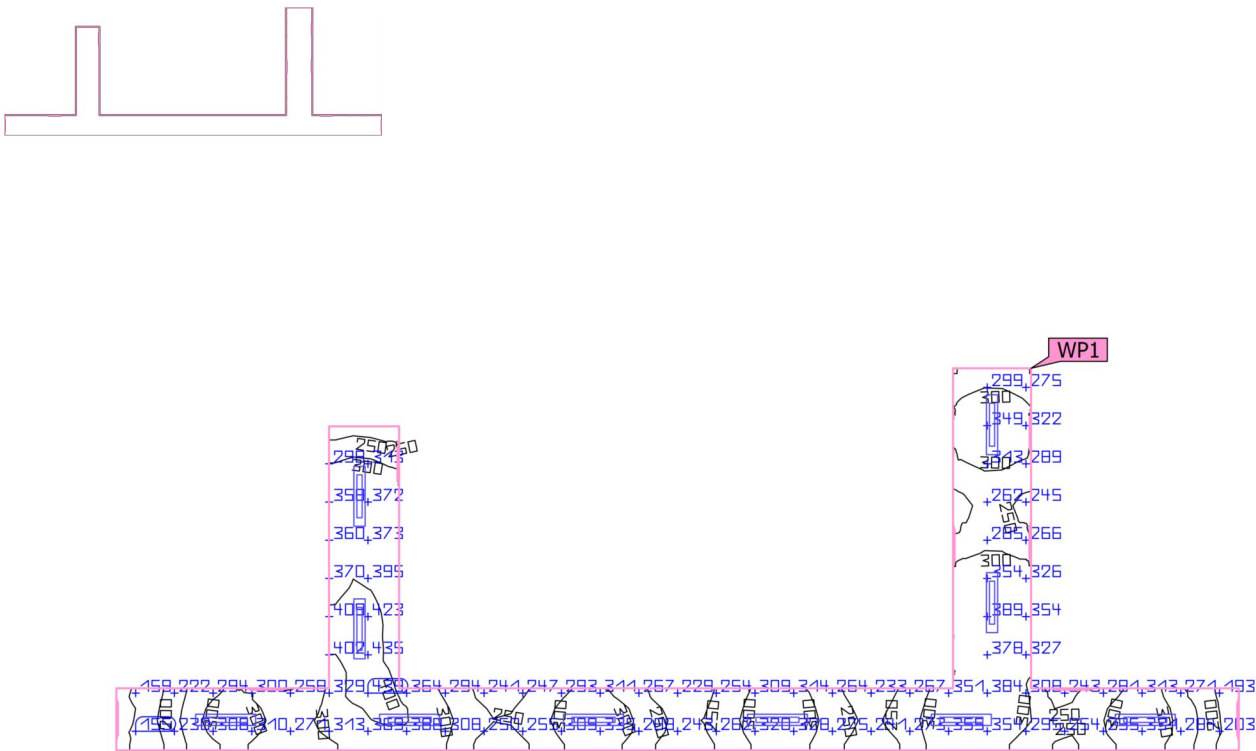
Glare plane 211 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 211 CORRIDOR (Light scene 1)

Working plane (211 CORRIDOR)

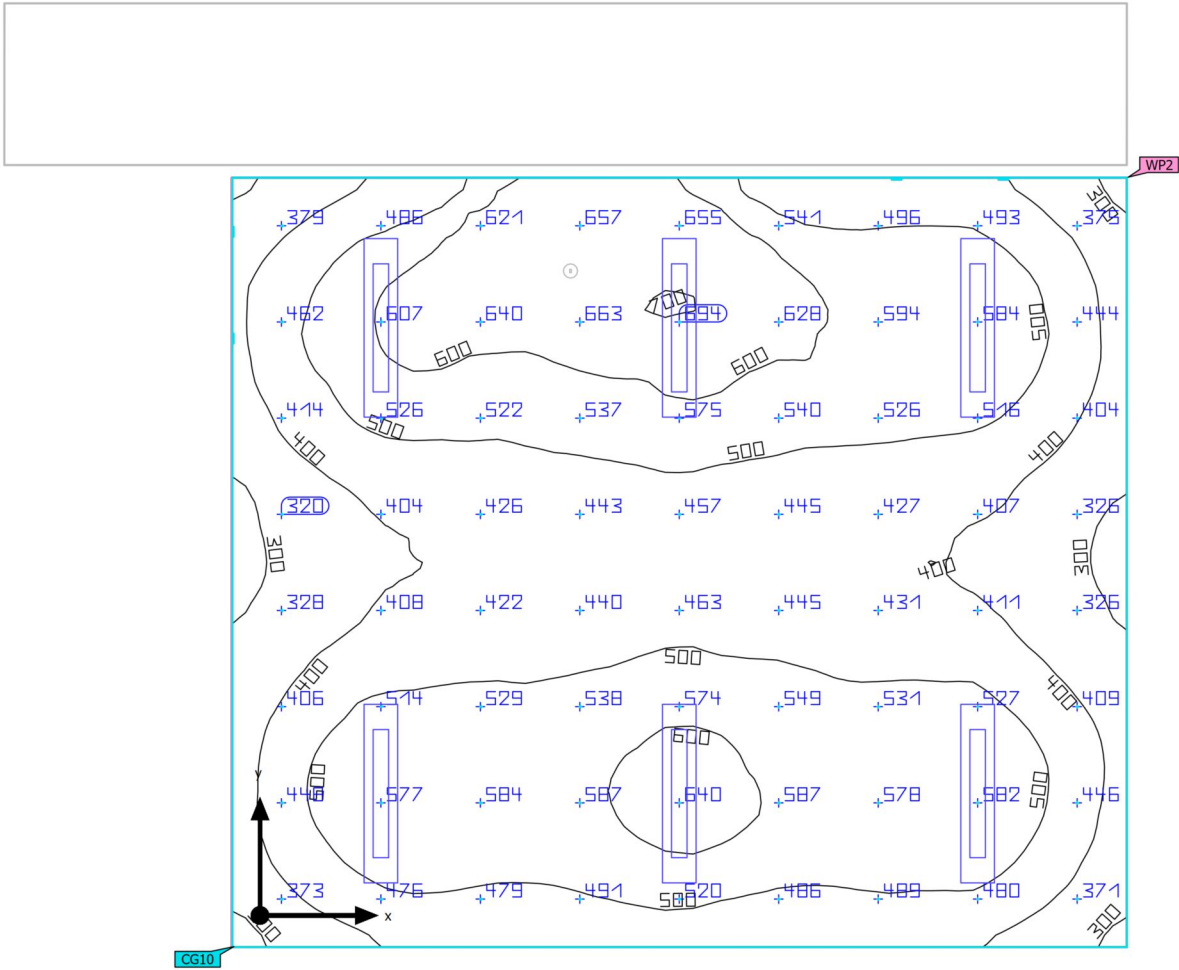


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (211 CORRIDOR)	307 lx	125 lx	448 lx	0.41	0.28	WP1
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.000 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

Summary



Ground area	41.15 m ²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	492 lx	≥ 500 lx	✗	WP2
	$U_o (g_1)$	0.55	≥ 0.60	✗	WP2
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	520 kWh/a	max. 1450 kWh/a	✓	
Space	Lighting power density	5.10 W/m ²	–		
		1.04 W/m ² /100 lx	–		

(1) Based on a rectangular space of 6.919 m x 5.947 m and SHR of 0.25.

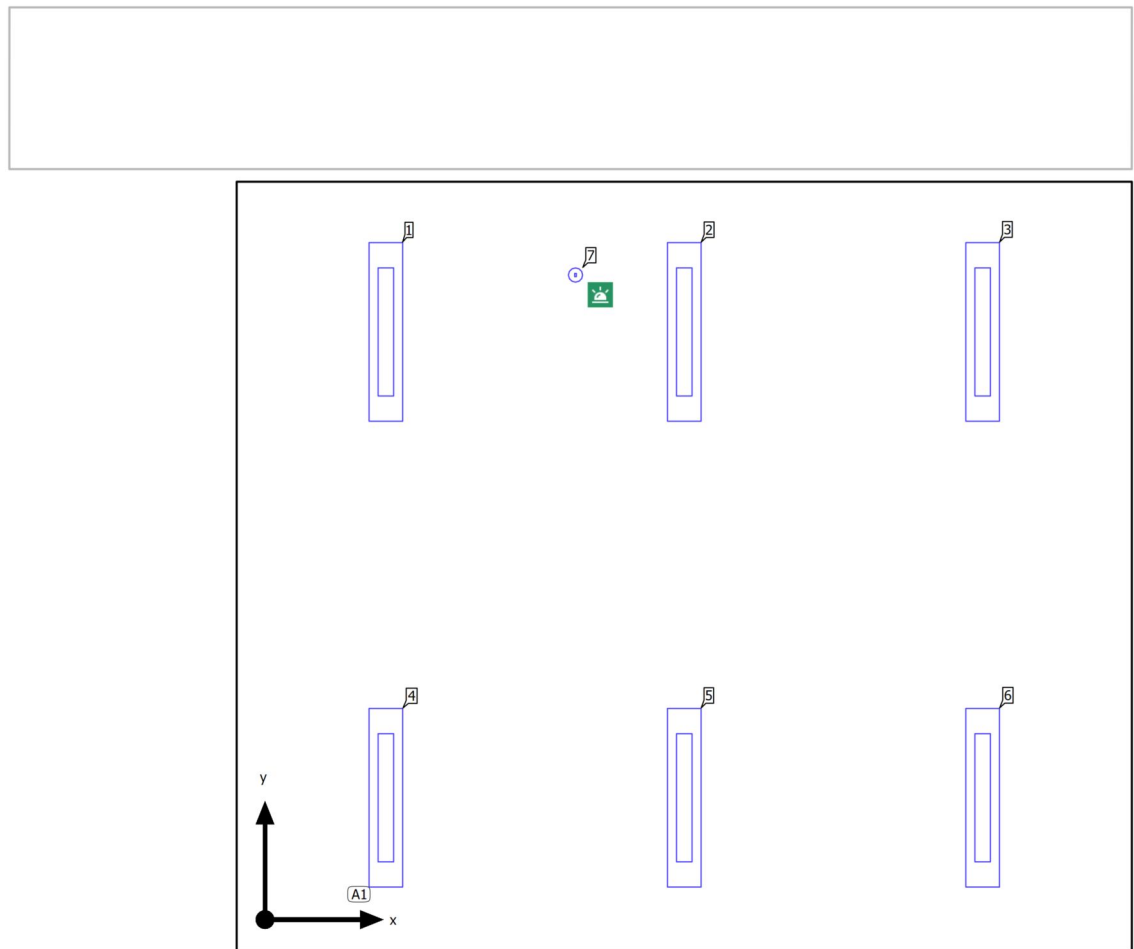
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

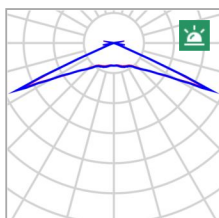
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
6	ETAP	-	R31111/LEDW455DX1 (_LTRRevAC)	16	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 212 COMMON

Luminaire layout plan

Building A · FIRST FLOOR · 212 COMMON

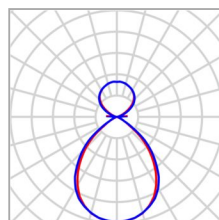
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
2.400 m	4.981 m	3.368 m	7

Building A · FIRST FLOOR · 212 COMMON

Luminaire layout plan

Manufacturer	ETAP	P	35.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDW455DX1 (_LTRRevAC)		
Fitting	1x LED		

6 x ETAP R3111/LEDW455DX1 (_LTRRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	5.547 m / 4.542 m / 2.800 m	0.934 m	4.542 m	2.800 m	1
X-direction	3 pcs., Centre - centre, Distances not equal	3.241 m	4.542 m	2.800 m	2
		5.547 m	4.542 m	2.800 m	3
Y-direction	2 pcs., Centre - centre, Distances not equal	0.934 m	0.942 m	2.800 m	4
		3.241 m	0.942 m	2.800 m	5
Arrangement	A1	5.547 m	0.942 m	2.800 m	6

Building A · FIRST FLOOR · 212 COMMON

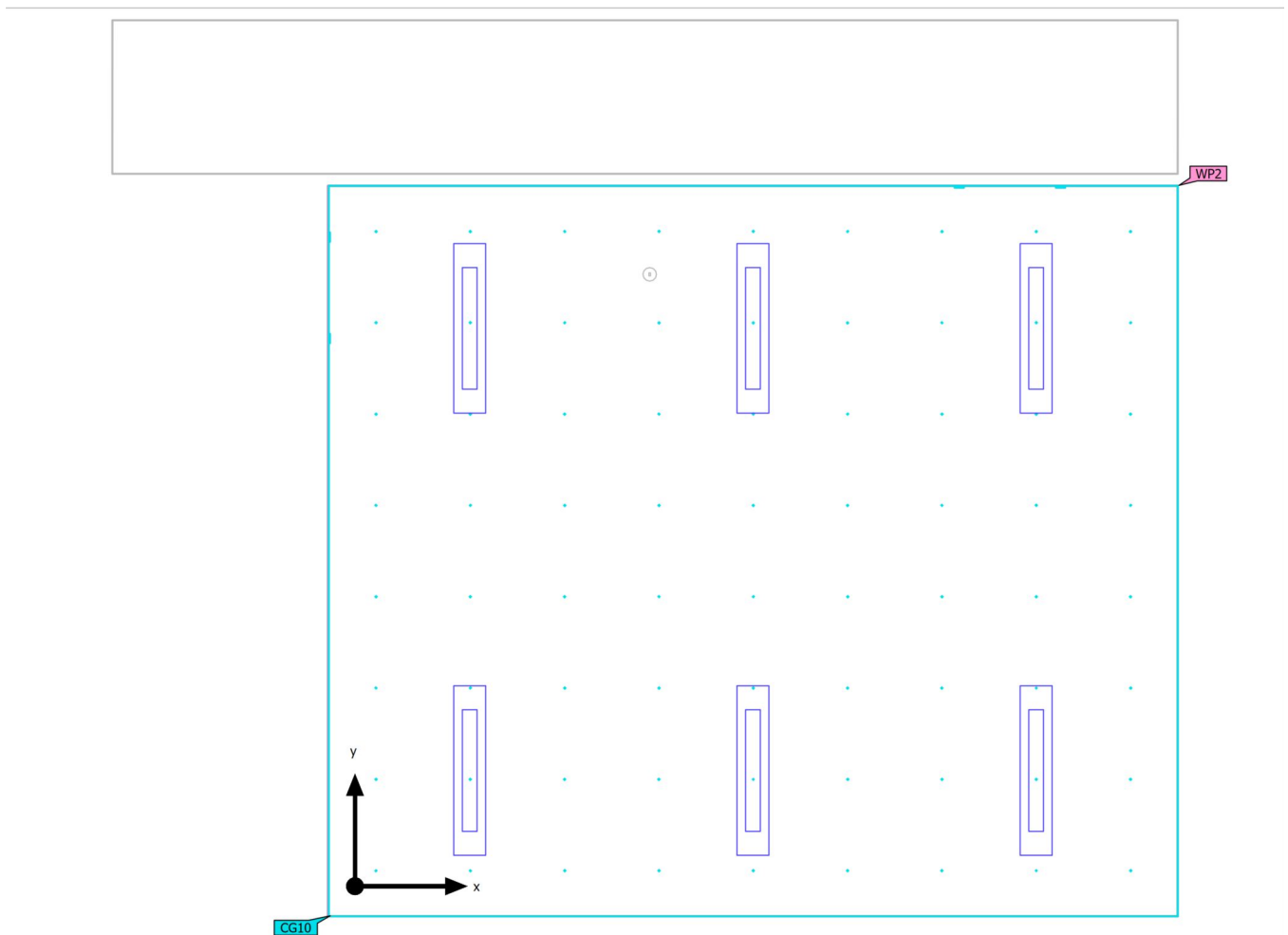
Luminaire list

Φ_{total} 31800 lm	P_{total} 210.9 W	Luminous efficacy 150.8 lm/W	$\Phi_{\text{Emergency lighting}}$ 300 lm	$P_{\text{Emergency lighting}}$ 0.9 W
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pcs.	Manufacturer	Article No.	Article name		P	Φ	Luminous efficacy
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)		0.9 W	300 lm	340.9 lm/W
					0.9 W	300 lm (100 %)	-
6	ETAP	-	R3111/LEDW455DX1 (_LTRRevAC)		35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (212 COMMON) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	492 lx (≥ 500 lx) ✗	272 lx	702 lx	0.55 (≥ 0.60) ✗	0.39	WP2

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

Calculation objects

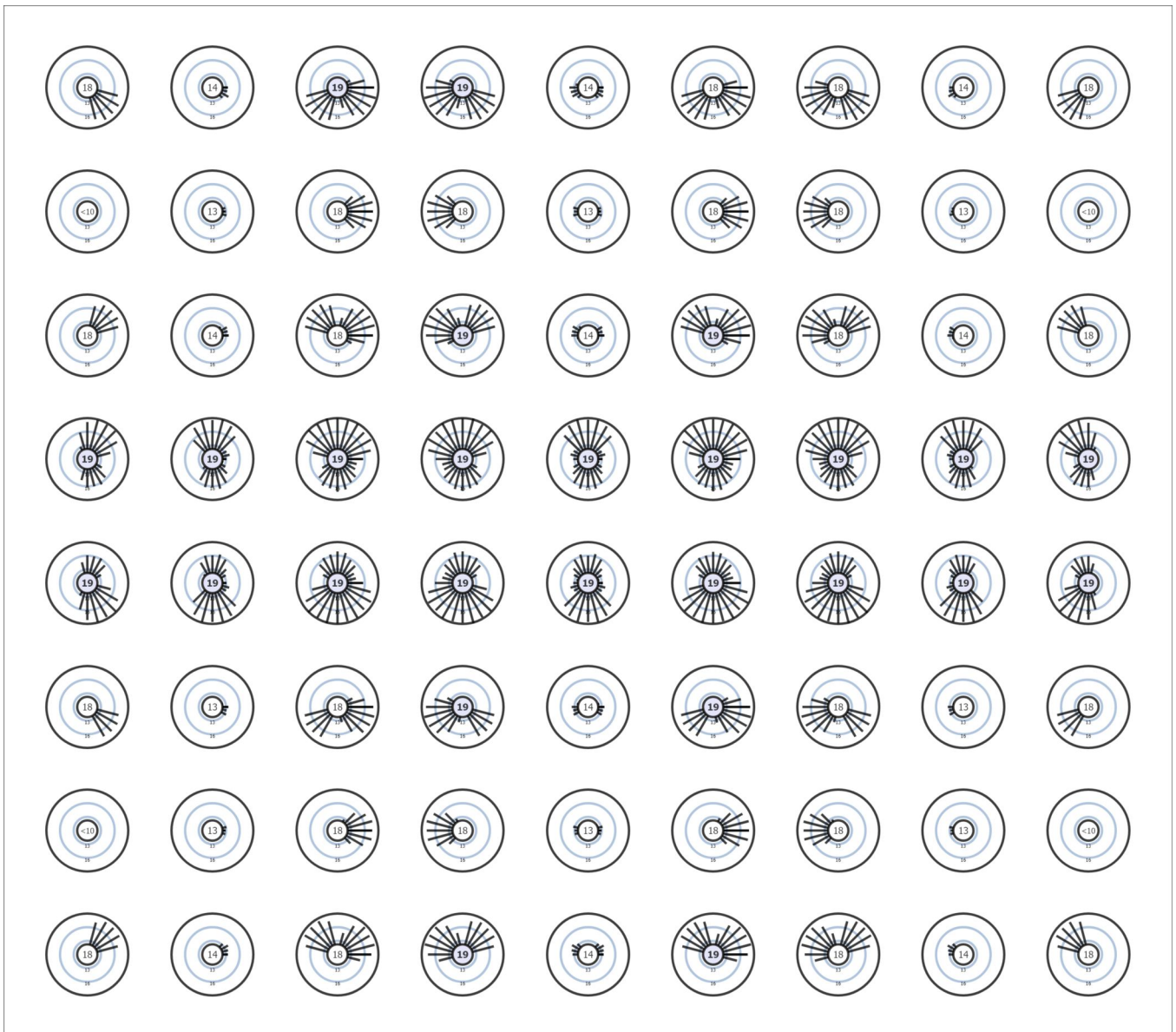
Glare plane 212 (RUG)

Strongest glare at	105°
max	19.2
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG10

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

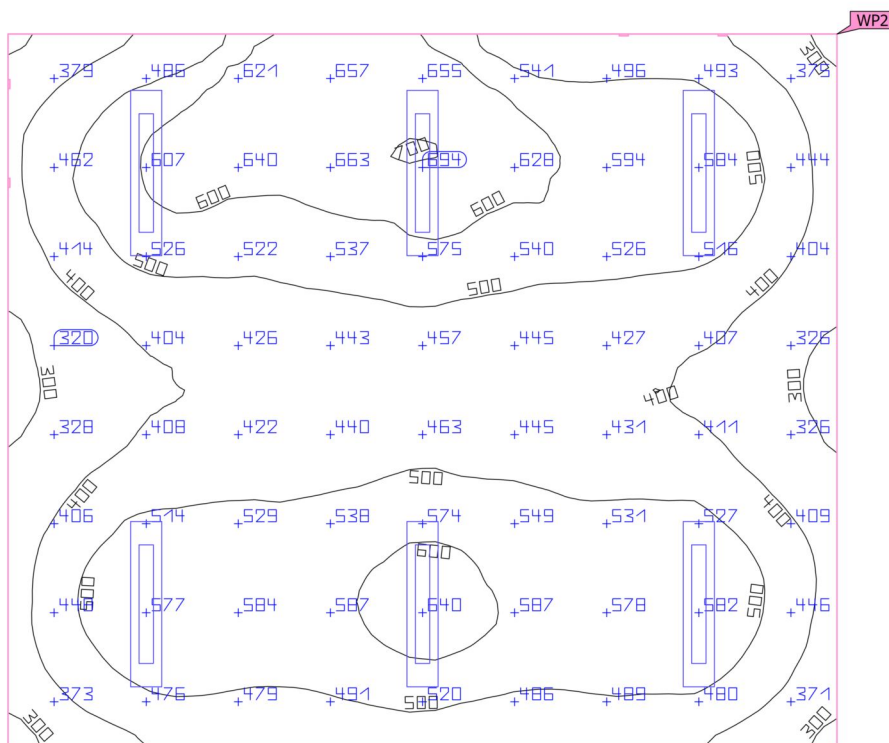
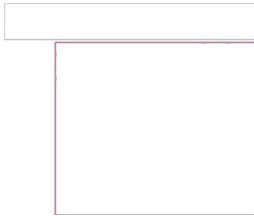
Calculation objects

Glare plane 212 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 212 COMMON (Light scene 1)

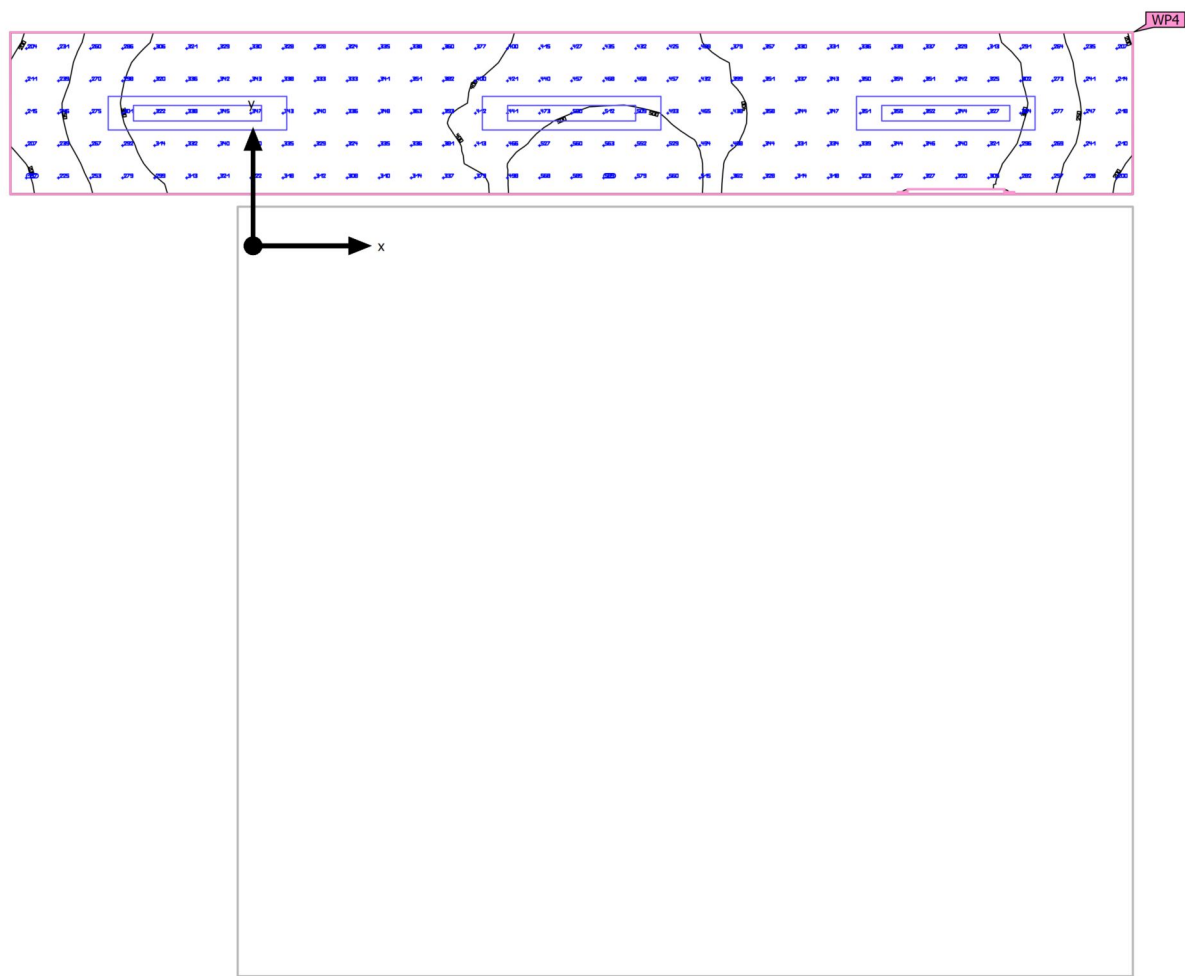
Working plane (212 COMMON)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (212 COMMON)	492 lx	272 lx	702 lx	0.55	0.39	WP2
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 213 BALCONY (Light scene 1)

Summary



Ground area	10.84 m ²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	3.300 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 213 BALCONY (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	350 lx	$\geq 500 \text{ lx}$	✗	WP4
	$U_o (g_1)$	0.54	≥ 0.60	✗	WP4
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	260 kWh/a	max. 400 kWh/a	✓	
Space	Lighting power density	9.68 W/m ²	–		
		2.76 W/m ² /100 lx	–		

(1) Based on a rectangular space of 8.676 m x 1.250 m and SHR of 0.25.

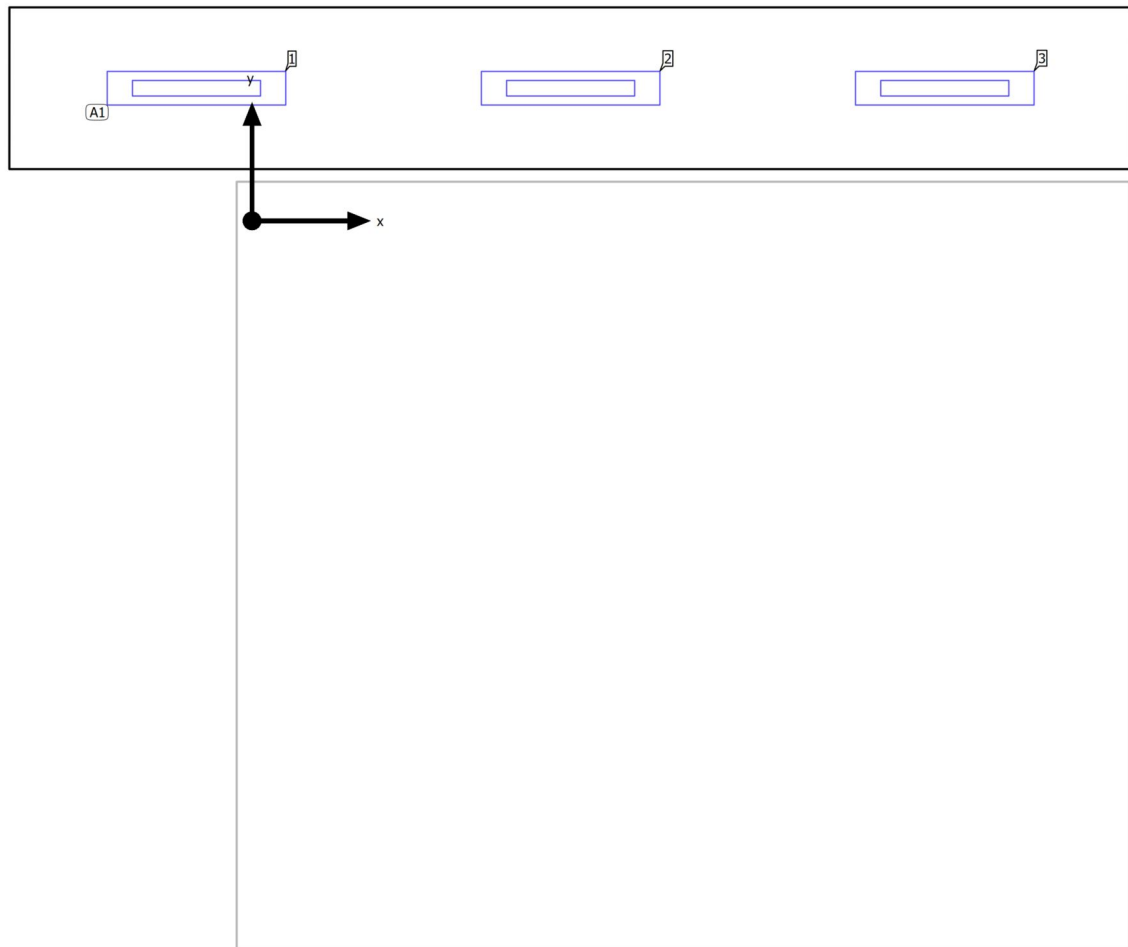
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

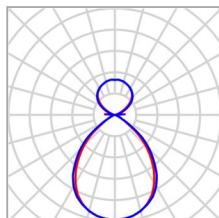
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
3	ETAP	-	R31111/LEDW455DX1 (_LTRRevAC)	16	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 213 BALCONY

Luminaire layout plan

Building A · FIRST FLOOR · 213 BALCONY

Luminaire layout plan

Manufacturer	ETAP	P	35.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDW455DX1 (_LTRRevAC)		
Fitting	1x LED		

3 x ETAP R3111/LEDW455DX1 (_LTRRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	-0.430 m / 1.025 m / 3.300 m	-0.430 m	1.025 m	3.300 m	1
X-direction	3 pcs., Centre - centre, 2.892 m	2.462 m	1.025 m	3.300 m	2
Y-direction	1 pcs., Centre - centre, 1.250 m	5.354 m	1.025 m	3.300 m	3
Arrangement	A1				

Building A · FIRST FLOOR · 213 BALCONY

Luminaire list Φ_{total}

15750 lm

 P_{total}

105.0 W

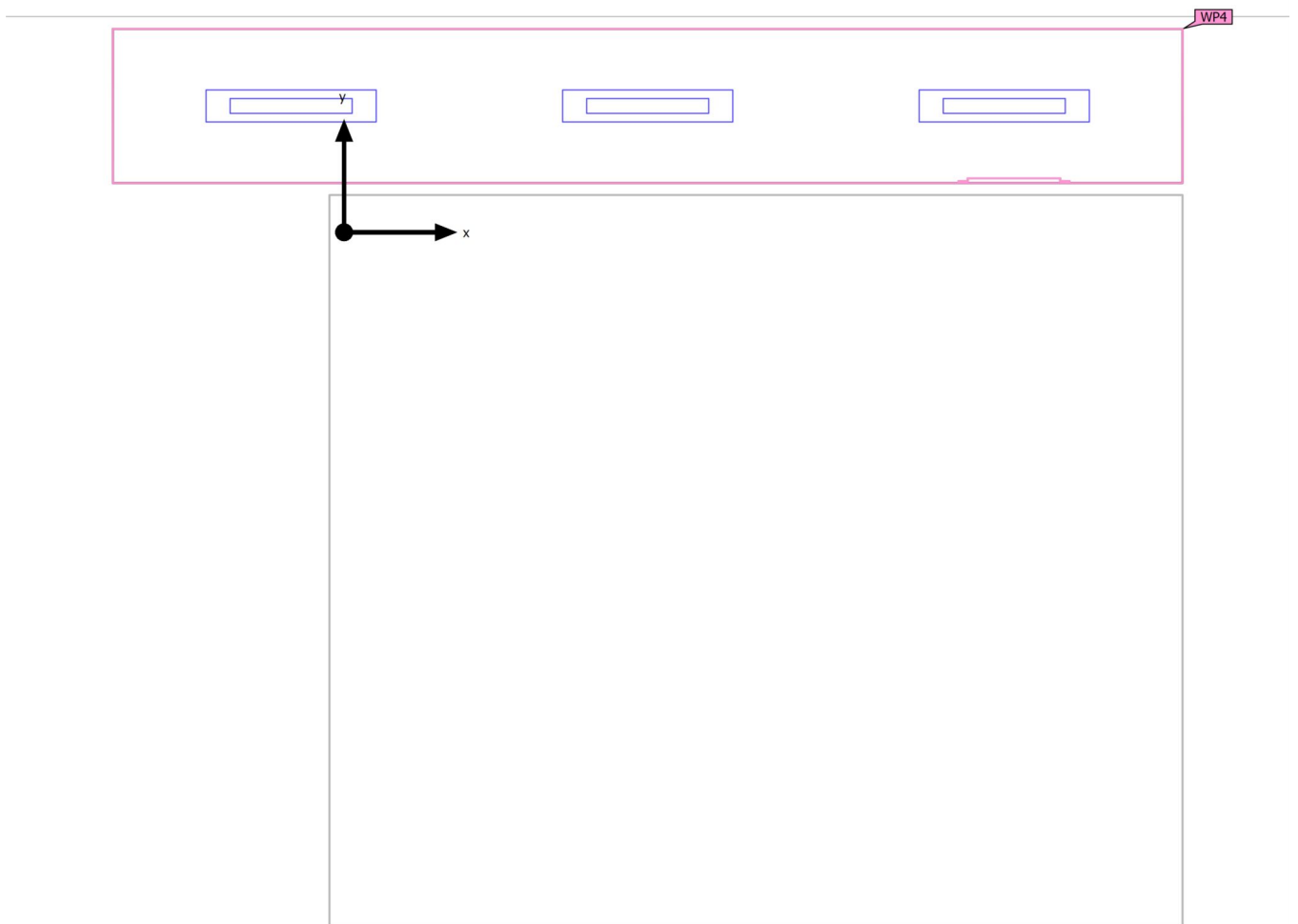
Luminous efficacy

150.0 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
3	ETAP	-	R3111/LEDW455DX1 (_LTRevAC)	35.0 W	5239 lm	149.7 lm/W

Building A · FIRST FLOOR · 213 BALCONY (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 213 BALCONY (Light scene 1)

Calculation objects

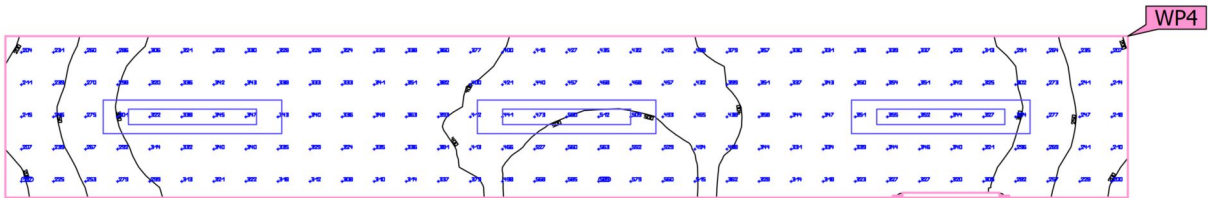
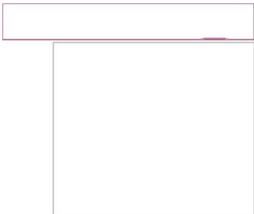
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (213 BALCONY)	350 lx	188 lx	588 lx	0.54	0.32	WP4
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 213 BALCONY (Light scene 1)

Working plane (213 BALCONY)

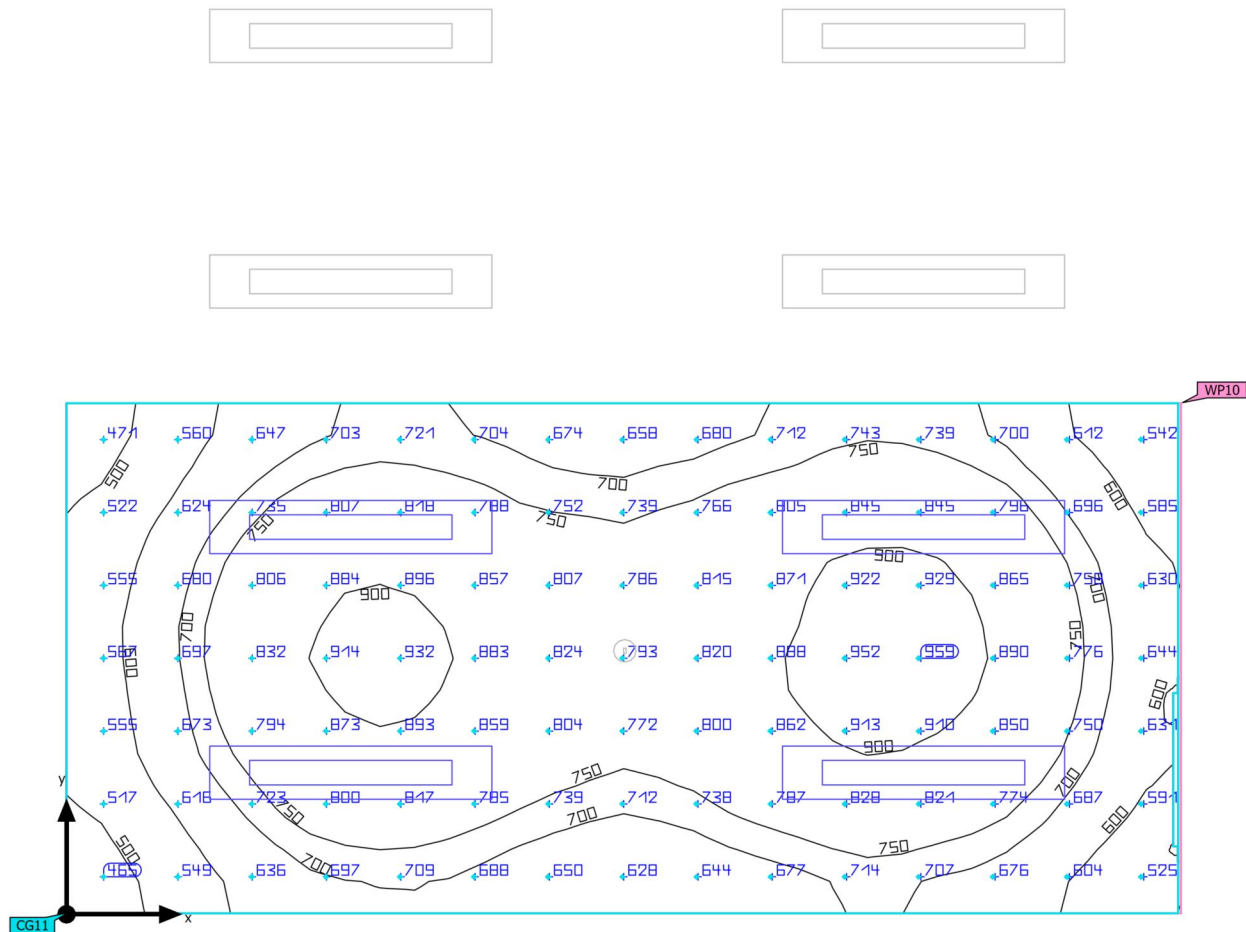


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (213 BALCONY)	350 lx	188 lx	588 lx	0.54	0.32	WP4
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.700 m, Wall zone: 0.000 m	✗			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Summary



Ground area	13.60 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.700 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	743 lx	$\geq 500 \text{ lx}$	✓	WP10
	$U_o (g_1)$	0.59	≥ 0.60	✗	WP10
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	327 kWh/a	max. 500 kWh/a	✓	
Space	Lighting power density	9.71 W/m ²	–		
		1.31 W/m ² /100 lx	–		

(1) Based on a rectangular space of 5.450 m x 2.495 m and SHR of 0.25.

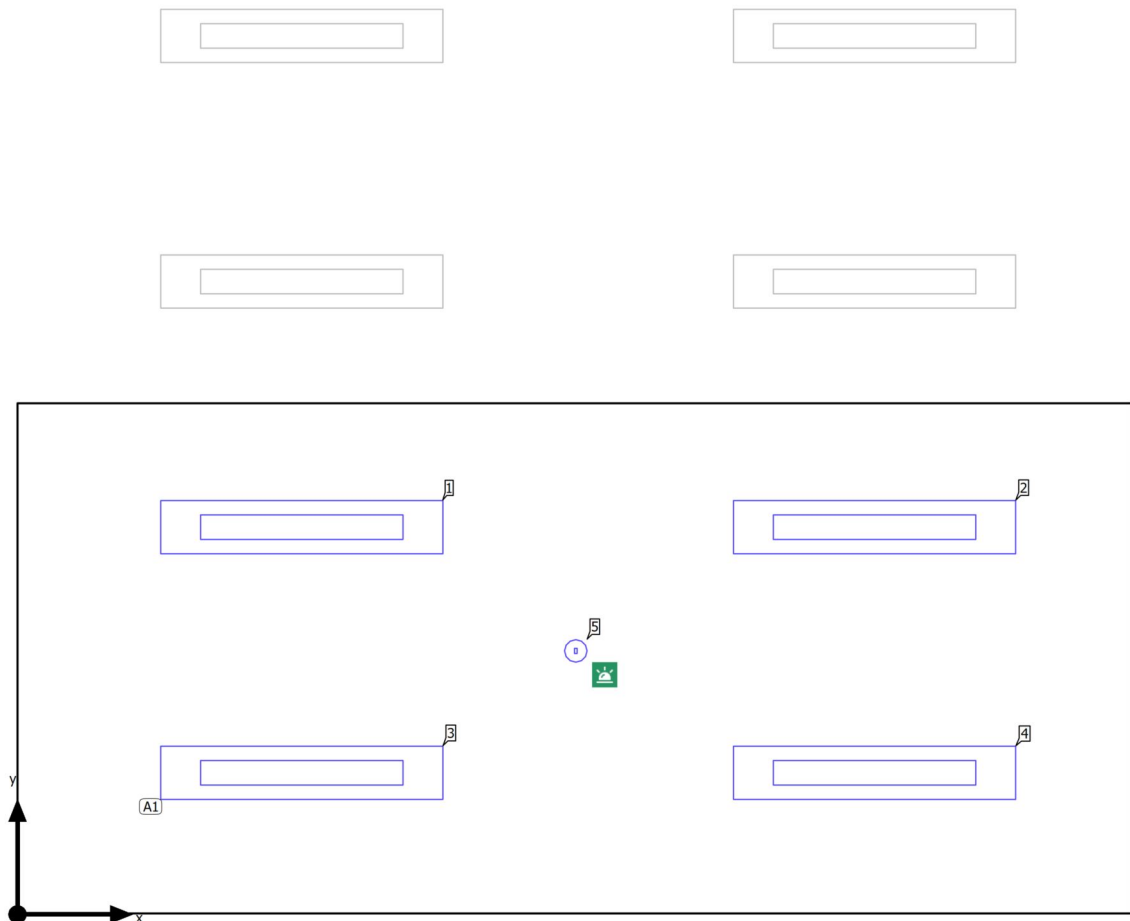
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

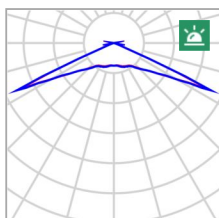
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 214 OFFICE

Luminaire layout plan

Building A · FIRST FLOOR · 214 OFFICE

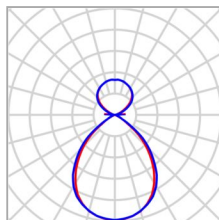
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
2.730 m	1.289 m	3.368 m	5

Building A · FIRST FLOOR · 214 OFFICE

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

8 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	4.192 m / 1.894 m / 2.800 m	1.390 m	1.894 m	2.800 m	1
X-direction	2 pcs., Centre - centre, Distances not equal	4.192 m	1.894 m	2.800 m	2
		1.390 m	0.693 m	2.800 m	3
Y-direction	6 pcs., Centre - centre, Distances not equal	4.192 m	0.693 m	2.800 m	4
Arrangement	A1				

Building A · FIRST FLOOR · 214 OFFICE

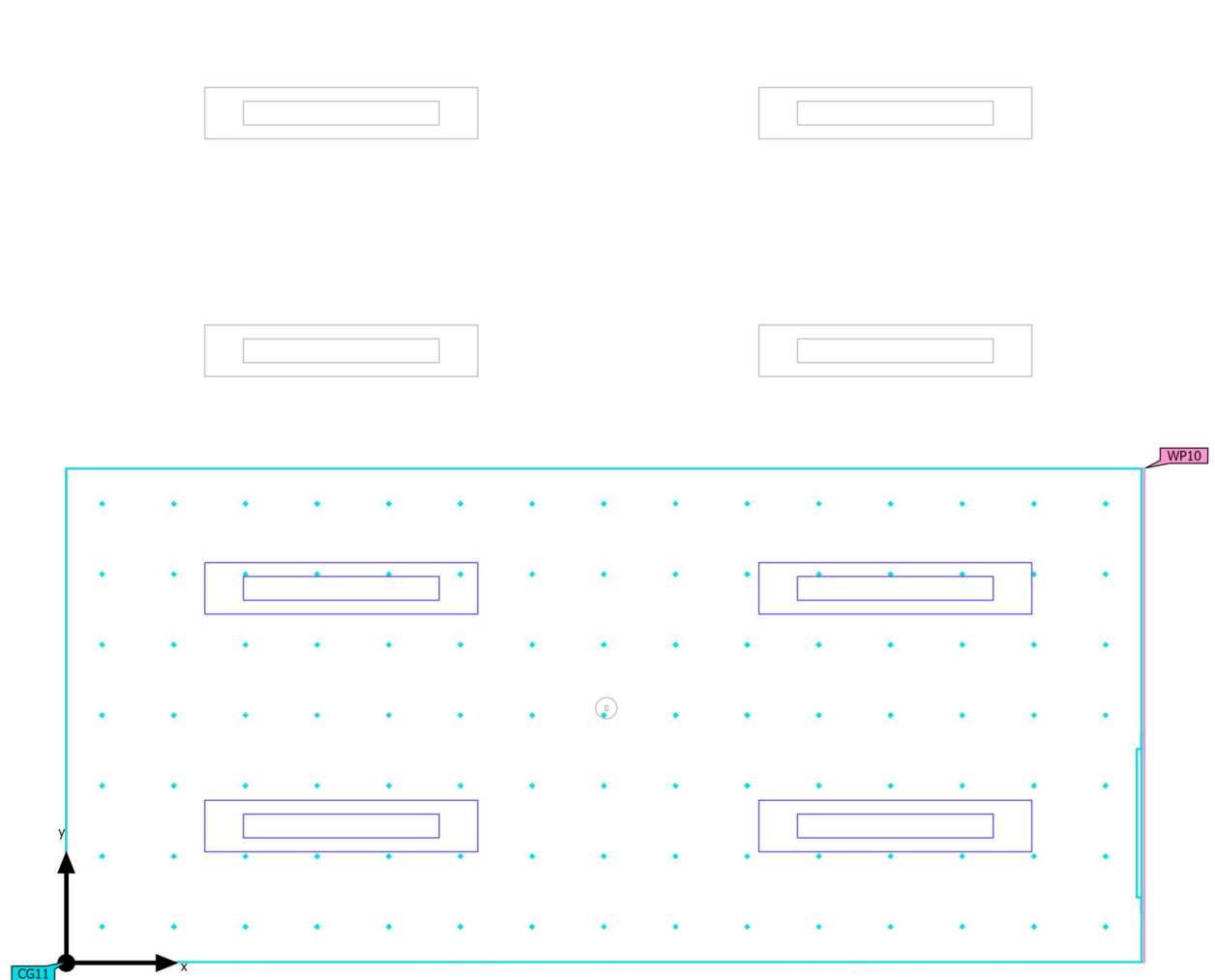
Luminaire list

Φ_{total} 21300 lm	P_{total} 132.9 W	Luminous efficacy 160.3 lm/W	$\Phi_{\text{Emergency lighting}}$ 300 lm	$P_{\text{Emergency lighting}}$ 0.9 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (214 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	743 lx (≥ 500 lx) ✓	440 lx	965 lx	0.59 (≥ 0.60) ✗	0.46	WP10

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Calculation objects

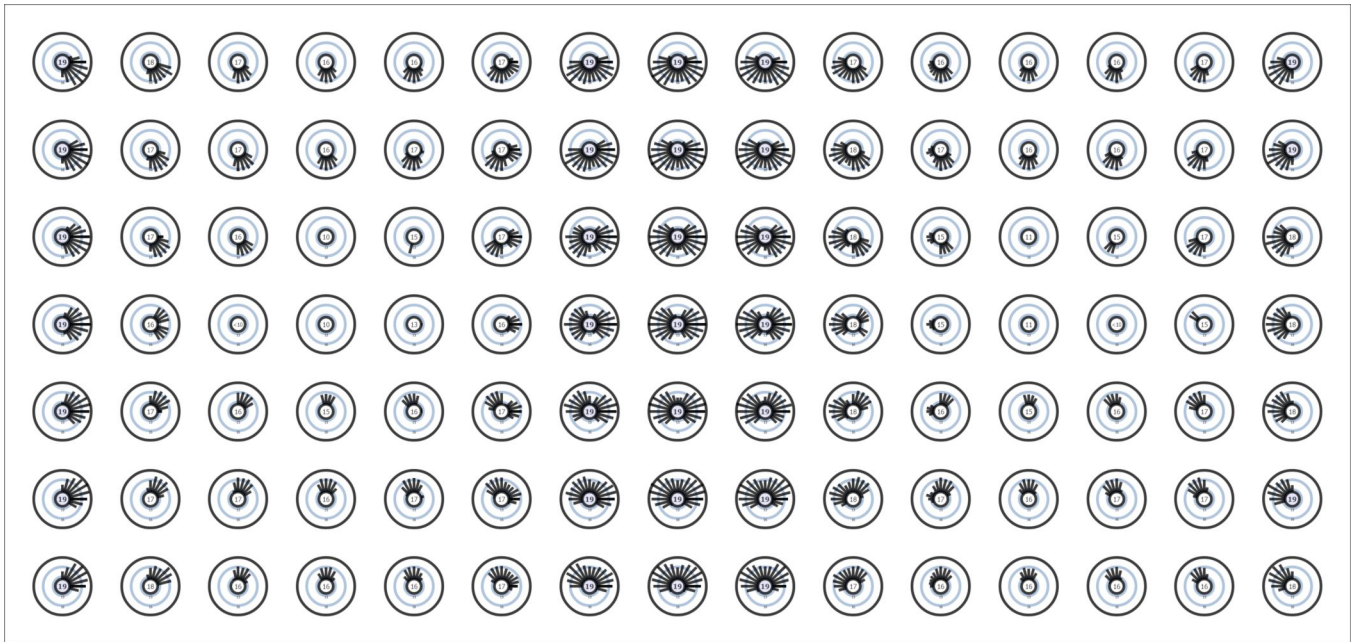
Glare plane 214 (RUG)

Strongest glare at	210°
max	19.4
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG11

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Calculation objects

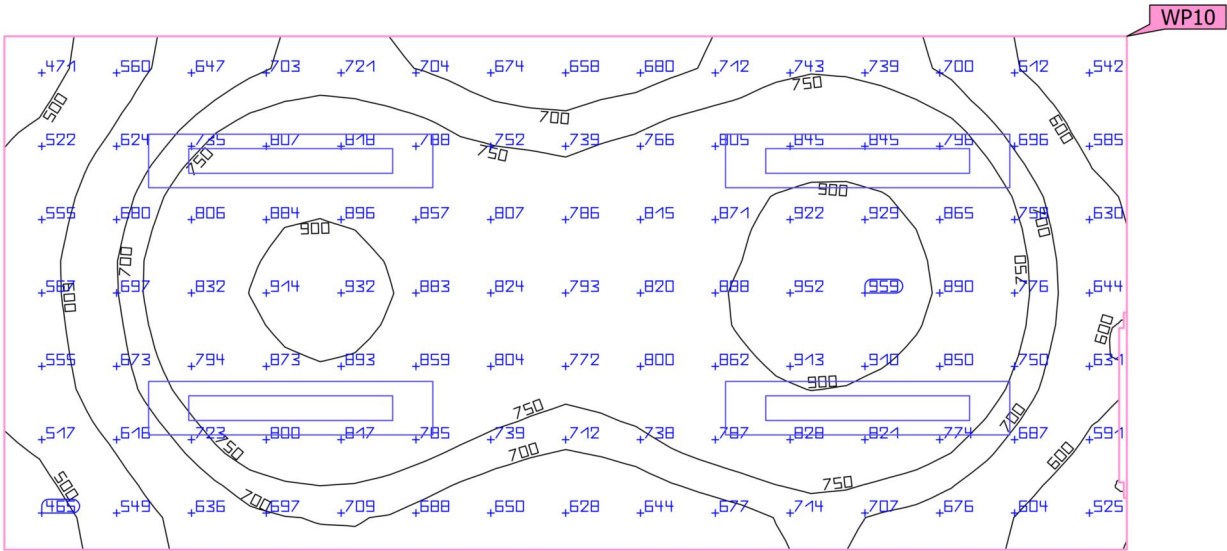
Glare plane 214 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 214 OFFICE (Light scene 1)

Working plane (214 OFFICE)

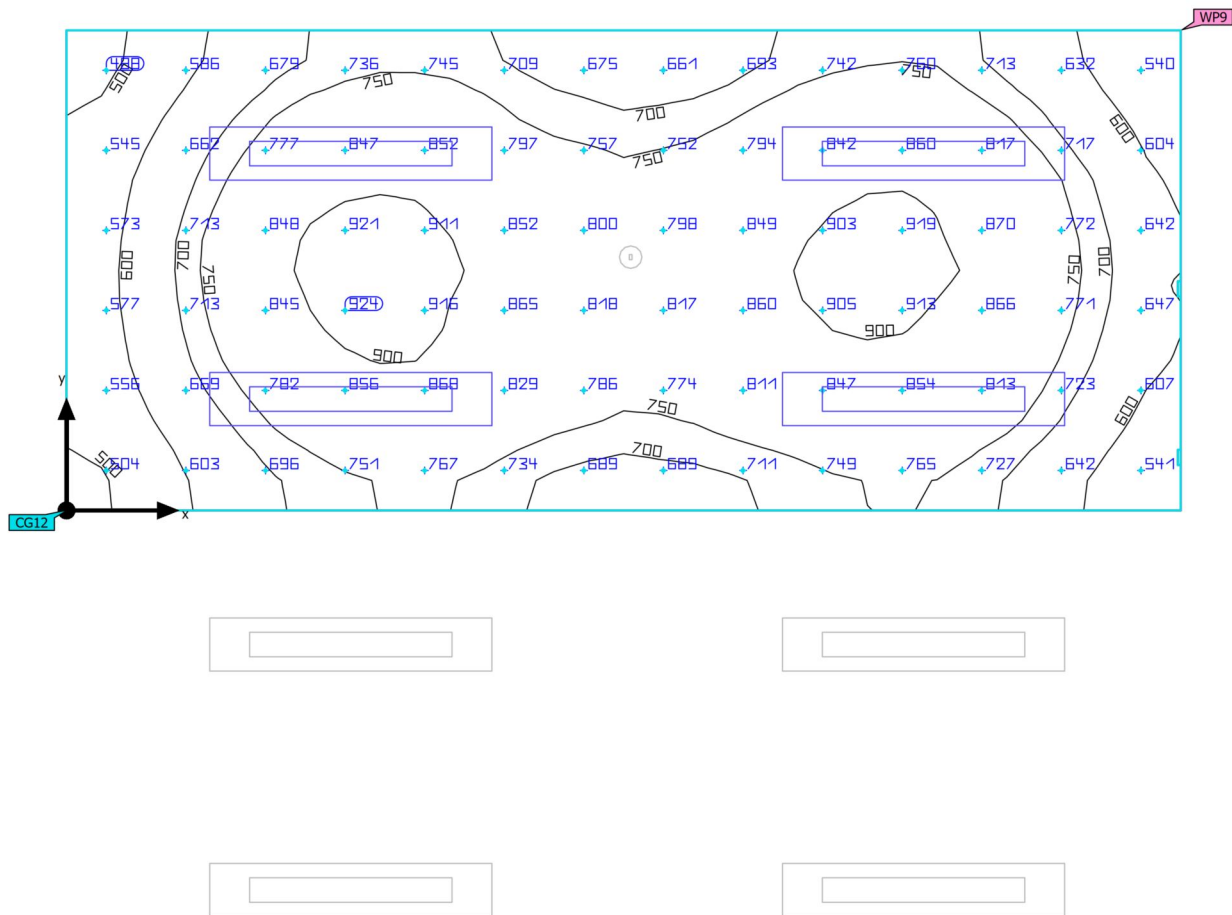


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (214 OFFICE)	743 lx	440 lx	965 lx	0.59	0.46	WP10
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Summary



Ground area	12.81 m²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	753 lx	≥ 500 lx	✓	WP9
	$U_o (g_1)$	0.61	≥ 0.60	✓	WP9
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	327 kWh/a	max. 450 kWh/a	✓	
Space	Lighting power density	10.31 W/m ²	–		
		1.37 W/m ² /100 lx	–		

(1) Based on a rectangular space of 5.450 m x 2.350 m and SHR of 0.25.

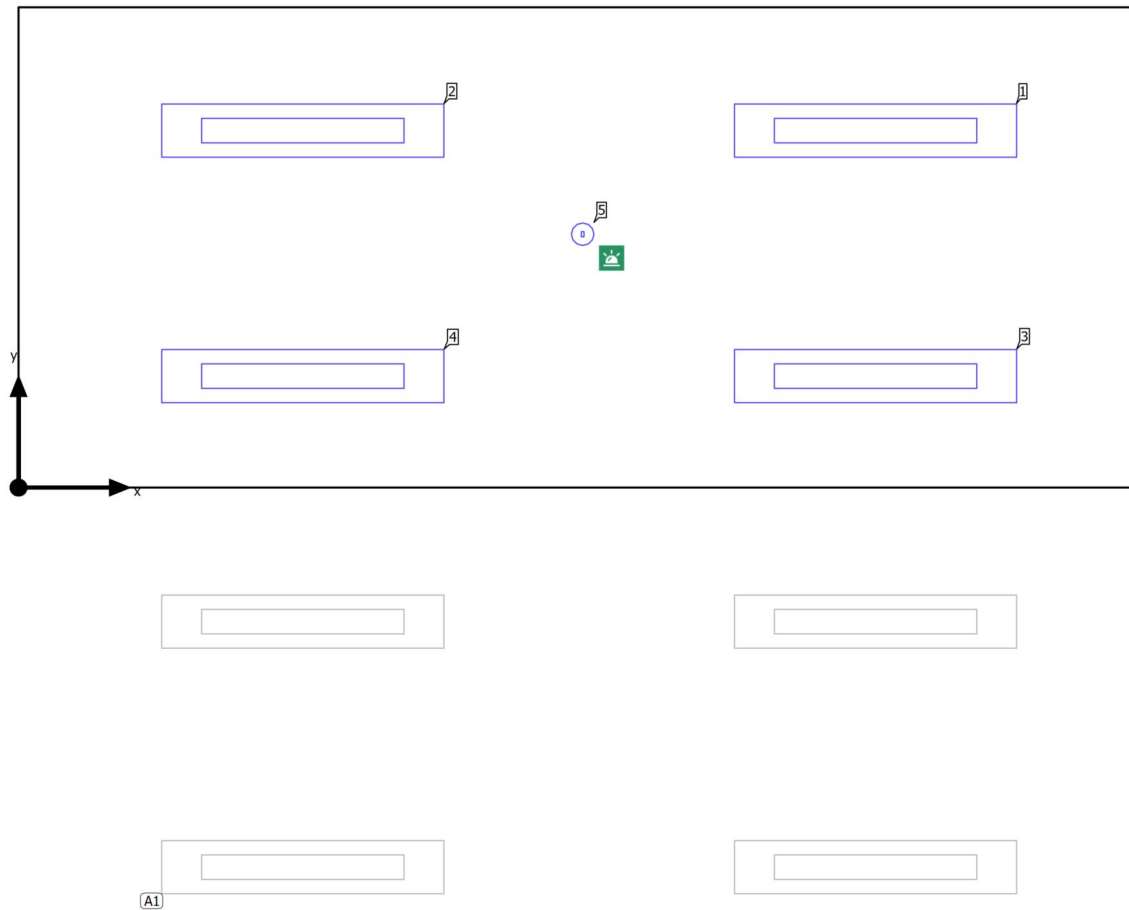
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

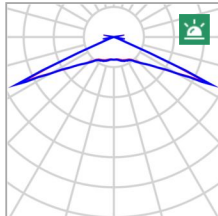
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R31111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 215 MEETING

Luminaire layout plan

Building A · FIRST FLOOR · 215 MEETING

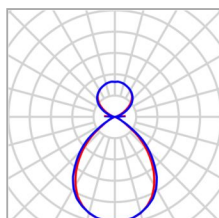
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
2.759 m	1.240 m	3.368 m	5

Building A · FIRST FLOOR · 215 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

8 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	4.192 m / 1.747 m / 2.800 m	4.192 m	1.747 m	2.800 m	1
X-direction	2 pcs., Centre - centre, Distances not equal	1.390 m	1.747 m	2.800 m	2
		4.192 m	0.545 m	2.800 m	3
Y-direction	6 pcs., Centre - centre, Distances not equal	1.390 m	0.545 m	2.800 m	4
Arrangement	A1				

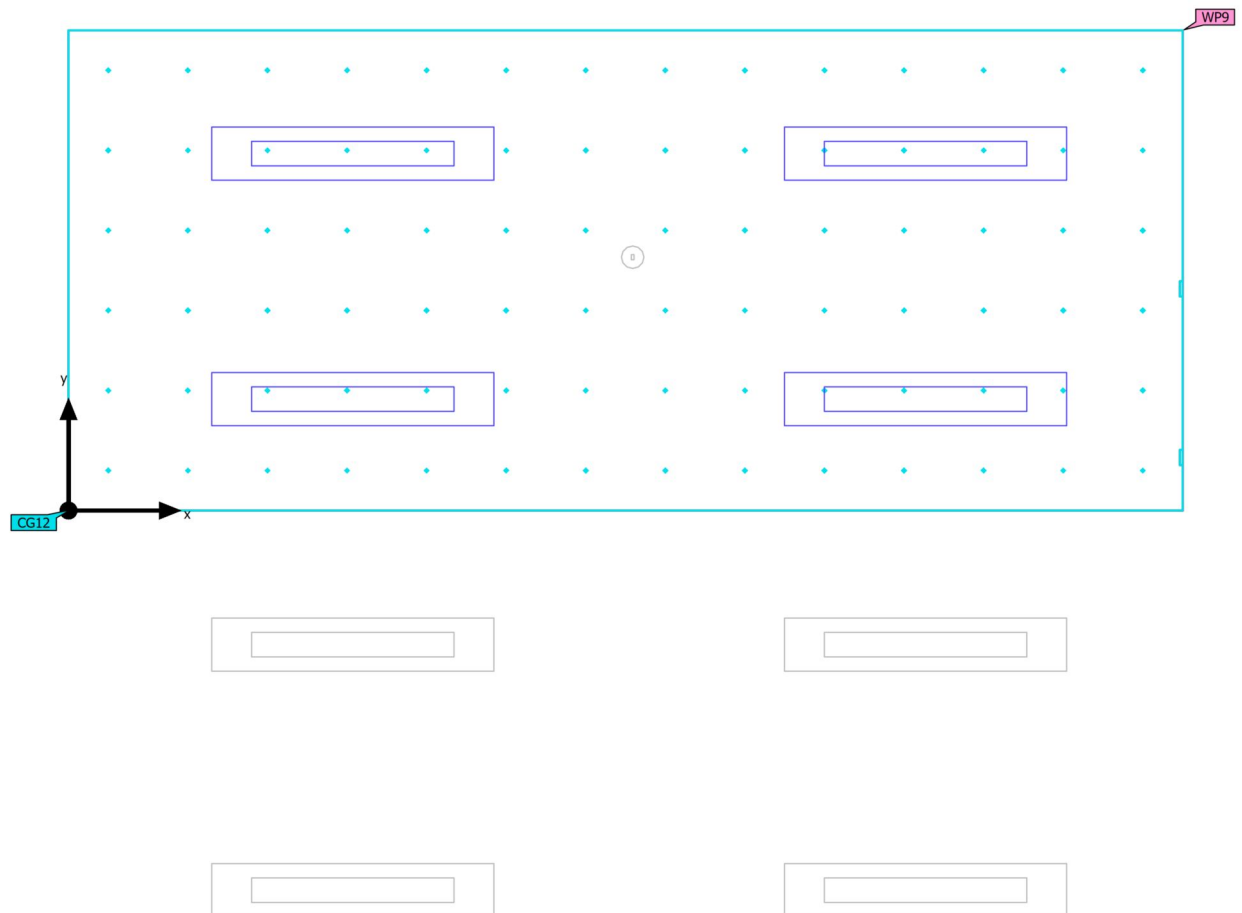
Building A · FIRST FLOOR · 215 MEETING

Luminaire list

Φ_{total} 21300 lm		P_{total} 132.9 W		Luminous efficacy 160.3 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)			0.9 W	300 lm	340.9 lm/W	
						0.9 W	300 lm (100 %)	-	
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (215 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	753 lx (≥ 500 lx) ✓	462 lx	940 lx	0.61 (≥ 0.60) ✓	0.49	WP9

Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Calculation objects

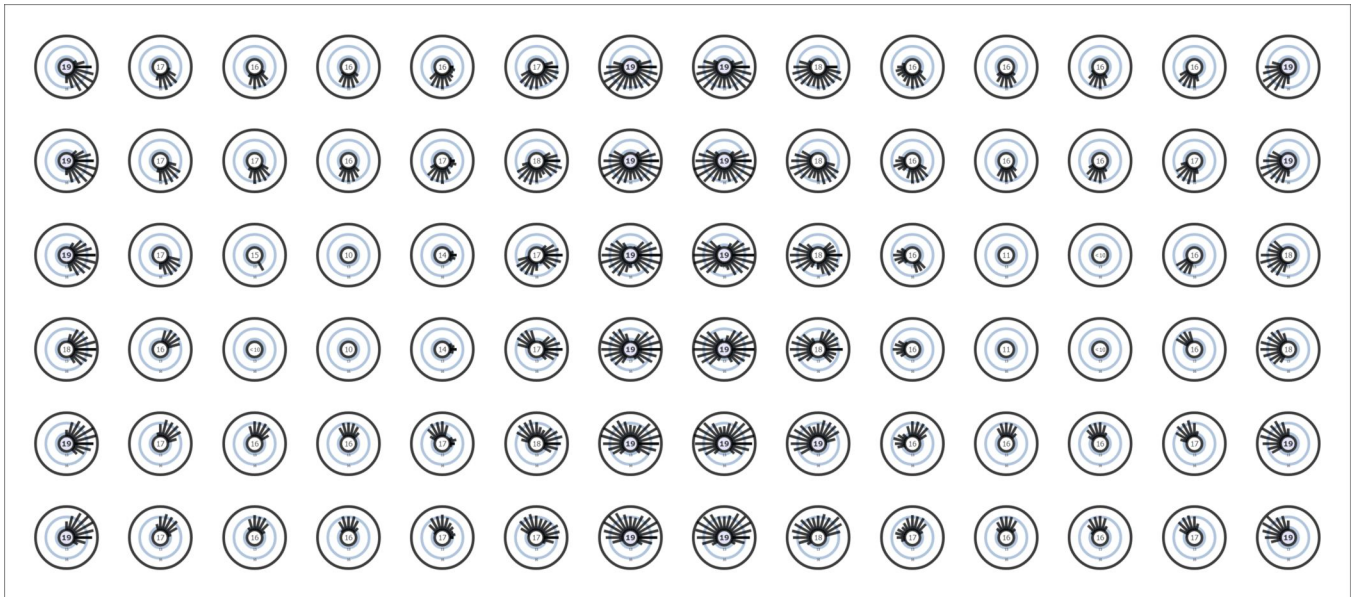
Glare plane 215 (RUG)

Strongest glare at	150°
max	19.1
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG12

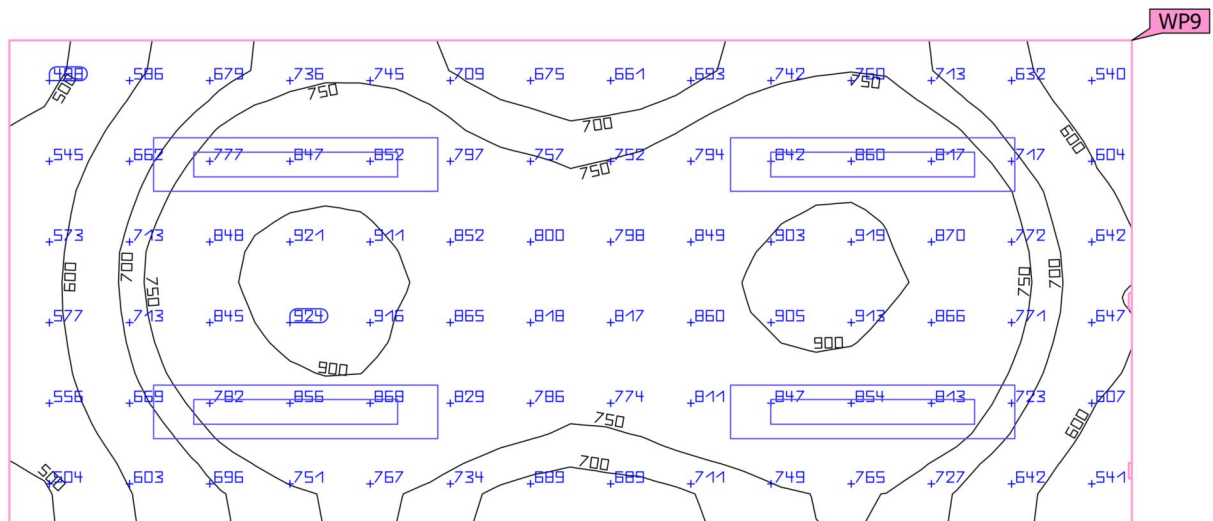
Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

Calculation objects

Glare plane 215 (RUG)



Building A · FIRST FLOOR · 215 MEETING (Light scene 1)

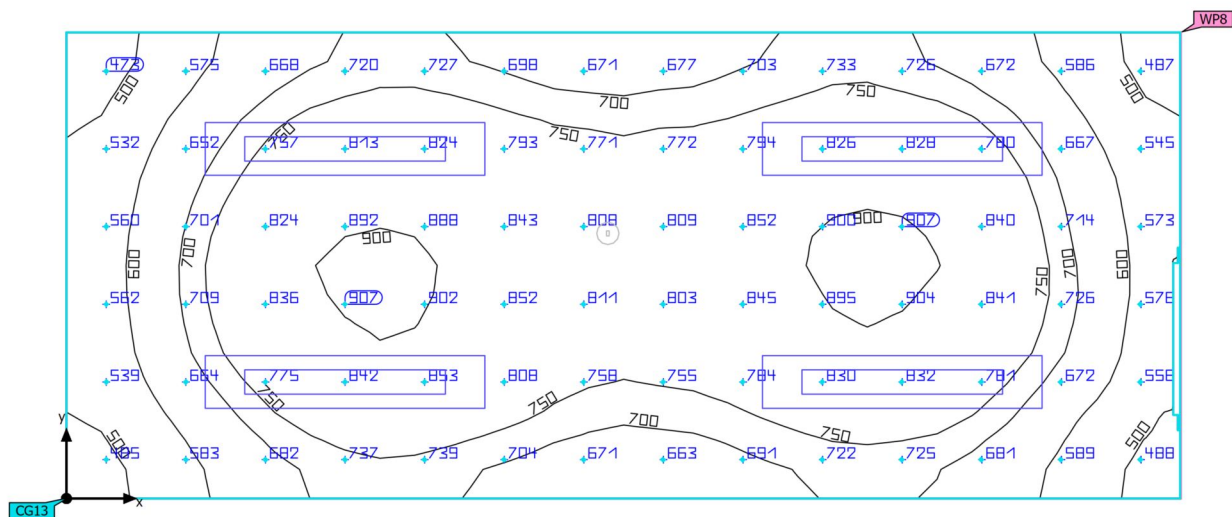
Working plane (215 MEETING)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (215 MEETING)	753 lx	462 lx	940 lx	0.61	0.49	WP9
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

Summary



Ground area	12.65 m²	Clearance height	3.300 m
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %	Mounting height	2.800 m
Maintenance factor	0.76 (fixed)	Height _{Working plane}	0.700 m
		Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	732 lx	≥ 500 lx	✓	WP8
	$U_o (g_1)$	0.58	≥ 0.60	✗	WP8
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[206 - 327] kWh/a	max. 450 kWh/a	✓	
Space	Lighting power density	10.43 W/m ²	–		
		1.43 W/m ² /100 lx	–		

(1) Based on a rectangular space of 5.500 m x 2.300 m and SHR of 0.25.

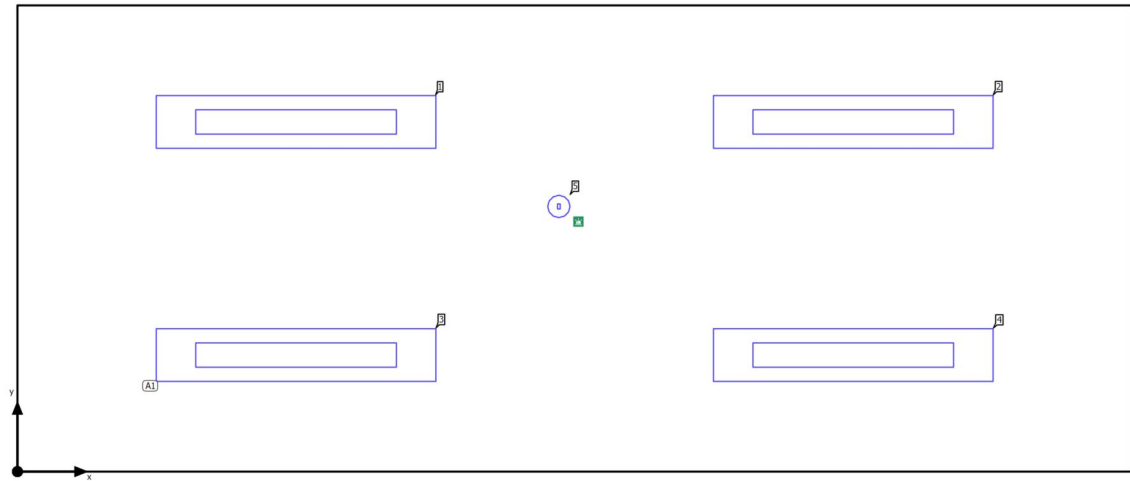
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

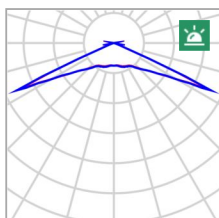
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 216 MEETING

Luminaire layout plan

Building A · FIRST FLOOR · 216 MEETING

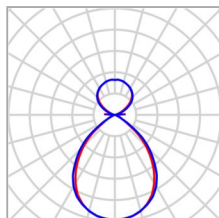
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
2.672 m	1.308 m	3.368 m	5

Building A · FIRST FLOOR · 216 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	Φ _{Luminaire}	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

4 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.375 m / 0.575 m / 2.800 m	1.375 m	1.725 m	2.800 m	1
X-direction	2 pcs., Centre - centre, 2.750 m	4.125 m	1.725 m	2.800 m	2
Y-direction	2 pcs., Centre - centre, 1.150 m	1.375 m	0.575 m	2.800 m	3
		4.125 m	0.575 m	2.800 m	4
Arrangement	A1				

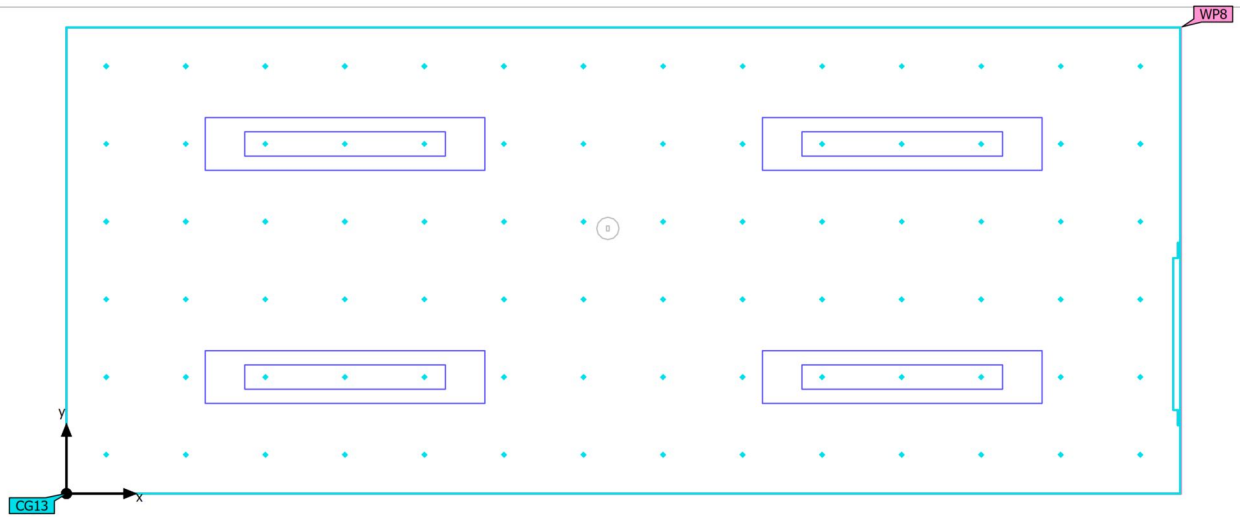
Building A · FIRST FLOOR · 216 MEETING

Luminaire list

Φ_{total} 21300 lm		P_{total} 132.9 W		Luminous efficacy 160.3 lm/W		$\Phi_{\text{Emergency lighting}}$ 300 lm		$P_{\text{Emergency lighting}}$ 0.9 W	
pcs.	Manufacturer	Article No.	Article name			P	Φ	Luminous efficacy	
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)			0.9 W	300 lm	340.9 lm/W	
						0.9 W	300 lm (100 %)	-	
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)			33.0 W	5239 lm	158.8 lm/W	

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (216 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	732 lx (≥ 500 lx) ✓	427 lx	928 lx	0.58 (≥ 0.60) ✗	0.46	WP8

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

Calculation objects

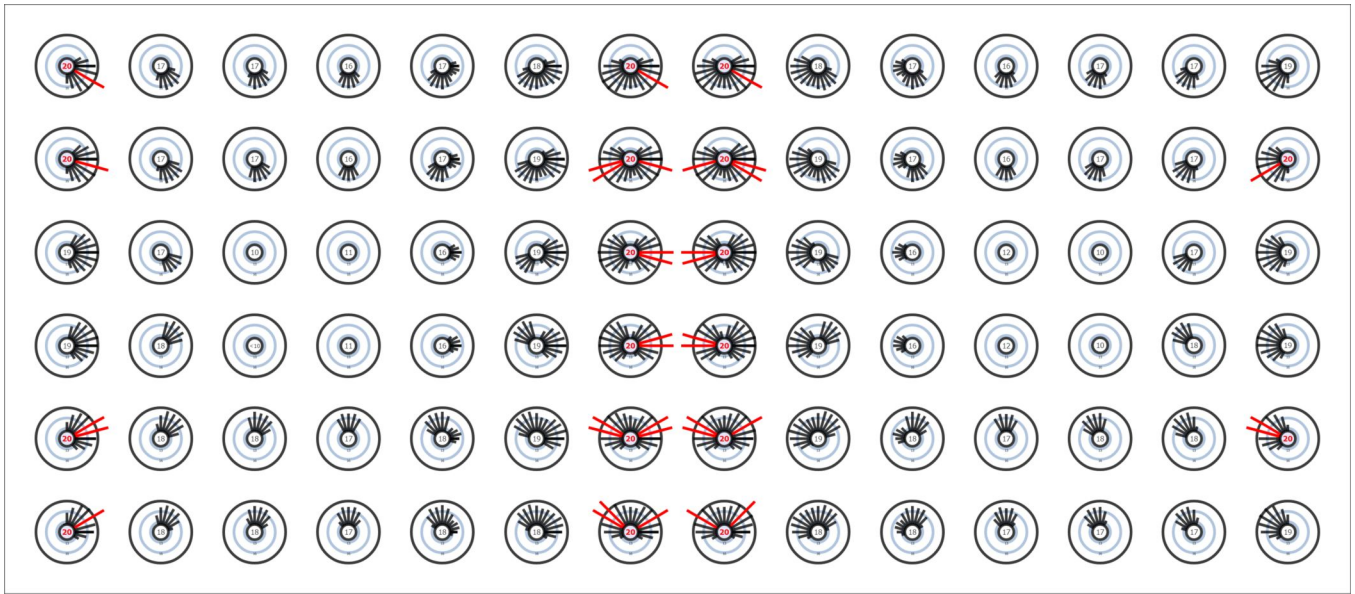
Glare plane 216 (RUG)

Strongest glare at	345°
max	20.0
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG13

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

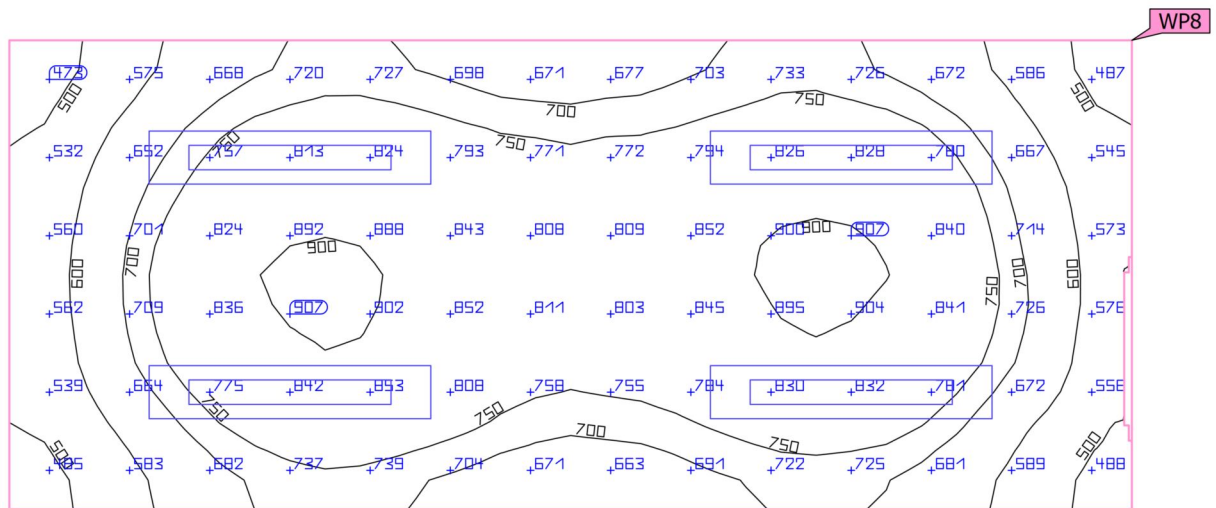
Calculation objects

Glare plane 216 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 216 MEETING (Light scene 1)

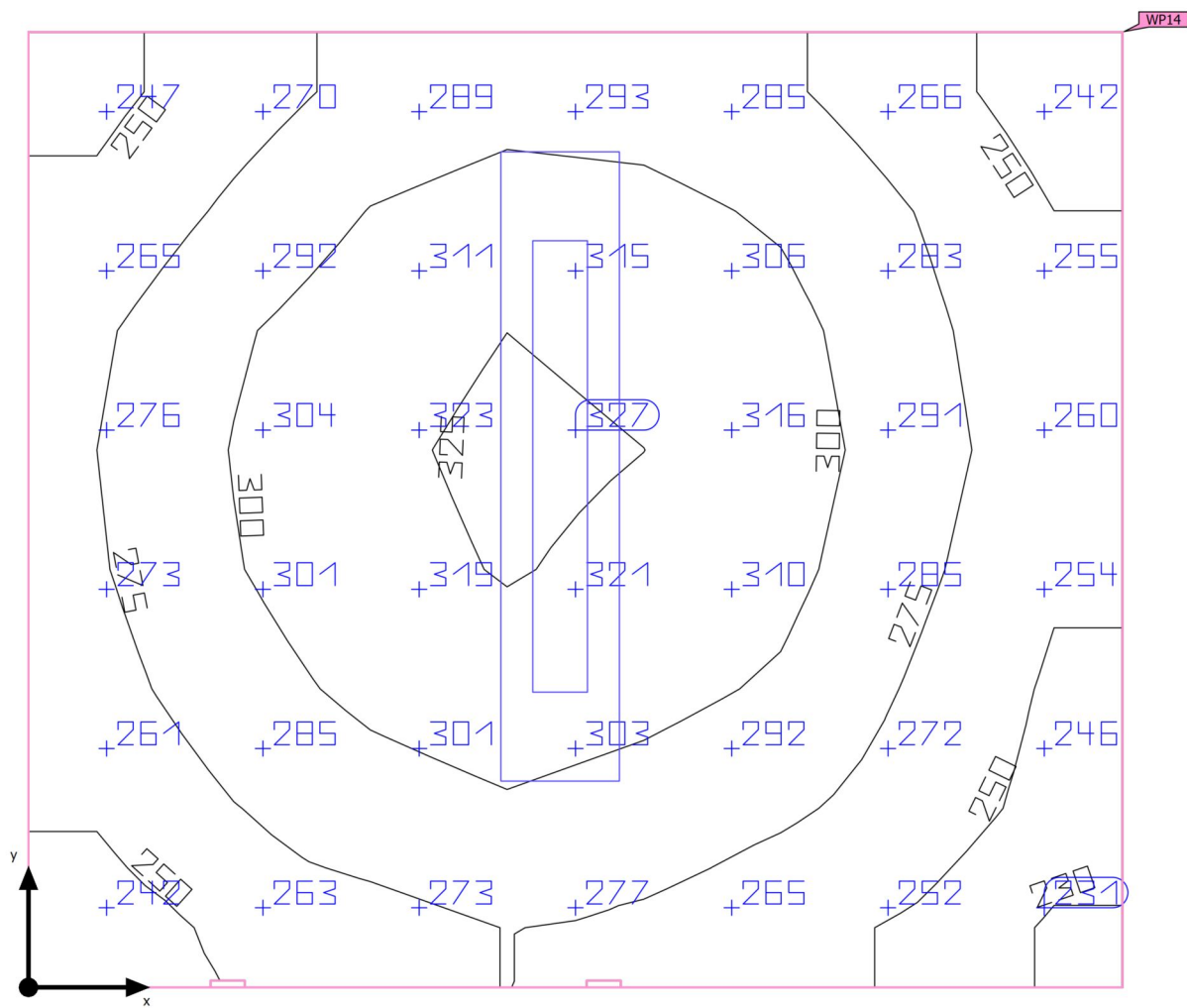
Working plane (216 MEETING)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (216 MEETING)	732 lx	427 lx	928 lx	0.58	0.46	WP8
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 217 TOILET (Light scene 1)

Summary



Ground area	5.03 m²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.000 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 217 TOILET (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	283 lx	≥ 500 lx	✗	WP14
	$U_o (g_1)$	0.80	≥ 0.60	✓	WP14
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	81.7 kWh/a	max. 200 kWh/a	✓	
Space	Lighting power density	6.56 W/m ²	–		
		2.32 W/m ² /100 lx	–		

(1) Based on a rectangular space of 2.095 m x 2.400 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

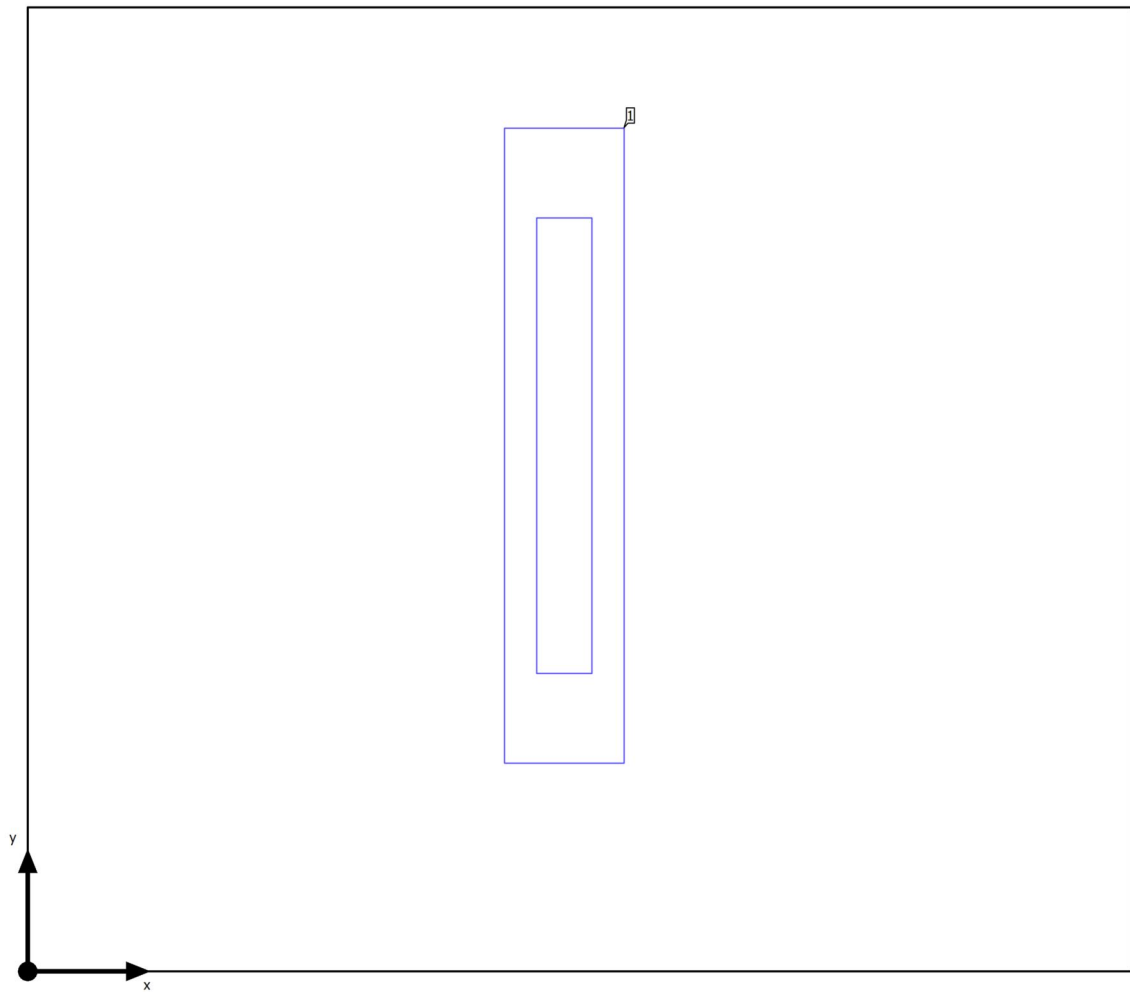
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

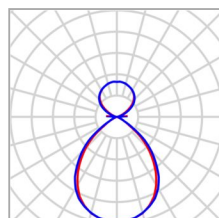
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 217 TOILET

Luminaire layout plan



Building A · FIRST FLOOR · 217 TOILET

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

Individual luminaires

X	Y	Mounting height	Luminaire
1.166 m	1.142 m	2.800 m	1

Building A · FIRST FLOOR · 217 TOILET

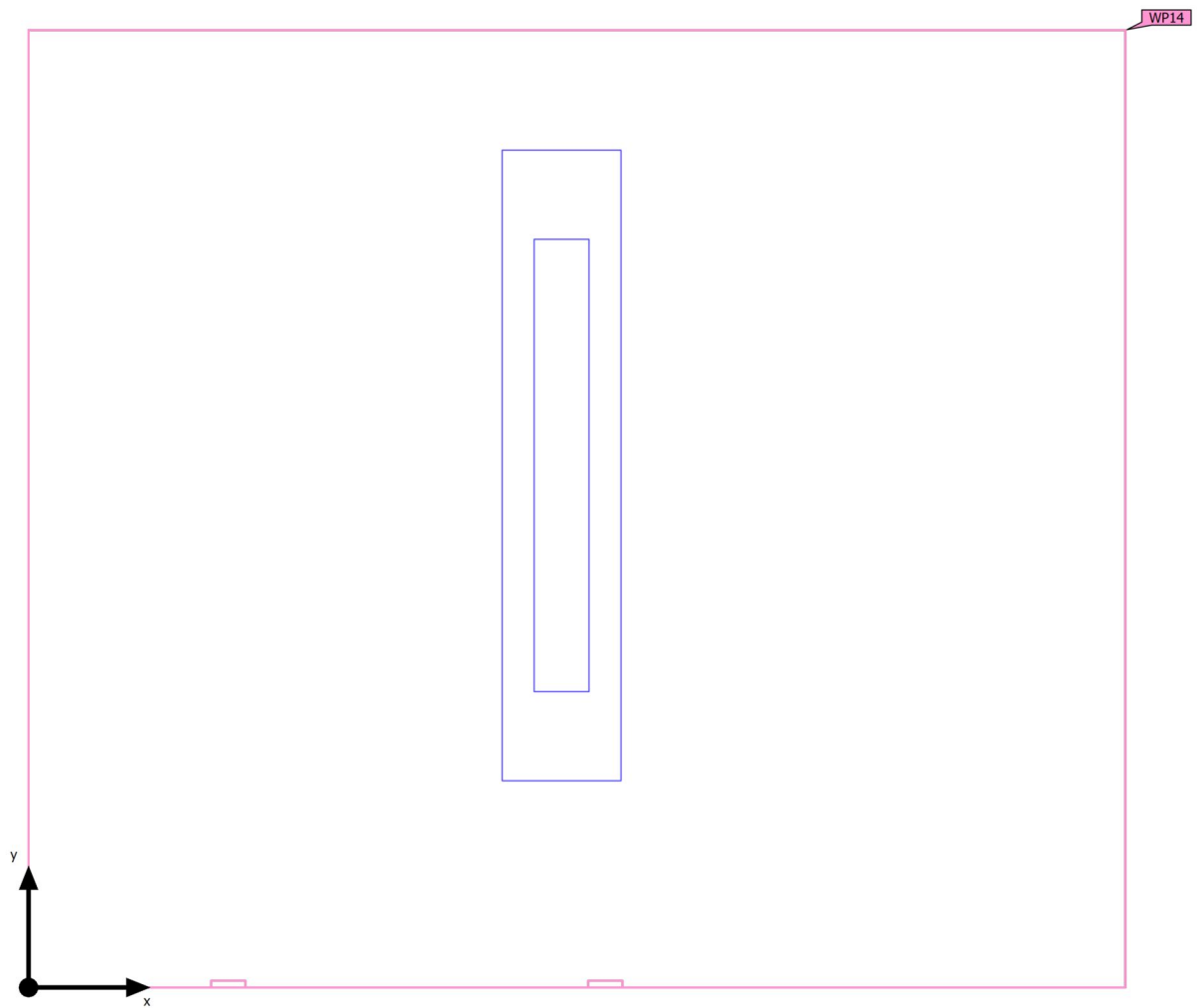
Luminaire list

Φ_{total} 5250 lm	P_{total} 33.0 W	Luminous efficacy 159.1 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 217 TOILET (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 217 TOILET (Light scene 1)

Calculation objects

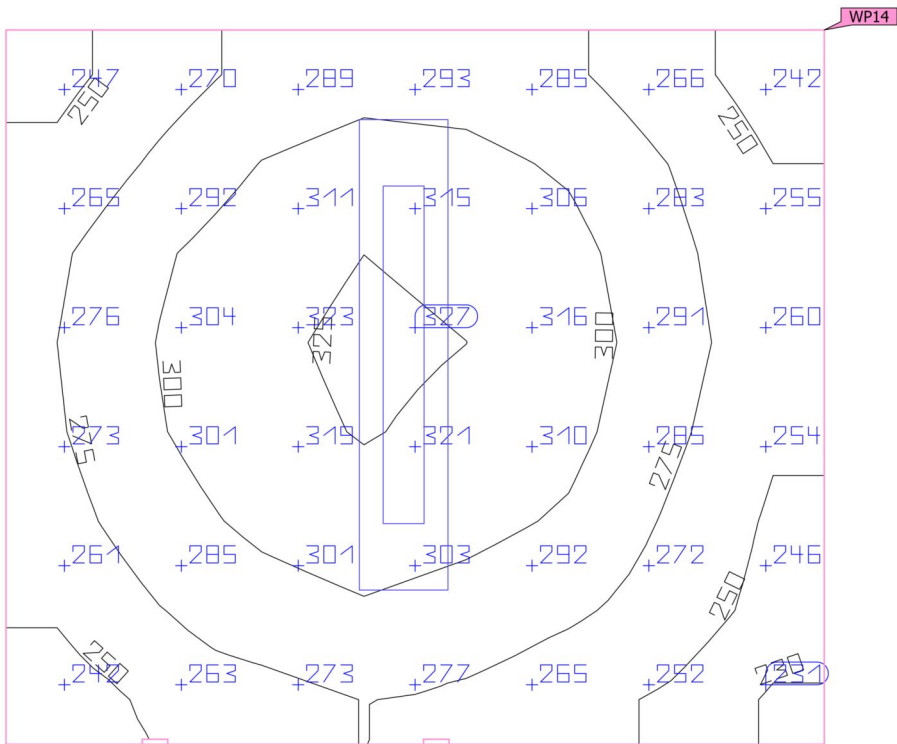
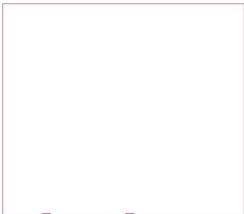
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (217 TOILET)	283 lx	227 lx	327 lx	0.80	0.69	WP14
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.000 m, Wall zone: 0.000 m	✗			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 217 TOILET (Light scene 1)

Working plane (217 TOILET)

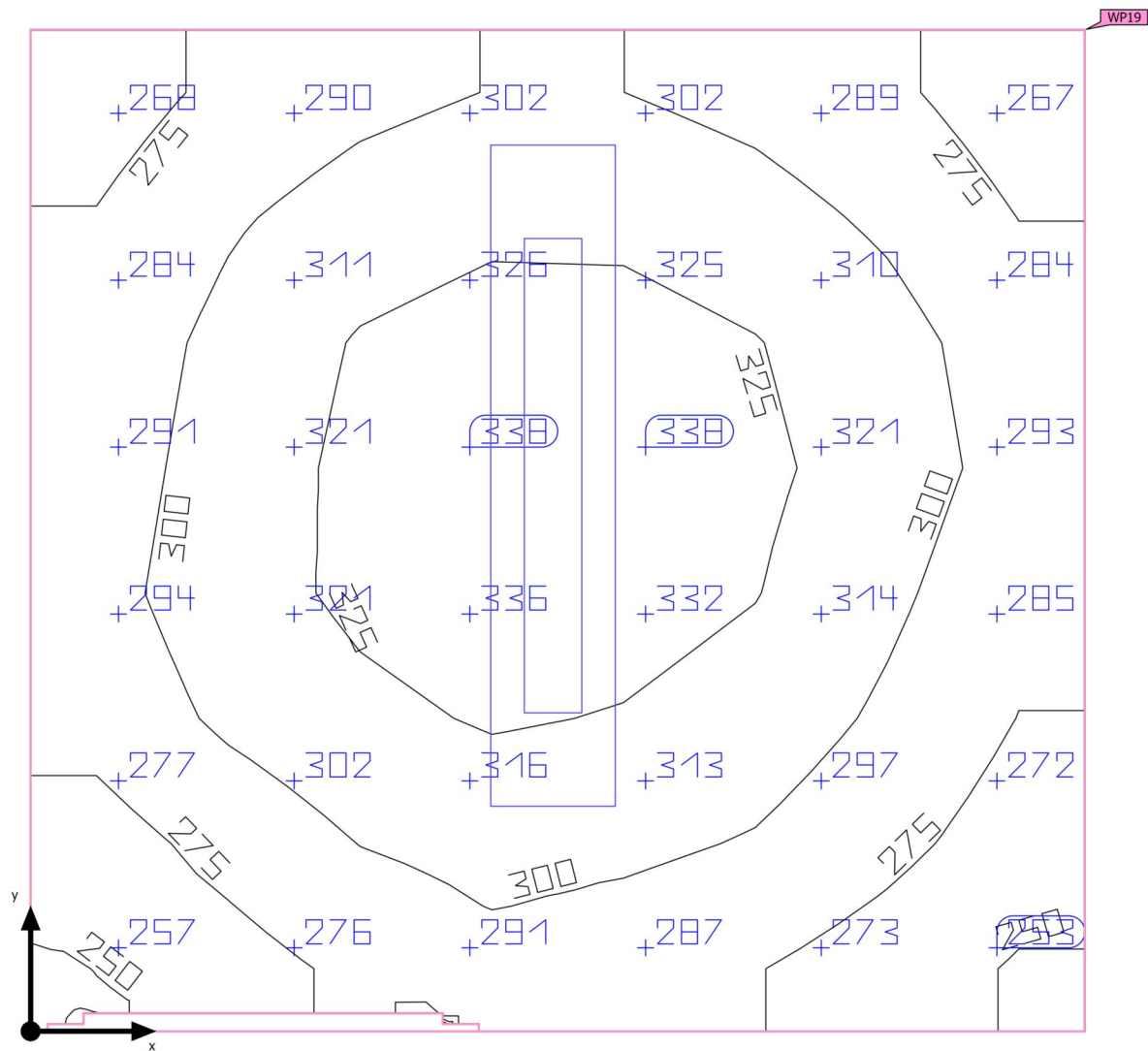


Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (217 TOILET)	283 lx	227 lx	327 lx	0.80	0.69	WP14
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.000 m, Wall zone: 0.000 m	✗			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 218 TOILET (Light scene 1)

Summary



Ground area	4.60 m ²
Reflection factors	Ceiling: 70.0 %, Walls: 50.0 %, Floor: 20.0 %
Maintenance factor	0.76 (fixed)

Clearance height	3.300 m
Mounting height	2.800 m
Height _{Working plane}	0.000 m
Wall zone _{Working plane}	0.000 m

Building A · FIRST FLOOR · 218 TOILET (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	299 lx	$\geq 500 \text{ lx}$	✗	WP19
	$U_o (g_1)$	0.72	≥ 0.60	✓	WP19
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	81.7 kWh/a	max. 200 kWh/a	✓	
Space	Lighting power density	7.18 W/m ²	–		
		2.40 W/m ² /100 lx	–		

(1) Based on a rectangular space of 2.090 m x 2.200 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

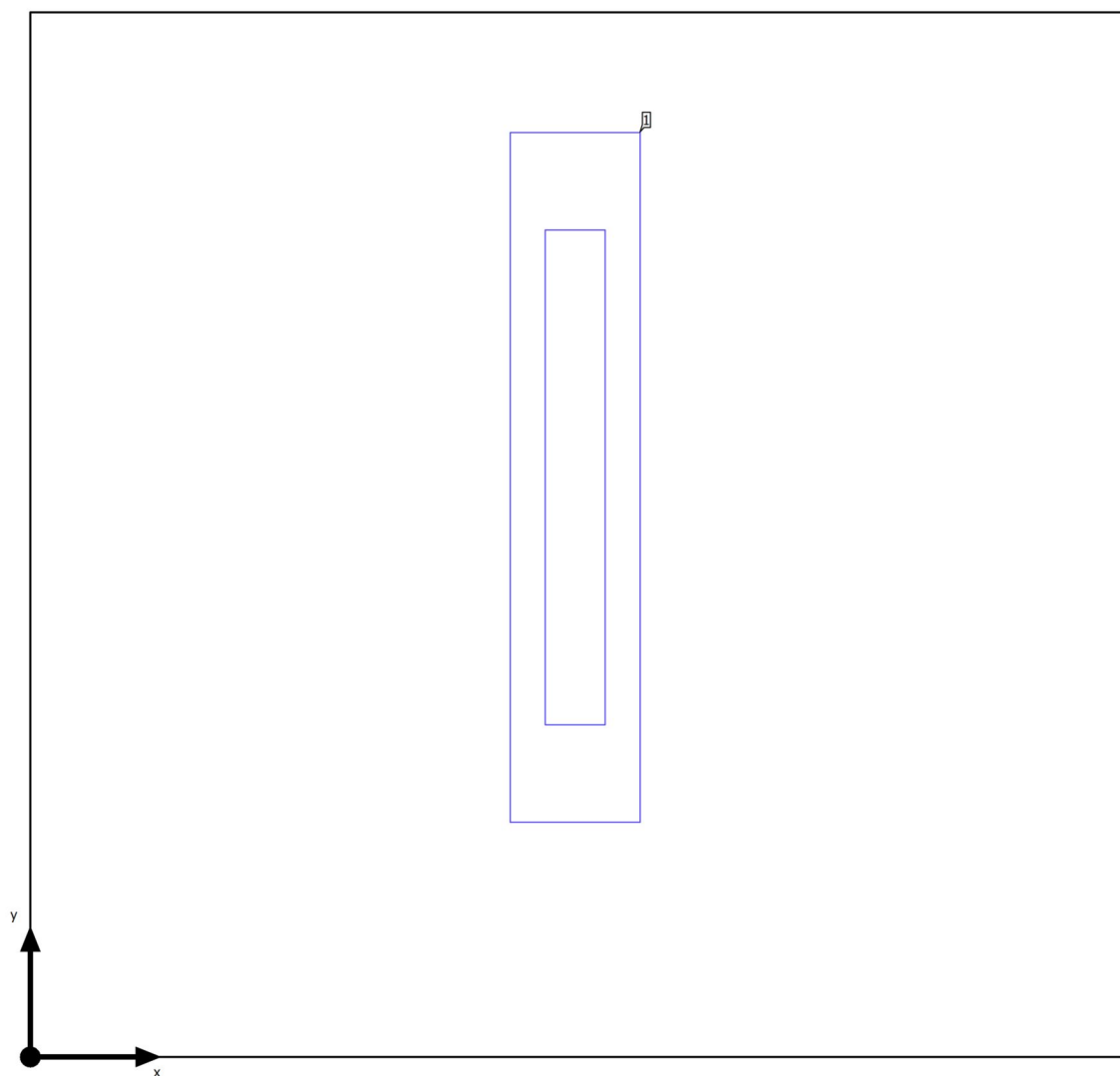
Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Luminaire list

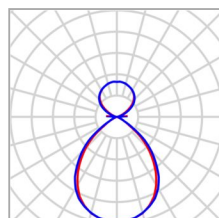
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 218 TOILET

Luminaire layout plan



Building A · FIRST FLOOR · 218 TOILET

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

Individual luminaires

X	Y	Mounting height	Luminaire
1.090 m	1.160 m	2.800 m	1

Building A · FIRST FLOOR · 218 TOILET

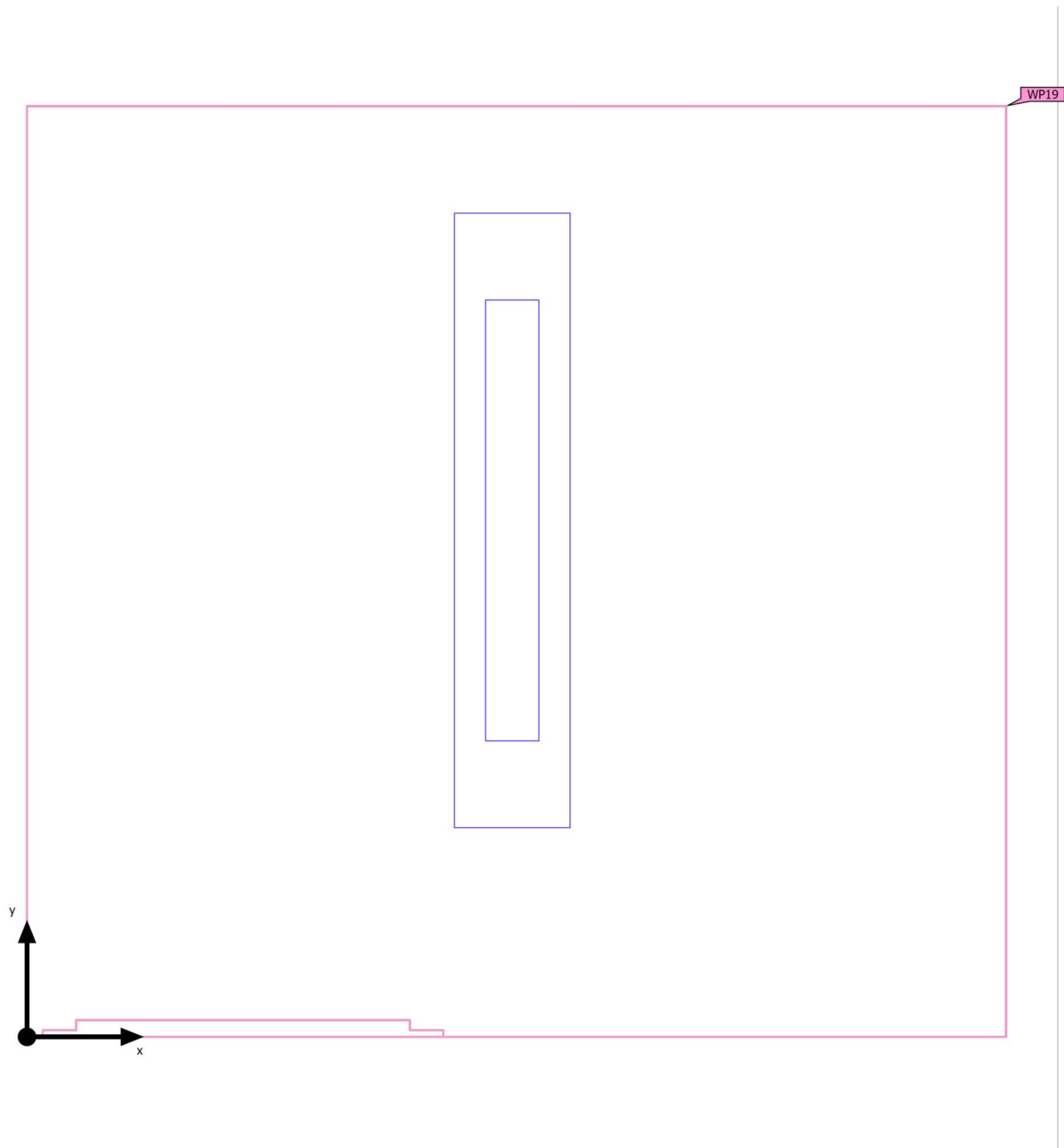
Luminaire list

Φ_{total} 5250 lm	P_{total} 33.0 W	Luminous efficacy 159.1 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
1	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 218 TOILET (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 218 TOILET (Light scene 1)

Calculation objects

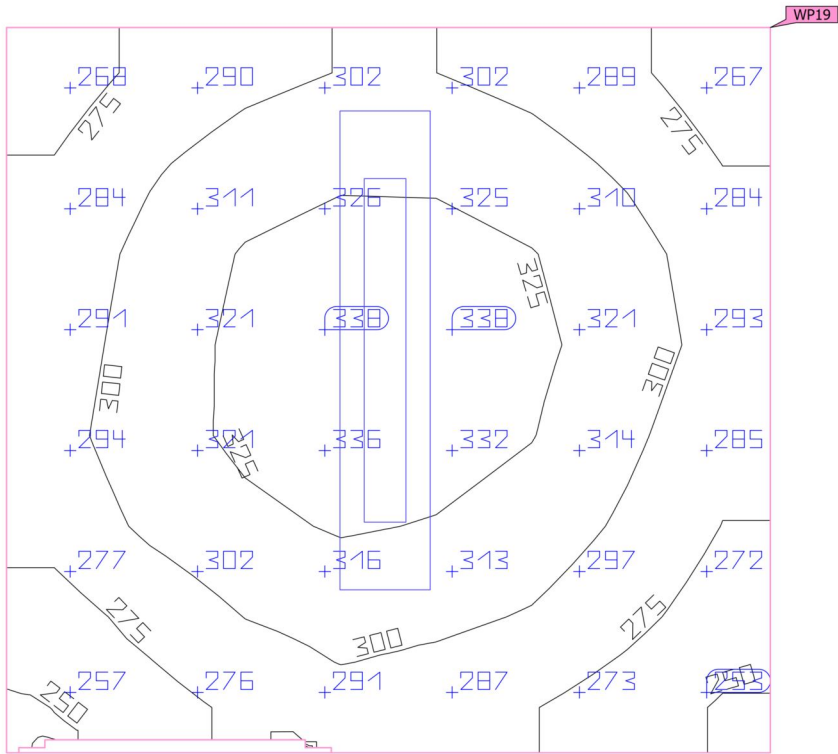
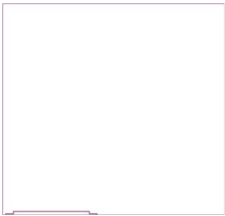
Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (218 TOILET)	299 lx	215 lx	340 lx	0.72	0.63	WP19
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.000 m, Wall zone: 0.000 m	✗			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

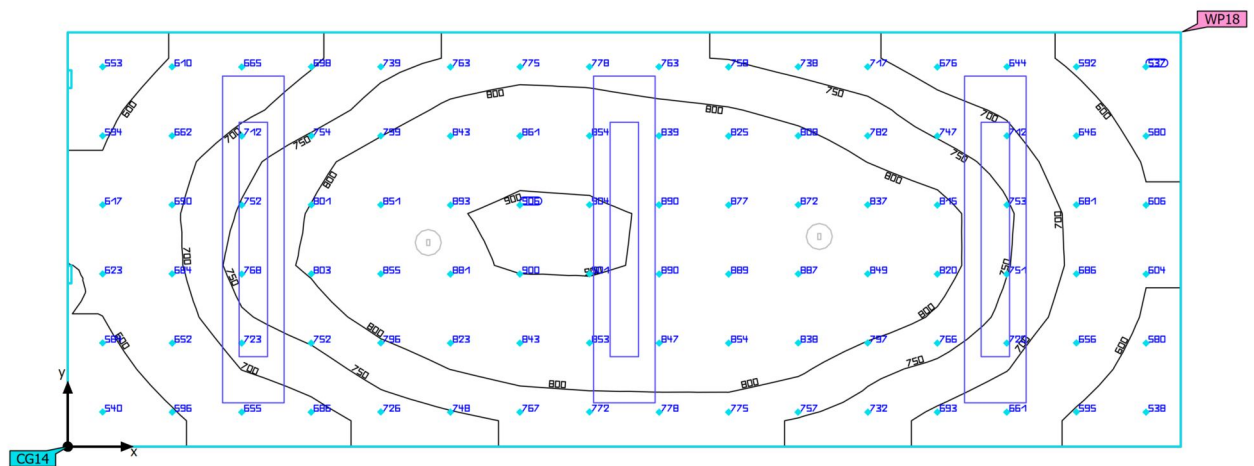
Building A · FIRST FLOOR · 218 TOILET (Light scene 1)

Working plane (218 TOILET)



Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (218 TOILET)	299 lx	215 lx	340 lx	0.72	0.63	WP19
Perpendicular illuminance (adaptive)	≥ 500 lx			≥ 0.60		
Height: 0.000 m, Wall zone: 0.000 m	✗			✓		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))



Building A · FIRST FLOOR · 219 MEETING (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	747 lx	$\geq 500 \text{ lx}$	✓	WP18
	$U_o (g_1)$	0.71	≥ 0.60	✓	WP18
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	245 kWh/a	max. 300 kWh/a	✓	
Space	Lighting power density	12.04 W/m ²	–		
		1.61 W/m ² /100 lx	–		

(1) Based on a rectangular space of 4.700 m x 1.750 m and SHR of 0.25.

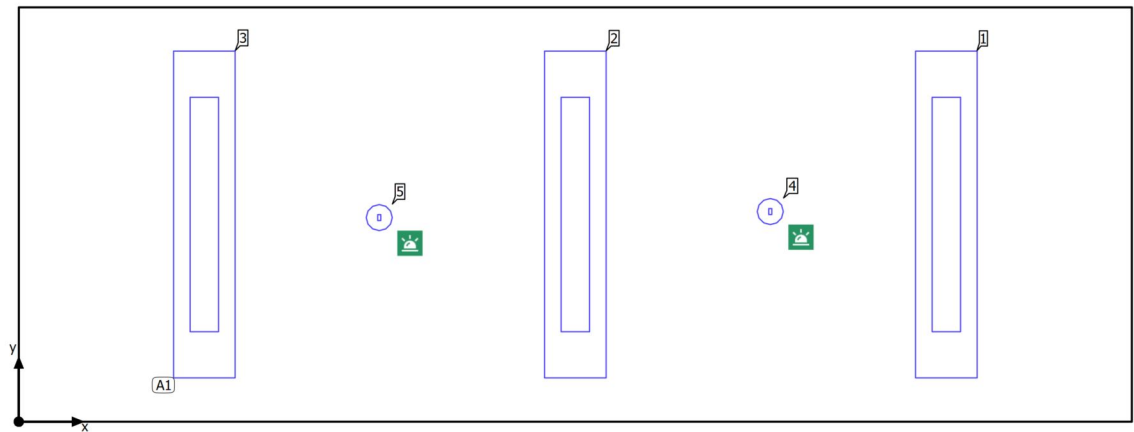
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

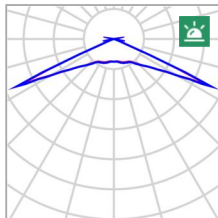
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
3	ETAP	-	R31111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 219 MEETING

Luminaire layout plan

Building A · FIRST FLOOR · 219 MEETING

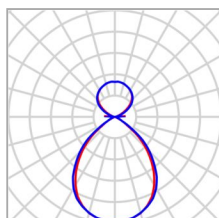
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
3.173 m	0.888 m	3.368 m	4
1.522 m	0.862 m	3.368 m	5

Building A · FIRST FLOOR · 219 MEETING

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

3 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	3.917 m / 0.875 m / 2.800 m	3.917 m	0.875 m	2.800 m	1
X-direction	3 pcs., Centre - centre, 1.567 m	2.350 m	0.875 m	2.800 m	2
Y-direction	1 pcs., Centre - centre, 1.750 m	0.783 m	0.875 m	2.800 m	3
Arrangement	A1				

Building A · FIRST FLOOR · 219 MEETING

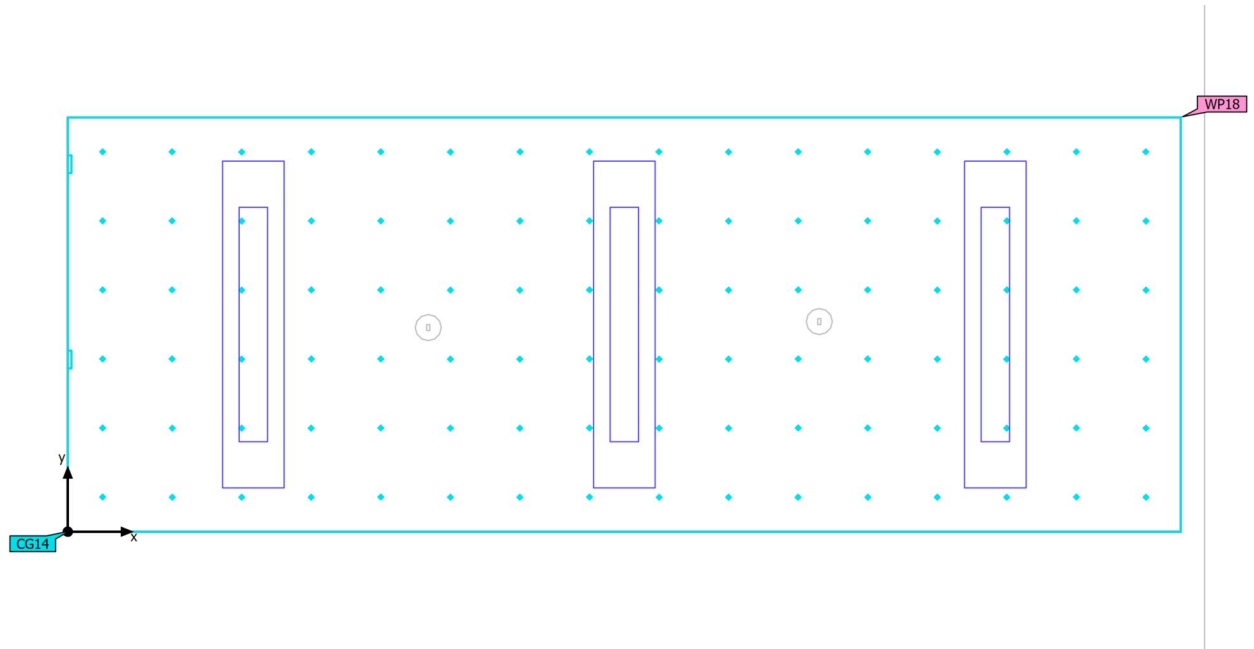
Luminaire list

Φ_{total} 16350 lm	P_{total} 100.8 W	Luminous efficacy 162.2 lm/W	$\Phi_{\text{Emergency lighting}}$ 600 lm	$P_{\text{Emergency lighting}}$ 1.8 W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	ETAP	-	K1R222/3-A-A630A630 (_LTRevAB)	0.9 W	300 lm	340.9 lm/W
				 0.9 W	300 lm (100 %)	-
3	ETAP	-	R3111/LEDN455DX1 (_LTRevAC)	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 219 MEETING (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 219 MEETING (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (219 MEETING) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	747 lx (≥ 500 lx) ✓	531 lx	908 lx	0.71 (≥ 0.60) ✓	0.58	WP18

Building A · FIRST FLOOR · 219 MEETING (Light scene 1)

Calculation objects

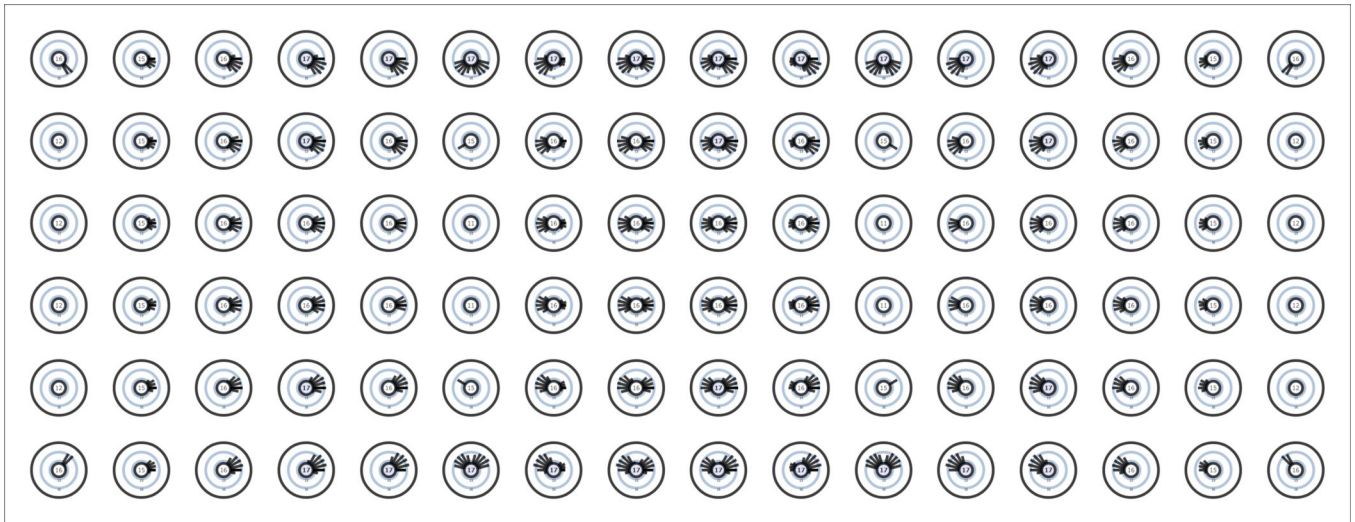
Glare plane 219 (RUG)

Strongest glare at	330°
max	17.0
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG14

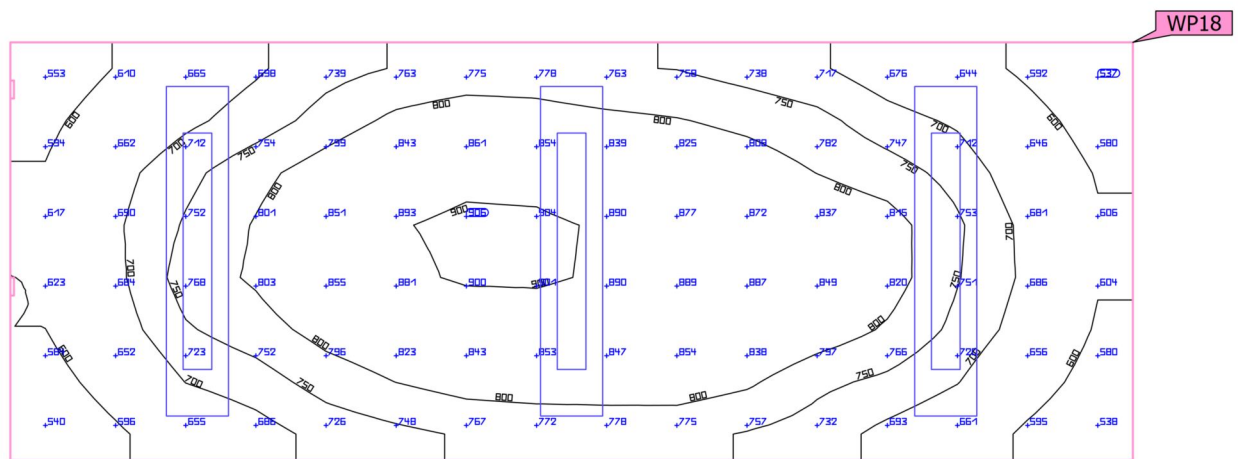
Building A · FIRST FLOOR · 219 MEETING (Light scene 1)

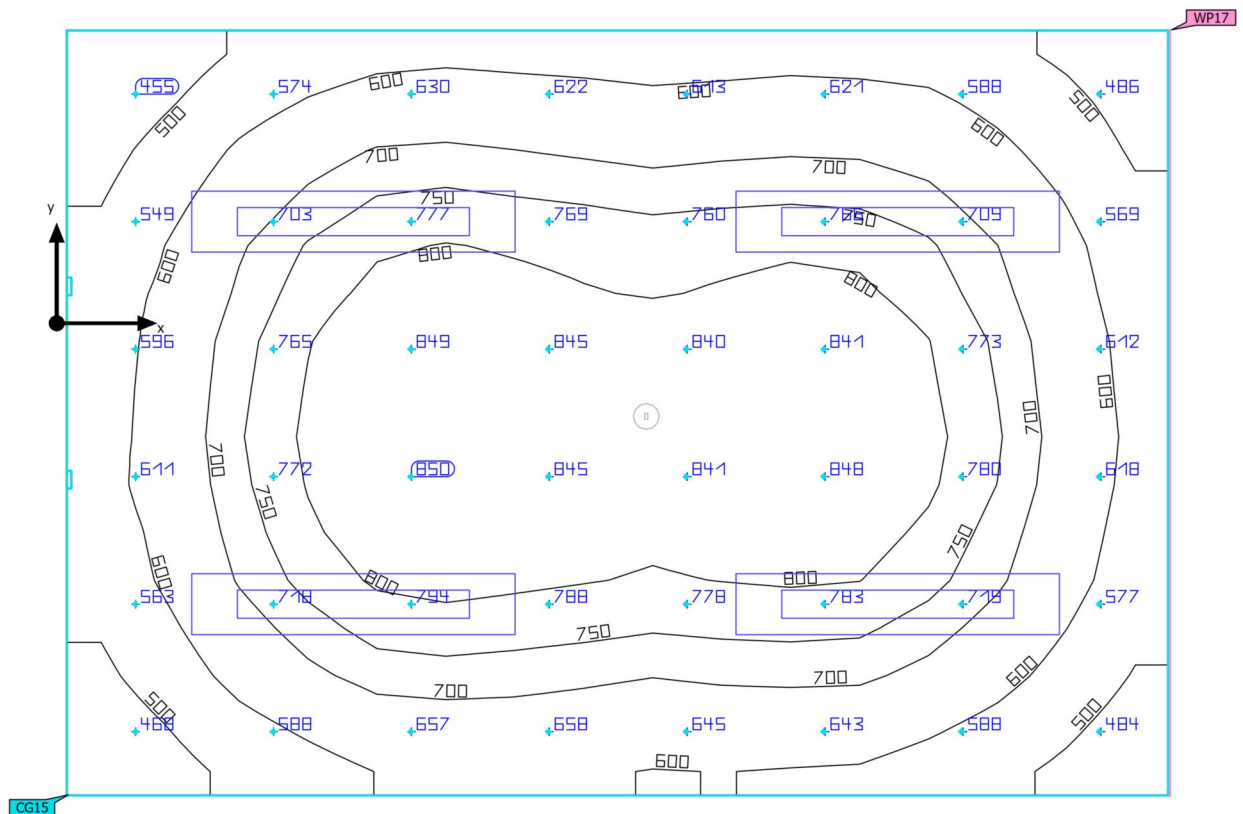
Calculation objects

Glare plane 219 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))





Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	684 lx	≥ 500 lx	✓	WP17
	$U_o (g_1)$	0.58	≥ 0.60	✗	WP17
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	16	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[206 - 327] kWh/a	max. 550 kWh/a	✓	
Space	Lighting power density	8.59 W/m ²	–		
		1.26 W/m ² /100 lx	–		

(1) Based on a rectangular space of 4.707 m x 3.265 m and SHR of 0.25.

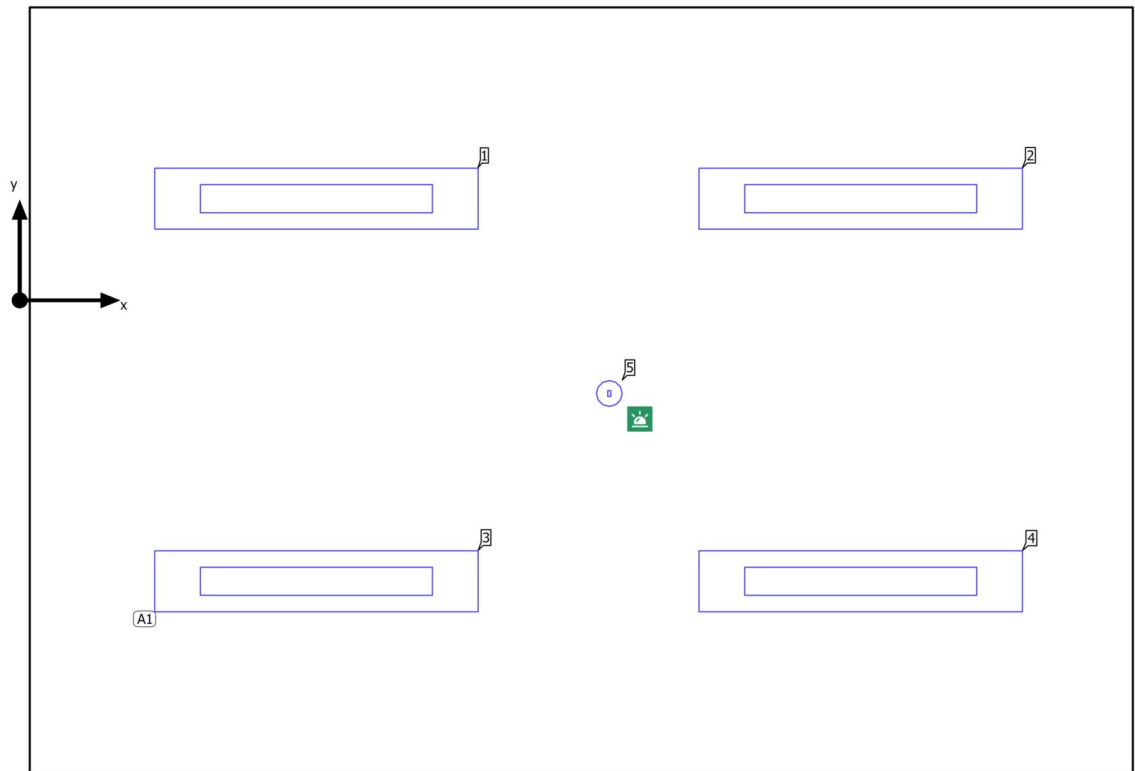
(2) Calculated using DIN:18599-4.

Utilisation profile: DIALux presetting (5.26.2 Standard (office))

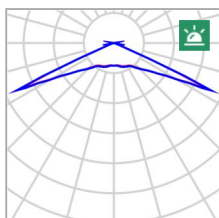
Luminaire list

pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)	16	33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 220 OFFICE

Luminaire layout plan

Building A · FIRST FLOOR · 220 OFFICE

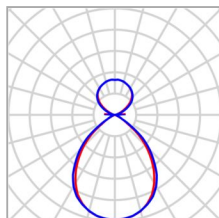
Luminaire layout plan

Manufacturer	ETAP	P	0.9 W
Article No.	-	P _{Emergency lighting}	0.9 W
Article name	K1R222/3-A-A630A630 (_LTRevAB)	Φ _{Luminaire}	300 lm
		Φ _{Emergency lighting}	300 lm
Fitting	1x light	ELF	100 %

Individual luminaires

X	Y	Mounting height	Luminaire
2.516 m	-0.397 m	3.368 m	5

Building A · FIRST FLOOR · 220 OFFICE

Luminaire layout plan

Manufacturer	ETAP	P	33.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	5239 lm
Article name	R3111/LEDN455DX1 (_LTRevAC)		
Fitting	1x LED		

4 x ETAP R3111/LEDN455DX1 (_LTRevAC)

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	3.589 m / 0.434 m / 2.800 m	1.266 m	0.434 m	2.800 m	1
X-direction	2 pcs., Centre - centre, 2.323 m	3.589 m	0.434 m	2.800 m	2
Y-direction	2 pcs., Centre - centre, 1.632 m	1.266 m	-1.199 m	2.800 m	3
		3.589 m	-1.199 m	2.800 m	4
Arrangement	A1				

Building A · FIRST FLOOR · 220 OFFICE

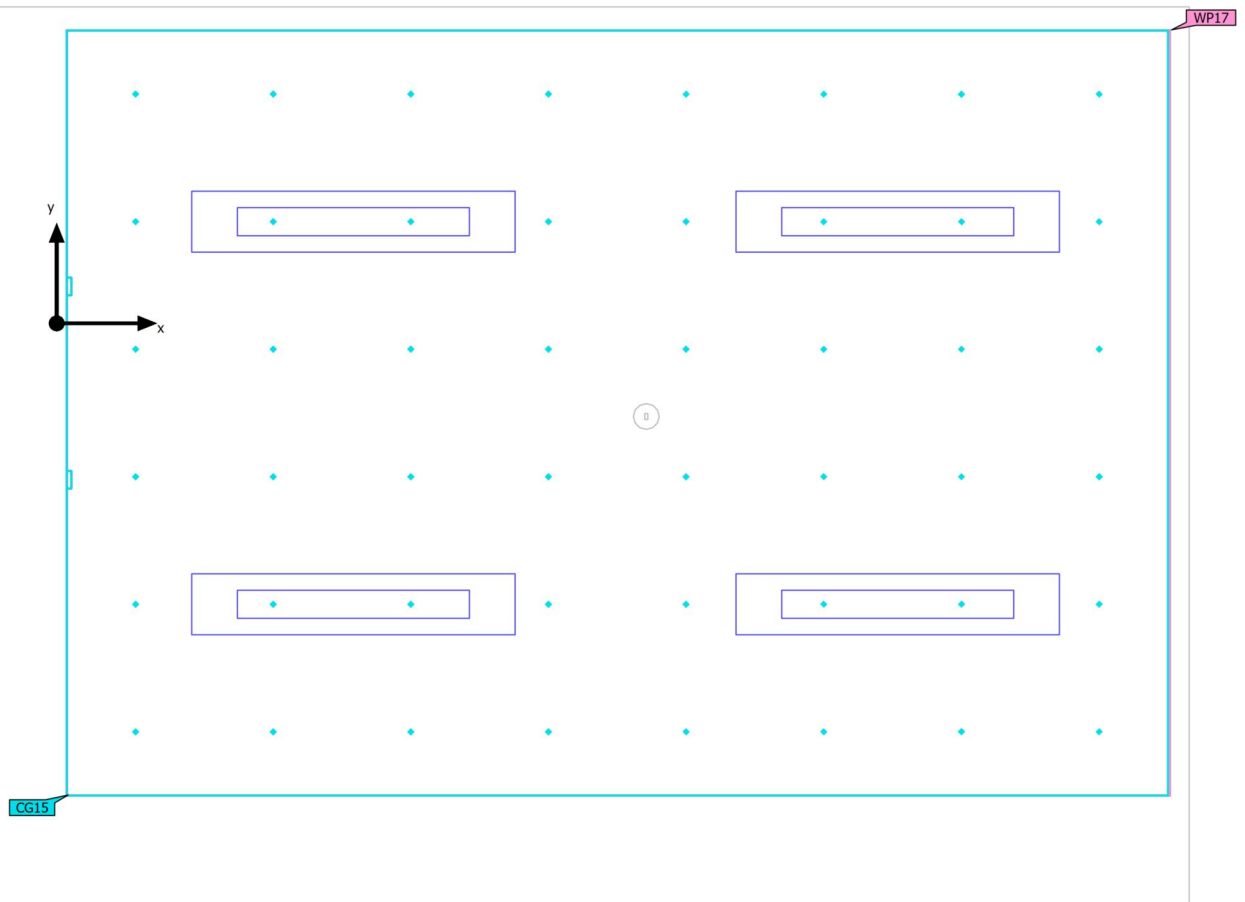
Luminaire list

Φ_{total} 21300 lm	P_{total} 132.9 W	Luminous efficacy 160.3 lm/W	$\Phi_{\text{Emergency lighting}}$ 300 lm	$P_{\text{Emergency lighting}}$ 0.9 W
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pcs.	Manufacturer	Article No.	Article name		P	Φ	Luminous efficacy
1	ETAP	-	K1R222/3-A-A630A630 (_LTRRevAB)		0.9 W	300 lm	340.9 lm/W
					0.9 W	300 lm (100 %)	-
4	ETAP	-	R3111/LEDN455DX1 (_LTRRevAC)		33.0 W	5239 lm	158.8 lm/W

Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

Calculation objects



Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (220 OFFICE) Perpendicular illuminance (adaptive) Height: 0.700 m, Wall zone: 0.000 m	684 lx (≥ 500 lx) ✓	400 lx	861 lx	0.58 (≥ 0.60) ✗	0.46	WP17

Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

Calculation objects

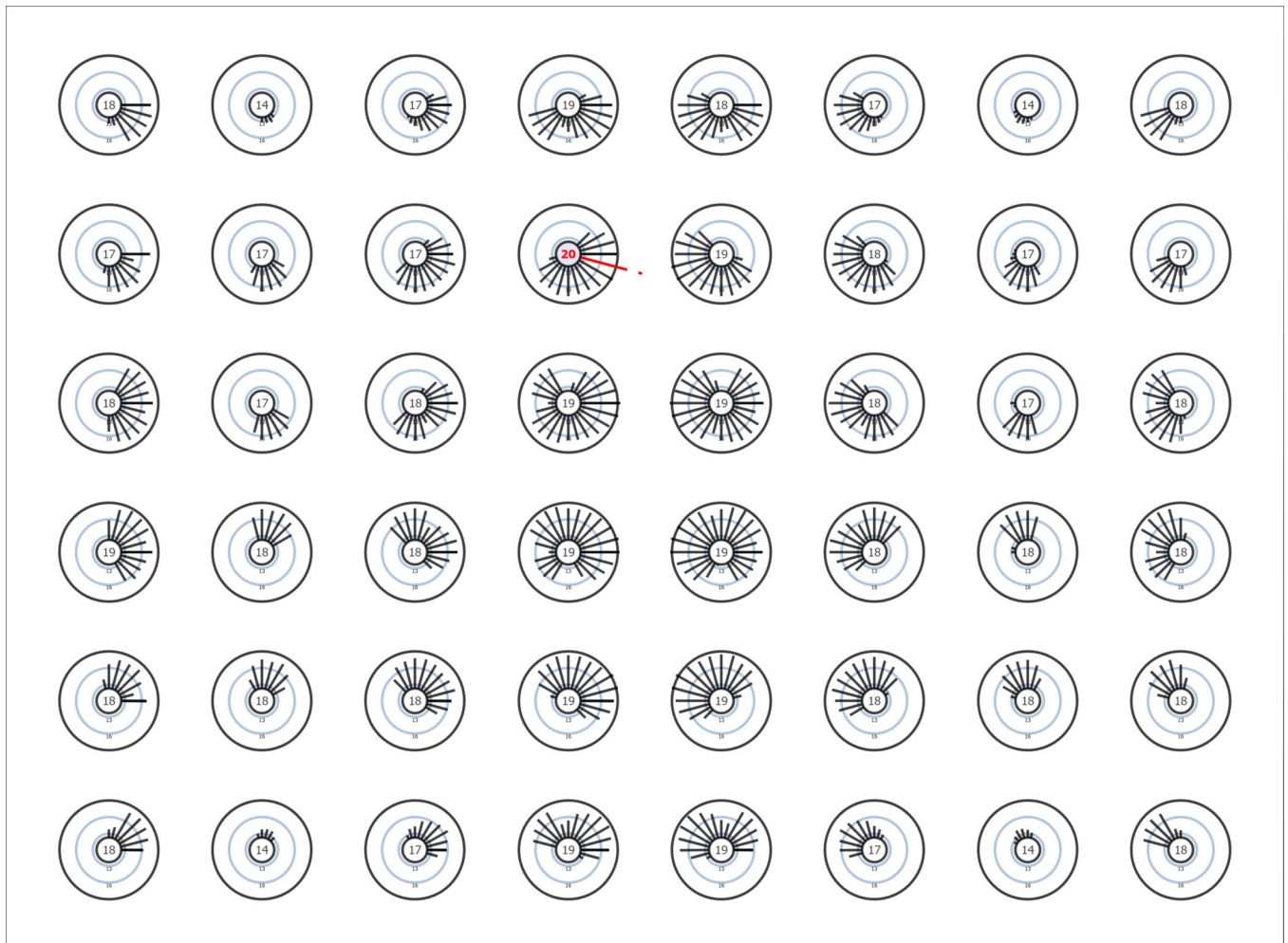
Glare plane 220 (RUG)

Strongest glare at	345°
max	19.5
Target	≤ 19.0
Viewing sector	0° - 360°
Step width	15°
Height	1.200 m
Index	CG15

Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

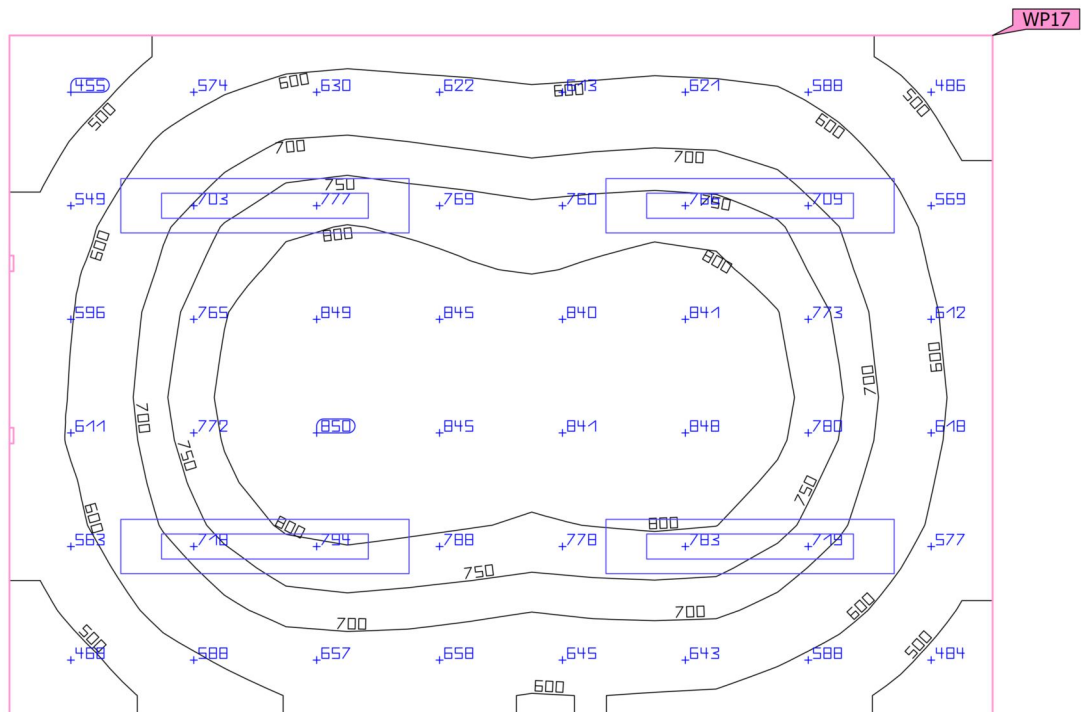
Calculation objects

Glare plane 220 (RUG)



Utilisation profile: DIALux presetting (5.26.2 Standard (office))

Building A · FIRST FLOOR · 220 OFFICE (Light scene 1)

Working plane (220 OFFICE)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (220 OFFICE)	684 lx	400 lx	861 lx	0.58	0.46	WP17
Perpendicular illuminance (adaptive)	(≥ 500 lx)			(≥ 0.60)		
Height: 0.700 m, Wall zone: 0.000 m	✓			✗		

Utilisation profile: DIALux presetting (5.26.2 Standard (office))